

A Full DR11 Strong-Lens Search in the DESI Legacy Surveys: CLAUDENET v3blend8 Finding with LENSJUDGE v3 HSC Cross-Validation

Greg Benson* Xiaosheng Huang†

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Abstract

We deploy the contaminant-aware CLAUDENET v3blend8 ensemble and the LENSJUDGE v3 vetting cascade on the final DESI Legacy Surveys data release (DR11). The **DR11-south** (DECam, native *griz*) sweep scores all 5.38×10^7 galaxies with the five-member v2-lean stage and re-ranks the survivors with the eight-member v3blend8 ensemble; a DR11-native recalibration is required because the deeper DR11 reductions have fatter random-galaxy score tails than DR9, tightening the matched-FPR thresholds (known-lens grade-A recall at the 10^{-4} operating point falls 62% $\rightarrow$ 41% relative to DR10 — a genuine DR9 $\rightarrow$ DR11 domain shift, not a threshold artifact, so the DR11 survivor list under-recovers grade-A lenses and is not complete). Group-conformal selection remains *power-limited* at $m \approx 5.4 \times 10^7$ (zero certified at $\alpha \leq 0.25$ with an 8M-negative calibration), so the candidate list is the v3blend8-ranked top- N , as in the prior campaigns. The decisive step is LENSJUDGE v3’s new **HSC PDR3 tier-2**: vetting the top-500 new candidates as a cascade (cheap detection triage $\rightarrow$ agentic mimic adjudication $\rightarrow$ high-resolution escalation, \$40 total) moves **26** HSC-covered candidates from DESI-ambiguous ($p_{\text{lens}} \sim 0.03\text{--}0.6$) to **24 HSC tier-2 grade-A/B** under LENSJUDGE’s automated grader (p_{lens} up to 0.97; a single automated grader, not a human committee or spectroscopy). An independent SuGOHI cross-match calibrates the engine: v3blend8 recovers **79** SuGOHI HSC committee lenses into the survivors at $2.7\times$ the survivor-median score, and **15 of the 24** graded A/B are SuGOHI lenses (a sensitivity check — the automated tier-2 grader’s precision on the genuinely-new candidates is not established here). The remaining **9** (8 distinct systems) are new to the DESI lens-finder catalogs and SuGOHI; a post-hoc literature crosscheck (Huang+2020, omitted from the campaign crossmatch, + NED/SIMBAD; App. A) then shows **five of these eight systems are previously-published lenses** (mostly Huang+2020; one also SOGRAS), leaving **3** genuinely new — automated-grader candidates pending follow-up. Beyond the HSC-overlap set, the honest-residual cascade also grades **108** candidates A/B at DESI resolution *alone* — with no HSC-SSP or Euclid Q1 coverage (App. B); a literature crosscheck finds **69 are previously-published lenses** (mostly DES, whose footprint this DECaLS-south region heavily overlaps) and **39 are new** (36 grade B and 3 grade A). These cannot be validated beyond DECaLS $1''$ — an unknown and likely large fraction are lrg+companion mimics — and the set is residual- and nondeterminism-sensitive (larger under the honest residual, but hotter rather than more complete), so it is a follow-up target list, not

*benson@cs.usfca.edu; Department of Computer Science, University of San Francisco, CA 94117, USA

†Department of Physics & Astronomy, University of San Francisco; and Physics Division, Lawrence Berkeley National Laboratory

a catalogue. The honest frame is unchanged from the v3 program: the model recovers known lenses and ranks new candidates the higher-resolution HSC grader rates highly, but *no net-new lens is established from DECaLS pixels alone* (the new candidates depend on HSC resolution and remain unconfirmed), and net-new discovery beyond existing high-resolution catalogs is resolution-bounded — the value of the *resolution lever* (now HSC over $\sim 1,300 \text{ deg}^2$, far wider than Euclid Q1) is to *convert* overlap candidates, not to manufacture discoveries. The **DR11-north** (BASS/MzLS) arm — which needs instrument-native retraining (v3 is DECam-specific) — is fully staged but deferred by a 7-day Perlmutter maintenance.

1 Premise

The CLAUDENET v3 program concluded that for DECaLS-resolution lens finding the binding constraint is *lens-vs-mimic separation* and *vetting resolution*, not architecture: **v3blend8** (the SHIP-gated eight-member ensemble: **effnet_B**, **zoobot_N**, and the v2-hard + v3-mimic-blend versions of **effnet_S2**, **effnet_B3**, **resnet46_C**) rejects common contaminants far better than v2 but is statistically tied with the originals at the hardest real-vs-mimic frontier, which is resolution-bounded (Inchausti Reyes et al., 2025). The productive mode is therefore *DESI pre-screen* $\rightarrow$ *high-resolution confirmation in the overlap*. This campaign operationalizes that on the final release (DR11) and exploits the new lever in LENSJUDGE v3: an **HSC PDR3** tier-2 grader (Aihara et al., 2022) ($0.168''/\text{px}$, $\sim 0.6''$ seeing) whose footprint (HSC-SSP Wide, $\sim 1,300 \text{ deg}^2$) is $\sim 20\times$ the Euclid Q1 area used previously, vastly widening the confirmable overlap.

2 The DR11-south sweep

DR11-south is the deepest DECam reduction to date: 1,600 sweep files (1.7 TB) yielding a parent of **53,809,040** galaxies after the published Inchausti cuts ($\text{TYPE} \in \{\text{SER}, \text{EXP}, \text{DEV}, \text{REX}\}$, $\text{NOBS}_{grz} \geq 3$, $m_z < 20$), 51.4M with native *i*. We shard into 32 footprint-pure, brick-disjoint parts, extract 101-px *grz* cutouts (100% success, 6.0 TB), and score every galaxy with the five v2-lean members.

DR11-native recalibration (required). The DR9-fit per-member 10^{-4} -FPR thresholds, when transferred to DR11 score-space, are *too tight*: DR11’s deeper imaging produces a fatter high-score tail of random galaxies (e.g. the **resnet46_C** isotonic score saturates, its 10^{-4} threshold moving $0.81 \rightarrow 1.00$, effectively reducing the union to four members). We recalibrate from the scored population itself (a 8M-galaxy random NegEval), yielding **95,104** survivors. Re-measuring recall of the published Inchausti 811 against these survivors gives grade-A 41% (37/90), versus 62% for the DR10-recalibrated DR10 sweep. We diagnose this drop below as dominated by the *stage-1 selection rule*, not a model failure: the denominator includes north and parent-cut lenses (so 41% is a conservative floor — the searchable recall is 54%), and the DR9-trained members still separate DR11 lenses from random DR11 galaxies near-perfectly.

Selection. Scoring the survivors with all eight **v3blend8** members and ranking by the mean calibrated score, the top-300 contain **102 known** lenses (recovered into the top of the list — a consistency check) and **198 new** candidates (top-300 **v3blend8** score 0.812–0.939). The discovery set is the **top-500 new** (**v3blend8** score 0.734–0.933); the broader survivor pool has 1,071 survivors above **v3blend8** 0.7 (726 new + 345 already-known).

Recall is recoverable: the union selector, not the model. A post-hoc diagnosis (scripts 380/381/382) decomposes the 41% into three causes and shows the model is not the bottleneck.

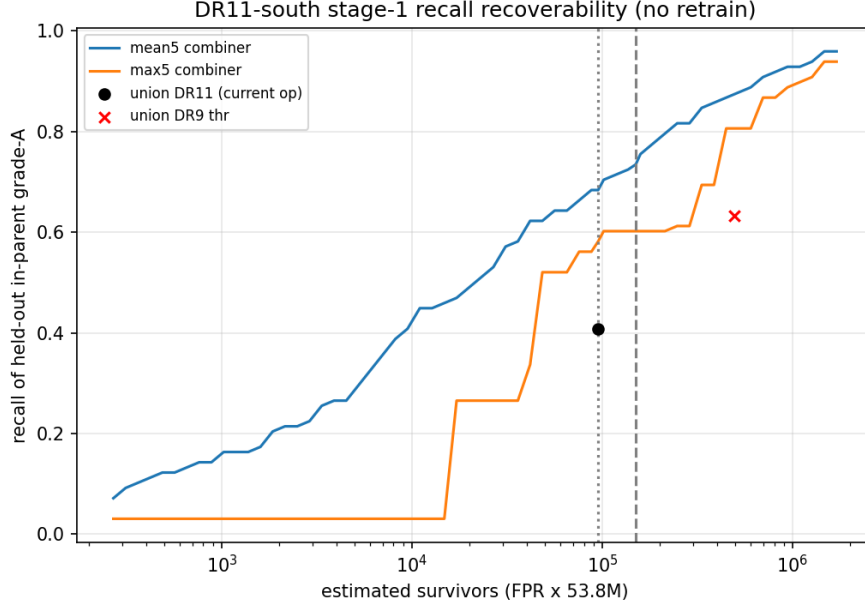


Figure 1: DR11-south stage-1 recall recoverability. Recall of held-out in-parent grade-A lenses against the estimated survivor budget (FPR $\times$ 53.8M) for the calibrated mean and max combiners; the per-member union operating point (black) sits far below the mean curve at the same budget, and the DR9 union thresholds (red) only reach comparable recall at $\sim 5\times$ the budget. The mean recovers the DR10 in-domain recall at under half the operating budget — the loss was the union selector, not a model domain shift.

(i) *Denominator*: 21/90 grade-A lie outside the searched footprint or are removed by the parent selection, so the *searchable* recall is 53.6% (37/69), not 41%. (ii) *Separation*: the threshold-free AUC of in-parent held-out grade-A versus a 2 M DR11 random NegEval is **0.9955** for the five-member calibrated mean — the DR9-trained members rank real DR11 lenses far above random DR11 galaxies, i.e. no feature-level domain collapse. (iii) *Selector*: stage-1 used a per-member 10^{-4} *union*, which on deeper imaging is dominated by single-member spikes and loses the saturated **resnet46_C** member. Replacing it with a calibrated-*mean* rank at the *same* 95k budget recovers Inchausti grade-A recall 54% $\rightarrow$ 75% (80% at the 150k budget, 88% on the held-out subset; Fig. 1) and reaches the DR10 in-domain 62% at < 43 k survivors — with *no* retraining or re-scoring, since the parent scores already exist. The calibrated mean is therefore now the default stage-1 selector (162/370 --**stage1-combiner mean**); the mean-selected DR11-south set retains 42.8k of the union survivors and adds 107k consensus-ranked candidates. The as-run vetting below stands on the union set; the mean selector is adopted going forward (DR11-north).

Certified-FDR (power-limited, reported honestly). Group-conformal selection (Jin and Candès, 2023) with the 8 M NegEval certifies **zero** candidates at $\alpha = 0.05/0.10/0.25$: the per-group p -value floor $1/(n_{\text{cal}}+1) \approx 2.5 \times 10^{-7}$ is $\sim 50\times$ short of what full- m Benjamini-Hochberg needs at $m = 5.38 \times 10^7$ to select even one. This is the same power-floor diagnosed at DR9/DR10 scale — rigorous full-population FDR needs an infeasibly large calibration set — so the candidate list is the v3blend8-ranked top- N , not an FDR-certified set. The arithmetic self-test (14/14) and the byte-level scoring-path check pass.

Table 1: DR11-south cascade outcome (500 new candidates, `pass_frac=0.7`).

quantity	value	note
graded (stage-1 triage)	500	\$0.012 each
escalated (agentic stage-2)	350	v2 mimic rubric
reached HSC tier-2	26	HSC-SSP Wide coverage
grade A / B / C / D	27 / 76 / 125 / 272	final (DESI- or HSC-grade)
HSC tier-2 grade-A/B	24 (19 A, 5 B)	automated; 8 more A are DESI-only
total cost	\$40.38	

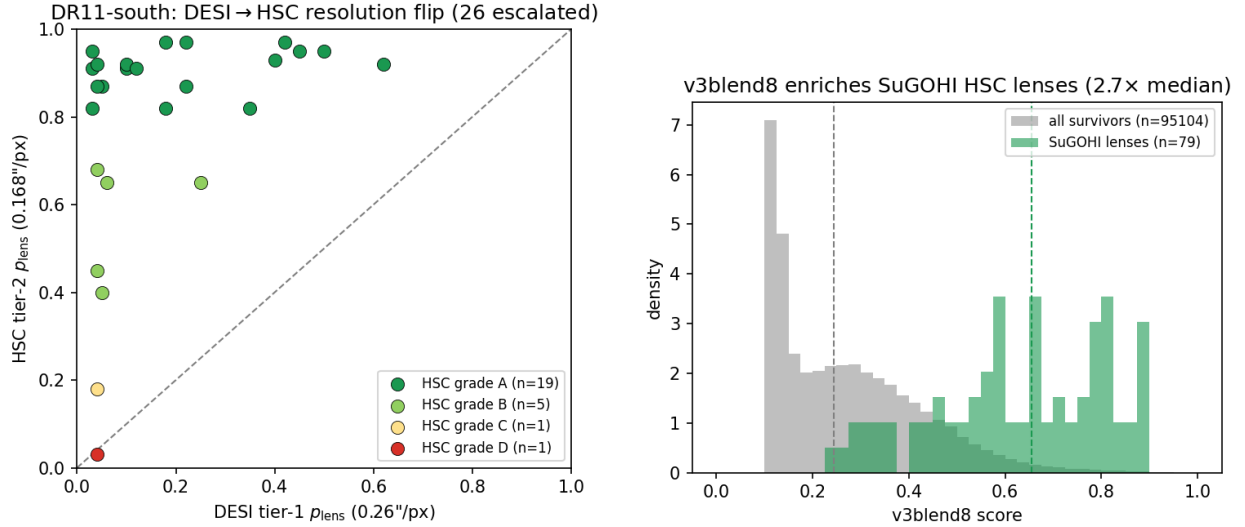


Figure 2: Left: the DESI→HSC resolution flip — the 26 HSC-covered candidates move from DESI-ambiguous (tier-1 p_{lens}) to the HSC tier-2 grade (automated grader), 24 landing A/B above the $y=x$ line. Right: `v3blend8` enriches the independent SuGOHI HSC lenses to $2.7\times$ the survivor-median score (median lines dashed).

3 LensJudge v3 cascade vetting + HSC cross-validation

The DESI viewer exposes no public DR11 layer (and is unreachable from our host), so we stage the 500 candidates’ *grz* cubes from Perlmutter as on-disk FITS, ensuring LENSJUDGE grades the exact pixels the CNN scored. We then run the B7 deploy cascade: a loop-free **direct** detection triage on all 500 ($\sim \$0.012$ each), escalating the top 70% by rank to the agentic v2 mimic-adjudication grader, which itself escalates to **HSC PDR3 tier-2** wherever HSC covers the position. Total cost **\$40.4** (campaign total \$43 of a \$250 budget).

The resolution lever converts. Every tier-2 grade-A/B candidate shows a large DESI→HSC flip: median tier-1 $p_{\text{lens}} \approx 0.1$ rising to tier-2 0.82–0.97 (e.g. $0.03 \rightarrow 0.95$, $0.04 \rightarrow 0.92$). At DECaLS $1''$ seeing these are ambiguous (lrg+companion $\approx$ lens); at HSC $0.6''$ the arc/ring morphology resolves — the flip is the grader’s response to resolution, not an observational confirmation.

Table 2: The 9 HSC tier-2 grade-A/B candidate detections (8 distinct systems). A literature crosscheck (Huang+2020 + NED/SIMBAD; App. A) finds **5 systems are previously-published** (*known*: H20 = Huang+2020; one also SOGRAS) and **3** genuinely new. p_1, p_2 = LENSJUDGE tier-1 (DESI) / tier-2 (HSC) lens probability.

name	RA	Dec	grade	$p_1 \rightarrow p_2$	v3blend8	known
s_340513_3688	16.332	+1.749	A	0.22 $\rightarrow$ 0.97	0.77	Huang+2020
s_327526_9059	9.722	−0.409	A	0.04 $\rightarrow$ 0.92	0.74	new
s_351303_461	194.534	+3.536	A	0.62 $\rightarrow$ 0.92	0.89	Huang+2020
s_345385_1515 [†]	154.312	+2.489	A	0.10 $\rightarrow$ 0.92	0.81	Huang+2020
s_340971_2860	130.857	+1.784	A	0.12 $\rightarrow$ 0.91	0.76	new
s_345385_1514 [†]	154.312	+2.488	A	0.05 $\rightarrow$ 0.87	0.86	Huang+2020
s_326089_1785	10.288	−0.730	A	0.03 $\rightarrow$ 0.82	0.83	H20+SOGRAS
s_325163_2981	138.848	−0.923	A	0.18 $\rightarrow$ 0.82	0.74	new
s_318043_1037	158.789	−2.304	B	0.04 $\rightarrow$ 0.45	0.86	Huang+2020

[†] s_345385_1514/1515 are adjacent ($\sim 1''$) detections of the same system, so the 9 detections are **8 distinct candidate systems** (8 A-detections / 7 distinct grade-A + 1 grade-B).

Independent SuGOHI validation. Against the SuGOHI/HOLISMOKES HSC committee lens catalog — an *independent* sample with expert grades and spectroscopic redshifts (Sonnenfeld et al., 2018) — v3blend8 recovers **79** lenses into the 95,104 survivors at a median v3blend8 score $2.7\times$ the survivor median (27 in the top-500), and **15 of the 24** tier-2 grade-A/B candidates are SuGOHI lenses. These 15 are a *sensitivity* check — the automated HSC grader recovers known committee lenses as grade-A/B — but they do not establish its *precision* on the genuinely-new candidates: no negative control was run through tier-2 and the certified-FDR is power-limited (§2), so the false-positive rate of the new set is unquantified.

The candidate catalog. Of the 24 tier-2 grade-A/B candidates, **0** are in the DESI lens-finder catalogs (by construction — the 500 were the new set after removing the 102 known), 15 are SuGOHI cross-matches, and **9** (8 distinct systems) are new to the DESI finders and SuGOHI (Table 2). A post-hoc literature crosscheck — Huang+2020 (omitted from the campaign’s lineage crossmatch) plus a NED/SIMBAD cone search (App. A) — then shows **five of these eight systems are previously-published lenses** (mostly Huang+2020; one also in the SOAR Gravitational Arc Survey), leaving **3** genuinely new (s_327526_9059, s_340971_2860, s_325163_2981); the recoveries are an external validation of the finder. The honest caveats: LENSJUDGE’s HSC grading is automated (a single Claude grader, not a human committee, no spectroscopy), and “new” here means new to the DESI finders, SuGOHI, *and* the NED/SIMBAD literature. They are candidates warranting human-expert and spectroscopic follow-up, not confirmed lenses.

Active-learning loop. The cascade’s agent-confirmed mimics (grade-D + named contaminant) export to **291** hard negatives in the CLAUDENET mimic-bank schema — lrg_companion 243-dominant, confirming that the high-v3blend8 new frontier is lrg+companion-dominated at DECaLS resolution (exactly the population the HSC tier-2 rejects). Folding these into the bank and a model re-iteration are Perlmutter GPU work, deferred (below).

Residual robustness (honest signed- χ residual). After this vetting was complete, LENSJUDGE’s lens-light *residual* view — one of the three images the grader sees per candidate — was

Table 3: Paired residual A/B on the same 500 DR11-south candidates: the prior (legacy) vetting run, a legacy-residual rerun (the LLM-nondeterminism control), and the new honest- χ -residual run. The honest residual runs $\sim 2\times$ hotter at the DESI tier-1 grade (raw A/B 90 $\rightarrow$ 135) but the HSC tier-2 gate absorbs it, so the confirmed catalogue (tier-2 A/B) is stable (24/24/27).

run	reached HSC tier-2	tier-2 A/B	raw A/B (all 500)	grade A/B/C/D	cost
Prior (legacy, Jun-17)	26	24	103	27/76/125/272	\$40.38
Legacy-now (control)	28	24	90	28/62/135/275	\$40.69
New (χ residual)	31	27	135	38/97/128/237	\$48.55

replaced: the legacy Gaussian high-pass (which discards the sign and imprints a four-lobe “butterfly” artifact on bright ellipticals that can mimic an arc) gave way to a signed, noise-normalised elliptical-model residual $\chi = (\text{data} - \text{model})/\sigma$, displayed per band on a fixed $\pm 5\sigma$ red-blue scale so a real arc must appear as red excess coherent across g and r (Fig. 3; full anatomy in the companion LENSJUDGE residual report). To test whether the DR11 catalogue *depends* on the old residual, we re-ran the *entire* 500-candidate cascade twice in a paired design — a legacy-residual rerun (an LLM-nondeterminism control) and the new χ residual — against the stored vetting run (Table 3). The new residual runs $\sim 2\times$ hotter at the DESI tier-1 grade (raw A/B 90 $\rightarrow$ 135), but the HSC tier-2 gate absorbs the inflation, so the confirmed catalogue is stable (tier-2 A/B 24 $\rightarrow$ 27). Crucially, **all 9 new-class candidate systems survive as grade A/B under the honest residual** (9/9; 13/15 SuGOHI cross-matches likewise; App. C, Table 5). The paired control is decisive: of five apparent new tier-2 promotions relative to the stored run, *four also appear in the legacy-now rerun* (run-to-run nondeterminism, which moves $\sim 18\%$ of raw grades), leaving only one genuinely new borderline-B candidate; conversely the honest residual *recovers* two known catalogue systems that nondeterminism had dropped. The two SuGOHI lenses that fall below A/B under the new residual are independently-confirmed committee lenses scored conservatively (one already lost in the control), not lost discoveries. The rewrite is therefore a scientific-integrity improvement that leaves the DR11 catalogue intact — consistent with the discrimination-neutral result the residual report measures on its own benchmark.

4 DR11-north: staged, maintenance-deferred

v3 is DECam-specific — it systematically under-scores BASS/MzLS north imaging — so the north arm requires instrument-native retraining. We built the full north parent (1.16×10^7 galaxies, *grz* only, no *i*), an 8-part manifest, and extracted **523** north-native training positives (297 BASS/MzLS curated + 226 non-leaked catalog lenses; 86 clean held-out; 33 v1 leaks tracked); the retrainer is patched to fine-tune the three swappable members from their `_b50` checkpoints on a north-native mimic blend. A full-system Perlmutter maintenance (2026-06-17 to 06-24) began mid-campaign before the north extraction could be scheduled, so the north sweep + retrain + validation gates are **deferred to post-maintenance**; all inputs are staged for a clean resume.

5 Conclusion

On the final DESI Legacy Surveys release, the contaminant-aware CLAUDENET finder plus the LENSJUDGE v3 HSC cascade is a working DESI $\times$ HSC cross-validation engine on the DECam-south

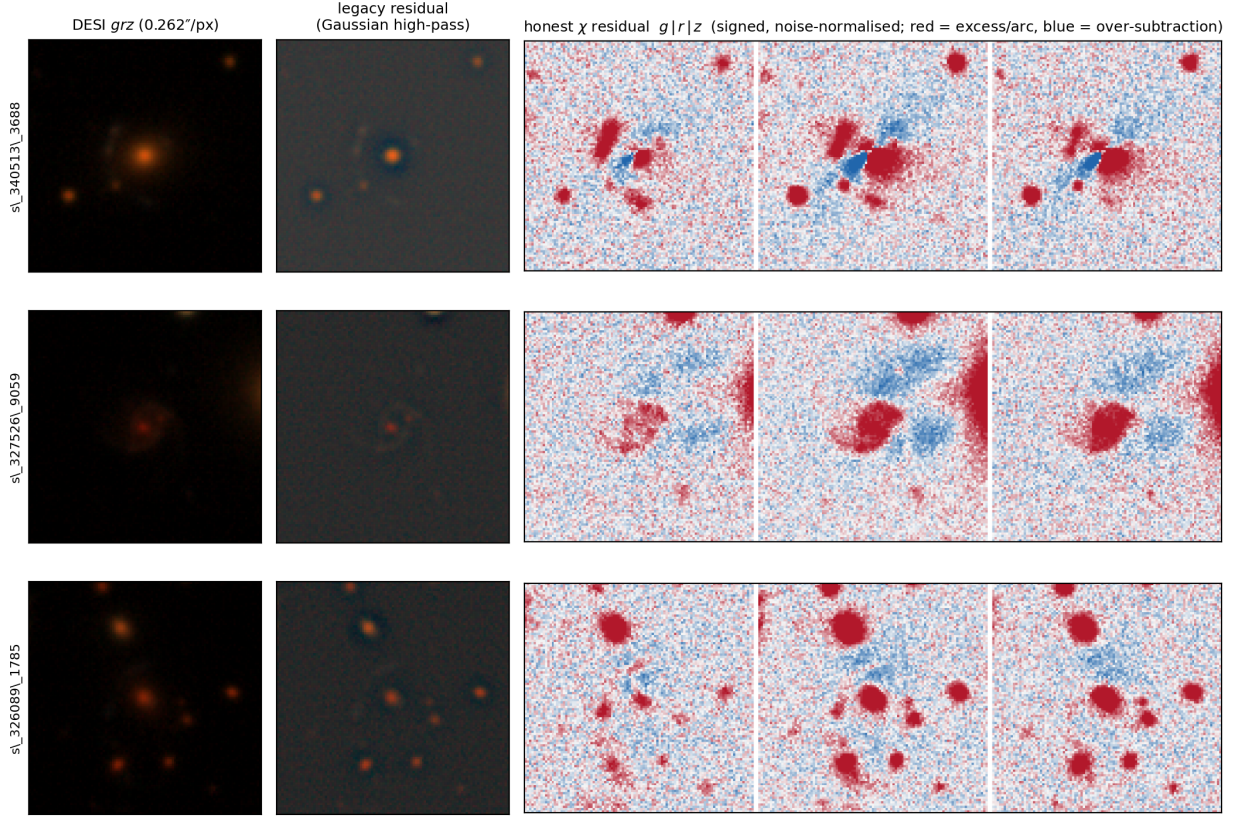


Figure 3: Legacy vs. honest residual for three confirmed NEW-class candidates. *Left:* the DESI *grz* cutout the finder scored (ambiguous at DECaLS 1"). *Centre:* the legacy Gaussian-high-pass residual — sign discarded, with a central dipole/butterfly on the bright lens that can read as structure. *Right:* the new signed $\chi = (\text{data} - \text{model})/\sigma$ residual as a $g|r|z$ montage on a fixed $\pm 5\sigma$ red–blue scale; the candidate arc shows as red excess *coherent across all three bands*, the real-vs-artifact discriminant. All three survive as grade A/B under the honest residual (Table 5).

footprint: it scores 5.4×10^7 galaxies, recalibrates honestly to the DR11 domain shift (with a real $\sim 34\%$ relative grade-A recall loss vs DR10 — the survivor list is incomplete), and — through the new HSC tier-2 over a $20\times$ -wider footprint than Euclid Q1 — moves 24 DESI-ambiguous candidates to HSC tier-2 grade-A/B under LENSJUDGE’s automated grader, of which 15 are SuGOHI committee lenses (a sensitivity check) and 9 (8 systems) are new to the DESI finders and SuGOHI — though a literature crosscheck (Huang+2020 + NED/SIMBAD; App. A) then shows 5 of those 8 systems are previously-published lenses, leaving **3** genuinely new. The certified-FDR is power-limited (zero) at full-population scale, the tier-2 grader is automated (no human committee or spectroscopy, precision on the new set unquantified), and *no net-new lens is established from DECaLS pixels alone* — the new candidate detections depend on HSC resolution and remain unconfirmed pending follow-up. This is the same honest ceiling the v3 program established — the resolution lever *converts* overlap candidates rather than manufacturing discoveries — now measured on DR11 with a much wider confirmable overlap. The north footprint, requiring instrument-native retraining, is staged for completion after the Perlmutter maintenance window.

References

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A The new candidate systems at DESI and HSC resolution

The figures below show the **8 distinct candidate systems** (the 9 detections of Table 2; the s_345385 pair is one system). A post-hoc literature crosscheck (below) finds **5 of the 8 are previously-published lenses** (Huang+2020; one also SOGRAS) and **3** are genuinely new. Each is rendered as: the CLAUDENET-seen DESI *grz* cutout ($0.262''/\text{px}$, Lupton RGB — what the finder scored, ambiguous at DECaLS resolution); the HSC PDR3 *grizy* cutout ($0.168''/\text{px}$ — the same view LENSJUDGE’s tier-2 grader inspected); and the HSC lens-light-subtracted view. The DESI→HSC contrast is the campaign’s mechanism made visual: the finder ranks these high from ambiguous DECaLS pixels, and the resolution lever reveals the arc/ring morphology (clearest in s_340513_3688; fainter and correspondingly less certain in others). These are *automated-grader candidates, not confirmed lenses* — they warrant human-expert and spectroscopic follow-up. *Literature crosscheck (added post-vetting)*. The campaign’s lineage crossmatch omitted the Huang+2020 DECaLS catalogue and did not query the wider literature. A full check — Huang+2020 plus a NED/SIMBAD cone search — finds that **five of these eight systems are previously-published lenses** (flagged *Known* below; mostly Huang+2020, one also in the SOAR Gravitational Arc Survey). Only s_327526_9059, s_340971_2860, and s_325163_2981 are genuinely new (absent from the lineage *and* the literature). The recoveries are an external validation of the v3+cascade finder; the net-new count here is **3**, not 8.

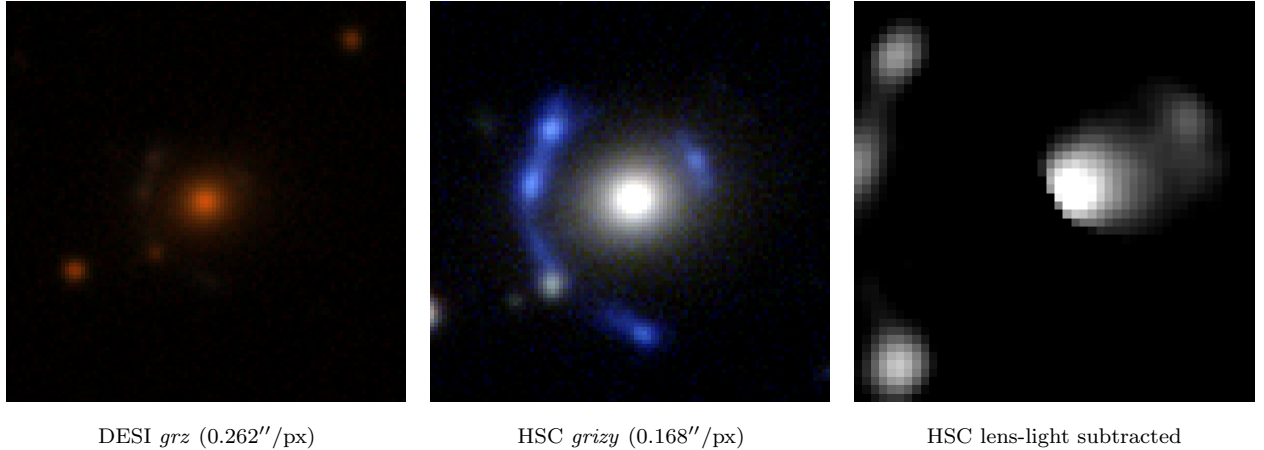


Figure 4: s_340513_3688 (RA 16.3319, Dec +1.7490) — HSC tier-2 grade **A**, p_{lens} 0.22→0.97, v3blend8 0.77. *Known:* Huang+2020 DESI-016.3319+01.7490 (grade A), $0.1''$.

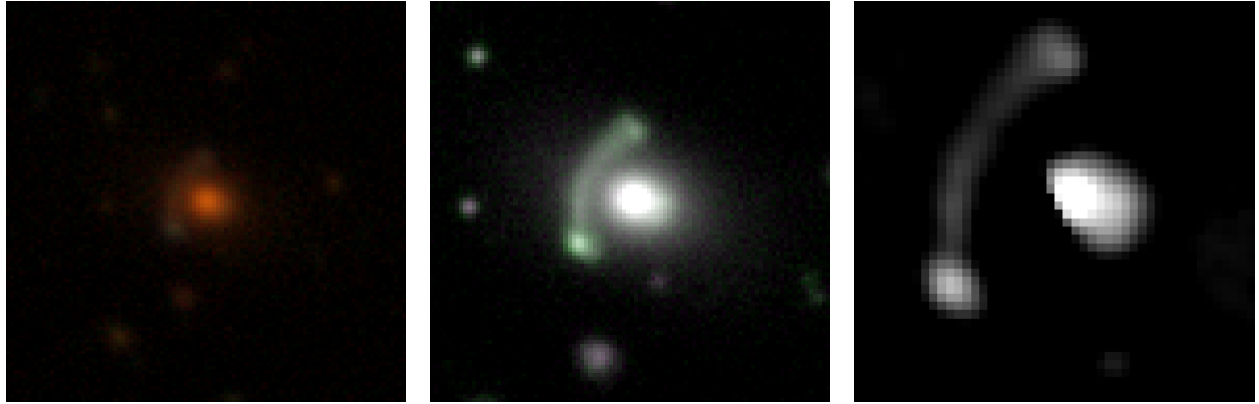


DESI *grz* (0.262''/px)

HSC *grizy* (0.168''/px)

HSC lens-light subtracted

Figure 5: s_327526_9059 (RA 9.7222, Dec -0.4093) — HSC tier-2 grade **A**, p_{lens} 0.04→0.92, v3blend8 0.74. *New*: absent from the lineage and from NED/SIMBAD.

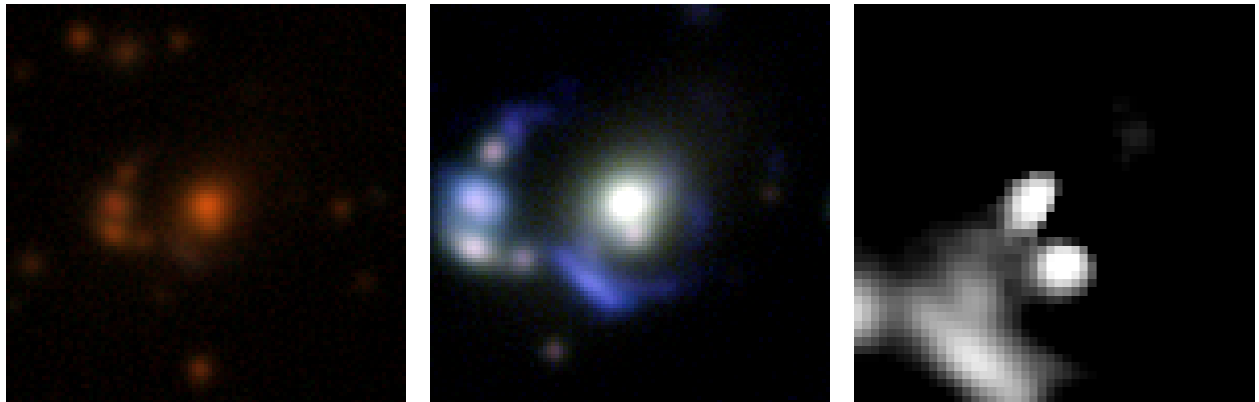


DESI *grz* (0.262''/px)

HSC *grizy* (0.168''/px)

HSC lens-light subtracted

Figure 6: s_351303_461 (RA 194.5345, Dec +3.5358) — HSC tier-2 grade **A**, p_{lens} 0.62→0.92, v3blend8 0.89. *Known*: Huang+2020 DESI-194.5344+03.5358 (grade A), 0.2''.



DESI *grz* (0.262''/px)

HSC *grizy* (0.168''/px)

HSC lens-light subtracted

Figure 7: s_345385_1515 (RA 154.3116, Dec +2.4885) — HSC tier-2 grade **A**, p_{lens} 0.10→0.92, v3blend8 0.81. *Known*: Huang+2020 DESI-154.3116+02.4885 (grade C), 0.1''. (= 8th detection s_345385_1514, the adjacent $\sim 1''$ pair, omitted)

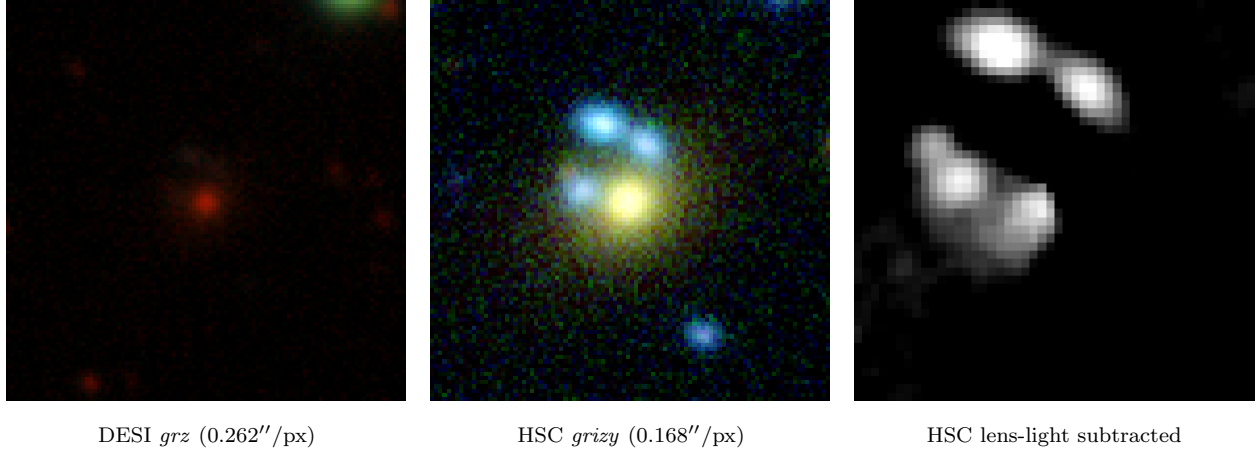


Figure 8: *s*_340971_2860 (RA 130.8571, Dec +1.7840) — HSC tier-2 grade **A**, p_{lens} 0.12→0.91, *v3blend8* 0.76. *New*: absent from the lineage and from NED/SIMBAD.

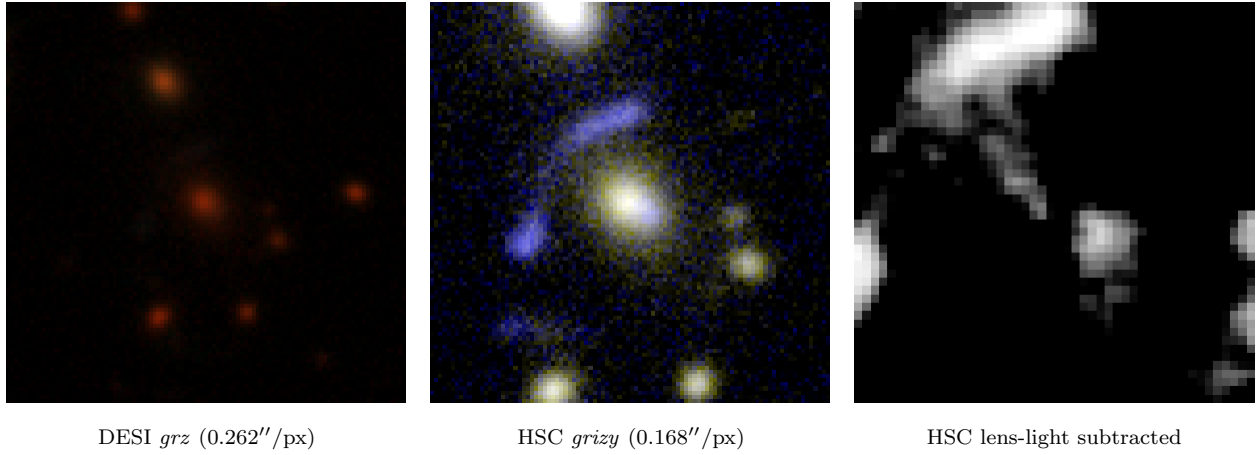


Figure 9: *s*_326089_1785 (RA 10.2876, Dec -0.7303) — HSC tier-2 grade **A**, p_{lens} 0.03→0.82, *v3blend8* 0.83. *Known*: Huang+2020 DESI-010.2876-00.7303 (grade B), also SOGRAS J0041-0043, 0.1''.

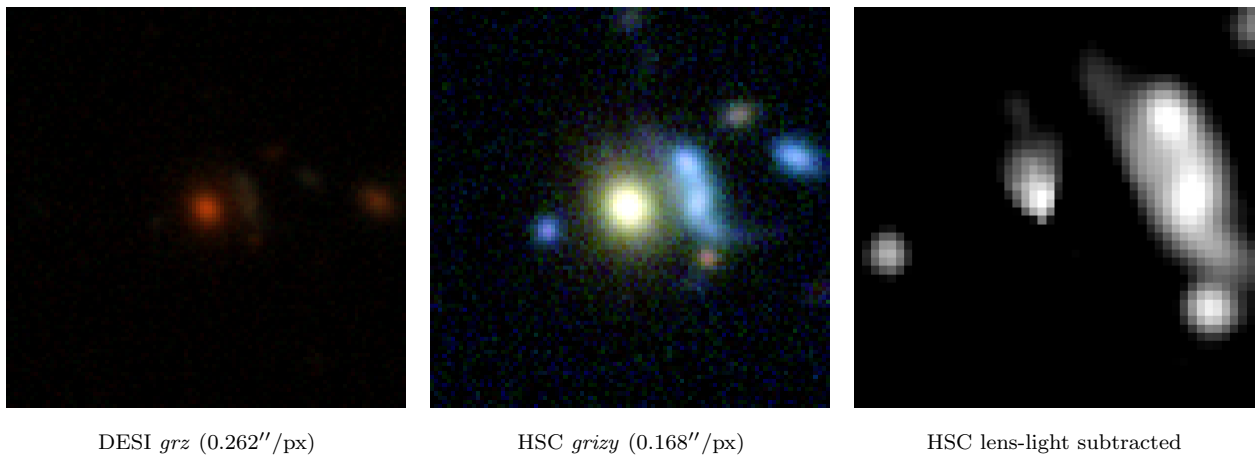


Figure 10: *s*_325163_2981 (RA 138.8482, Dec -0.9226) — HSC tier-2 grade **A**, p_{lens} 0.18→0.82, *v3blend8* 0.74. *New*: absent from the lineage and from NED/SIMBAD.

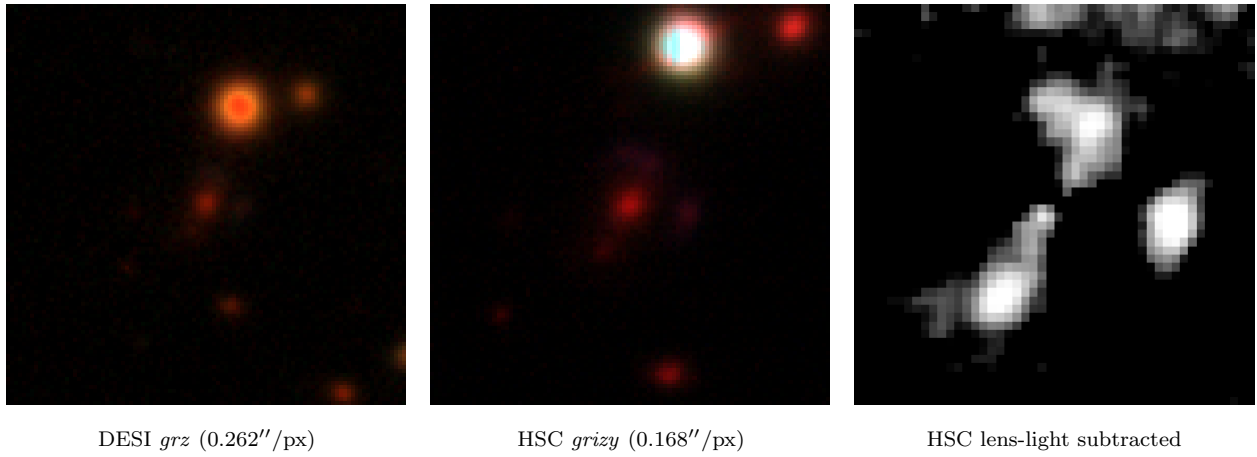


Figure 11: *s_318043_1037* (RA 158.7893, Dec -2.3037) — HSC tier-2 grade **B**, p_{lens} 0.04→0.45, *v3blend8* 0.86. *Known:* Huang+2020 DESI-158.7893-02.3037 (grade B), $0.1''$.

B DESI-resolution candidates without high-resolution coverage

Only $\sim 7\%$ of the DR11-south candidates fall in HSC-SSP (§3); the rest cannot be validated beyond DECaLS resolution with current high-res data. **This appendix is regenerated from the honest- χ -residual cascade run** (§3, “Residual robustness”): Table 4 lists all **108** candidates graded A/B at DECaLS $1''$ by the LensJudge cascade that are *not* in HSC-SSP *or* Euclid Q1 (17 A, 91 B), ranked by tier-1 p_{lens} — versus 78 under the legacy residual. This DESI-resolution-only set is the *least stable* part of the pipeline: with no high-resolution anchor its membership is sensitive both to the residual (the honest one runs $\sim 2\times$ hotter at tier-1) and to LLM nondeterminism (a legacy-residual rerun alone gives 66), so the larger list is hotter, not more complete, and its exact membership is not load-bearing. Each is then shown below as an expanded DESI *grz* triptych — *full* (the whole 101 px, $0.262''/\text{px}$ cutout the finder scored), *zoom* ($2.5\times$ center crop), and *residual* (the signed $\chi = (\text{data} - \text{model})/\sigma$ of the $r+z$ luminance after subtracting a robust elliptical-galaxy model, on a red–blue diverging scale at fixed $\pm 5\sigma$: red marks unmodelled excess such as an arc/ring, blue marks over-subtraction; the noise σ is estimated per band, so the residual is honest about significance rather than re-floored to hide its sign) — all at DECaLS resolution, carrying its full metadata and a KNOWN/NEW tag (KNOWN naming the previously-published lens it recovers). The panels are grouped into a KNOWN sub-section (previously-published lenses recovered) followed by a NEW sub-section (candidates new to the literature), ranked by tier-1 p_{lens} within each. A literature crosscheck (`crosscheck_candidates.py`; Huang/Storfer/Inchausti + NED/SIMBAD) finds **69 of the 108 are previously-published lenses** — 48 from the Dark Energy Survey lens searches, 11 Huang+2020, 6 AGEL, 1 CASSOWARY, and 3 via SIMBAD (the *known* column of Table 4) — because this DECaLS-south footprint heavily overlaps DES, which the campaign crossmatch omitted; the **39 genuinely new** are mostly grade B, with **3 grade A** that the honest residual newly surfaces (`s_230539_4159`, `s_194300_2796`, `s_320069_835`) — still DESI-resolution-only and unvalidatable. A NED cone search corroborates the new set (no NED lens match; reached 105/108, with 3 timeouts). At $1''$ a genuine lens and an lrg+companion mimic are not separable (the resolution ceiling of §3), so these are *candidates*, an unknown and likely large fraction of which are mimics — the per-candidate panels show both clear arc/ring morphologies and ambiguous lrg+companion configurations. They are listed in full for completeness and as a follow-up target set, awaiting Euclid DR1 / Rubin / Roman or targeted high-resolution and spectroscopic follow-up.

Table 4: The 108 DESI-resolution grade-A/B candidates (LensJudge `direct/escalate` grade at DECaLS $0.262''$) with *no* HSC-SSP or Euclid Q1 coverage, ranked by tier-1 p_{lens} . A literature crosscheck (`crosscheck_candidates.py`: the Huang/Storfer/Inchausti catalogues + a NED/SIMBAD cone search) finds **69 are previously-published lenses** — 48 DES, 11 Huang+2020, 6 AGEL, 3 SIMBAD, 1 CASSOWARY (*known* column: survey code; per-object matches in `data/dr11s_resolution_xmatch.csv`) — and **39 are genuinely new** (—; 3 grade A and 36 grade B). The high known fraction is because this DECaLS-south footprint heavily overlaps the Dark Energy Survey lens searches, which the campaign crossmatch omitted. The new ones cannot be validated beyond DECaLS resolution; at $1''$ a genuine lens and an lrg+companion mimic are not separable, so an unknown fraction are mimics. Per-candidate full/zoom/residual panels follow (App. B).

#	name	RA	Dec	grade	p_{lens}	v3blend8	known
1	s_441355_1188	177.1381	+19.5009	A	0.95	0.83	CSWA
2	s_98012_247	56.1462	-44.7956	A	0.92	0.86	DES
3	s_310364_6388	38.2078	-3.3906	A	0.88	0.82	H20

continued...

Table 4 continued

#	name	RA	Dec	grade	p_{lens}	v3blend8	known
4	s_301694_7426	26.2679	-4.9310	A	0.88	0.85	H20
5	s_181942_4122	61.6016	-26.7733	A	0.87	0.86	DES
6	s_137502_1047	26.4450	-35.6910	A	0.87	0.80	DES
7	s_230539_4159	222.0715	-17.8489	A	0.85	0.84	—
8	s_71042_1653	20.1760	-51.7314	A	0.85	0.82	DES
9	s_180593_5604	44.0974	-27.1218	A	0.85	0.88	DES
10	s_194300_2796	242.6052	-24.4155	A	0.83	0.74	—
11	s_304994_4662	133.5531	-4.4026	A	0.82	0.84	H20
12	s_61472_9399	359.5200	-54.6765	A	0.80	0.88	DES
13	s_91761_4679	350.3683	-46.5138	A	0.80	0.86	DES
14	s_91796_2958	2.9714	-46.2395	A	0.78	0.79	DES
15	s_320069_835	305.2638	-1.9376	A	0.78	0.89	—
16	s_292418_3907	216.8280	-6.7541	A	0.78	0.89	H20
17	s_81254_12458	57.3313	-48.9592	A	0.76	0.84	DES
18	s_124958_7349	58.1768	-38.4292	B	0.68	0.89	DES
19	s_59958_9730	64.5412	-54.9597	B	0.65	0.83	DES
20	s_197474_9756	30.7668	-23.6341	B	0.65	0.81	DES
21	s_313198_1471	27.5380	-3.0773	B	0.65	0.81	AGEL
22	s_232538_8716	25.2755	-17.2232	B	0.65	0.85	H20
23	s_280236_3353	25.8623	-8.8393	B	0.65	0.84	AGEL
24	s_234689_542	227.0427	-16.8771	B	0.65	0.80	—
25	s_502450_1680	161.1145	+31.2344	B	0.63	0.79	—
26	s_94959_7228	56.8054	-45.5850	B	0.62	0.84	DES
27	s_119342_4136	50.7164	-39.7694	B	0.62	0.90	DES
28	s_243378_3831	326.5170	-15.4964	B	0.62	0.84	—
29	s_240846_8850	30.2832	-15.8547	B	0.62	0.80	H20
30	s_188493_9685	83.4556	-25.6151	B	0.62	0.88	DES
31	s_94933_7964	47.5548	-45.5743	B	0.60	0.78	DES
32	s_95070_5466	96.2505	-45.4358	B	0.60	0.78	DES
33	s_369309_653	30.4465	+6.6551	B	0.60	0.74	—
34	s_34114_11363	353.7468	-64.0687	B	0.58	0.90	DES
35	s_102952_10164	334.8017	-43.8098	B	0.58	0.85	DES
36	s_191004_9637	56.9356	-24.9088	B	0.58	0.88	DES
37	s_43736_6056	67.3945	-60.1905	B	0.58	0.83	DES
38	s_137518_1133	31.3588	-35.6632	B	0.58	0.84	DES
39	s_129372_3768	16.4504	-37.4286	B	0.58	0.82	DES
40	s_98882_7820	1.6860	-44.4974	B	0.58	0.80	DES
41	s_218735_721	350.2028	-20.0497	B	0.58	0.79	—
42	s_360639_3636	13.4868	+5.3590	B	0.55	0.85	—
43	s_381188_1199	144.1511	+8.8633	B	0.55	0.90	H20
44	s_114736_1046	343.5127	-40.9303	B	0.55	0.86	DES
45	s_87091_1800	84.5193	-47.5872	B	0.55	0.89	DES
46	s_327710_4950	55.6649	-0.3820	B	0.55	0.74	—
47	s_289350_3716	164.8051	-7.1449	B	0.55	0.81	—

continued. . .

Table 4 continued

#	name	RA	Dec	grade	p_{lens}	v3blend8	known
48	s_384032_2232	143.3887	+9.3219	B	0.55	0.84	H20
49	s_73683_6069	357.3753	-51.2276	B	0.55	0.83	DES
50	s_80200_6872	15.4919	-49.2940	B	0.55	0.88	DES
51	s_58263_3751	42.7179	-55.4032	B	0.55	0.88	DES
52	s_491286_1438	180.3616	+29.0116	B	0.52	0.74	—
53	s_164140_7181	41.2060	-30.3214	B	0.50	0.79	DES
54	s_71871_246	353.9664	-51.8716	B	0.50	0.76	AGEL
55	s_61626_1035	65.1782	-54.3770	B	0.50	0.87	DES
56	s_262449_2840	188.7861	-11.9798	B	0.50	0.87	—
57	s_107312_5223	23.8451	-42.5398	B	0.50	0.85	DES
58	s_90920_9419	45.9517	-46.4407	B	0.50	0.85	DES
59	s_217461_3974	12.0451	-19.9206	B	0.50	0.78	—
60	s_260341_325	9.9798	-12.2097	B	0.50	0.84	—
61	s_207576_2272	248.1195	-22.0355	B	0.48	0.76	—
62	s_162970_1777	62.5554	-30.6238	B	0.48	0.82	DES
63	s_234043_6467	58.4427	-17.1109	B	0.47	0.88	AGEL
64	s_283917_2408	235.6200	-8.3490	B	0.45	0.89	—
65	s_50907_11900	303.5808	-57.9504	B	0.45	0.74	DES
66	s_61411_4481	332.9866	-54.6445	B	0.45	0.76	DES
67	s_165468_9648	64.7771	-29.9937	B	0.45	0.78	DES
68	s_148704_6732	183.8176	-33.6006	B	0.45	0.80	—
69	s_280264_9147	33.1051	-8.8697	B	0.45	0.84	literature (SIMBAD)
70	s_266046_1185	25.9989	-11.2770	B	0.45	0.76	—
71	s_251013_4202	137.4230	-13.9601	B	0.45	0.79	—
72	s_52008_8091	96.5963	-57.5083	B	0.45	0.89	DES
73	s_209853_1103	139.8053	-21.4604	B	0.45	0.85	—
74	s_504880_4027	152.0763	+31.7005	B	0.45	0.84	H20
75	s_181762_9592	11.4347	-26.6294	B	0.45	0.86	—
76	s_69087_268	307.2325	-52.5218	B	0.43	0.81	DES
77	s_256199_9097	29.6819	-13.0258	B	0.42	0.78	DES
78	s_36700_530	347.9904	-63.1176	B	0.42	0.78	literature (SIMBAD)
79	s_23764_4161	132.6603	-68.1876	B	0.42	0.79	—
80	s_29019_24056	248.9430	-66.0747	B	0.42	0.74	—
81	s_204169_8722	49.1618	-22.6093	B	0.42	0.77	DES
82	s_112158_2867	209.6741	-41.4932	B	0.42	0.83	—
83	s_139825_11098	18.9226	-35.3386	B	0.42	0.76	literature (SIMBAD)
84	s_60524_3113	309.7973	-54.9952	B	0.42	0.81	DES
85	s_38835_314	56.6543	-61.9792	B	0.42	0.84	DES
86	s_405220_1292	136.4655	+13.0307	B	0.42	0.89	—
87	s_68464_11443	52.7373	-52.4703	B	0.42	0.88	DES
88	s_206743_10995	24.2120	-22.0076	B	0.42	0.85	DES
89	s_41562_4180	42.2402	-60.9010	B	0.40	0.87	DES
90	s_188312_3322	33.2537	-25.3945	B	0.40	0.76	—
91	s_120781_3317	156.8525	-39.6093	B	0.40	0.83	—

continued. . .

Table 4 continued

#	name	RA	Dec	grade	p_{lens}	v3blend8	known
92	s_266620_918	172.1320	-11.3108	B	0.40	0.80	—
93	s_410935_2097	164.1204	+14.0606	B	0.38	0.83	—
94	s_32666_26056	248.6090	-64.5569	B	0.38	0.73	—
95	s_244220_4253	184.3042	-15.2610	B	0.38	0.80	—
96	s_119499_8020	101.8093	-39.8359	B	0.38	0.75	—
97	s_156771_3446	54.3219	-31.8704	B	0.38	0.89	AGEL
98	s_205319_6061	359.9772	-22.4596	B	0.38	0.88	—
99	s_416616_6090	189.5370	+15.0309	B	0.38	0.93	H20
100	s_364281_4525	207.3133	+5.6308	B	0.35	0.90	—
101	s_250611_2161	33.7898	-14.1185	B	0.35	0.77	—
102	s_212207_5081	50.6597	-20.9567	B	0.35	0.84	—
103	s_128091_3374	332.6355	-37.9263	B	0.35	0.85	—
104	s_194944_4494	59.2045	-24.1447	B	0.35	0.85	DES
105	s_94861_9433	21.9716	-45.5428	B	0.32	0.87	DES
106	s_235253_4230	14.4452	-16.7457	B	0.32	0.76	H20
107	s_394431_6727	261.7637	+11.0021	B	0.30	0.77	AGEL
108	s_369977_4434	198.5174	+6.6887	B	0.28	0.75	—

Known — previously-published lenses recovered (69 of 108)

These coincide ($\leq \text{few}''$) with a catalogued strong lens — 48 DES, 11 Huang+2020, 6 AGEL, 3 SIMBAD, 1 CASSOWARY — an external validation of the finder (the campaign crossmatch had omitted these catalogues). Ranked by tier-1 p_{lens} .

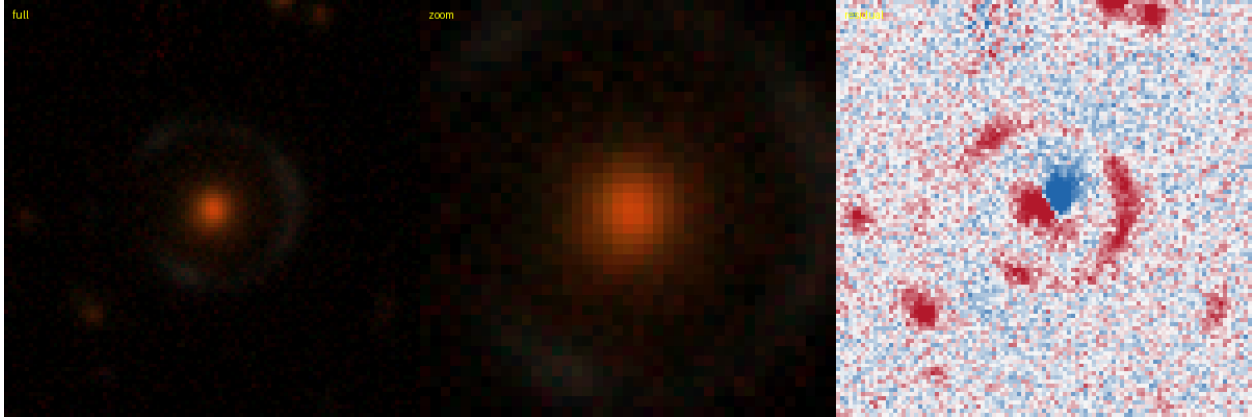


Figure 12: #1 s_441355_1188 — RA 177.1381, Dec +19.5009; grade **A**, tier-1 p_{lens} 0.95, v3blend8 0.83. **KNOWN** — CASSOWARY: CWA 1, 0.01'' (lineage).

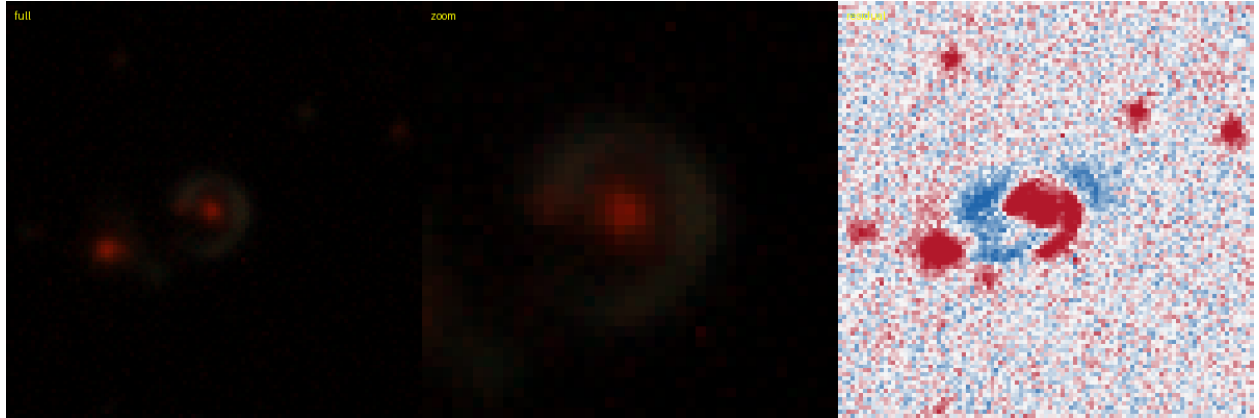


Figure 13: #2 s_98012_247 — RA 56.1462, Dec -44.7956; grade **A**, tier-1 p_{lens} 0.92, v3blend8 0.86. **KNOWN** — DES: DES J034435.09-444743.9, 0.18'' (SIMBAD).

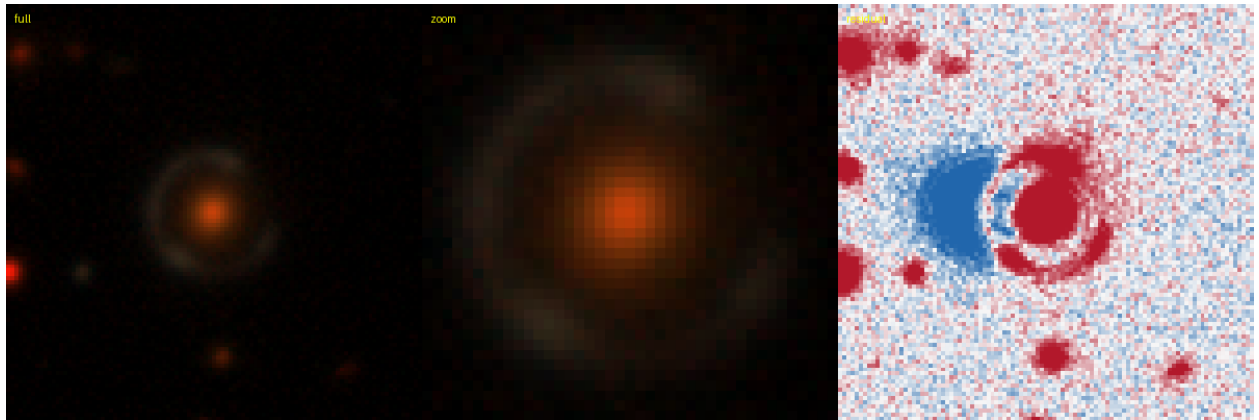


Figure 14: #3 s_310364_6388 — RA 38.2078, Dec -3.3906; grade **A**, tier-1 p_{lens} 0.88, v3blend8 0.82. **KNOWN** — Huang+2020: DESI-038.2078-03.3906, 0.16'' (lineage).

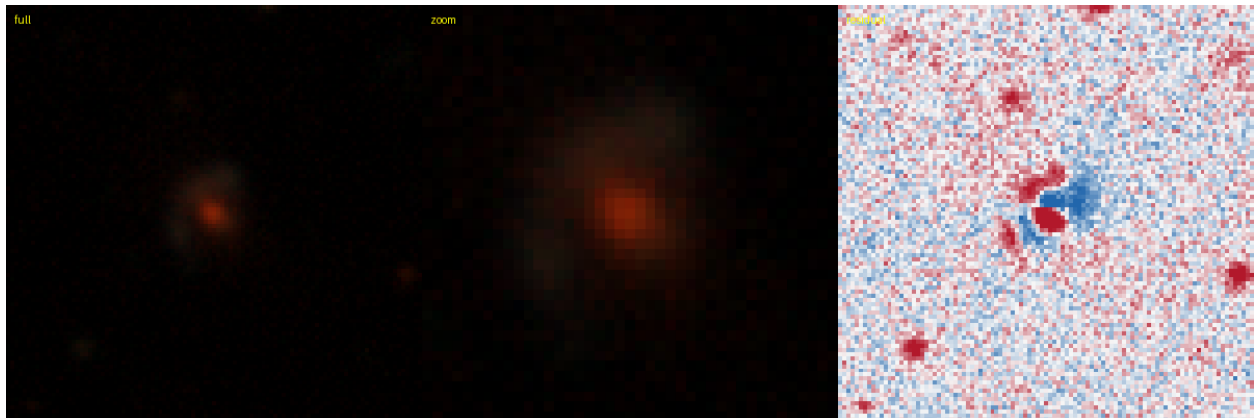


Figure 15: #4 s_301694_7426 — RA 26.2679, Dec -4.9310; grade **A**, tier-1 p_{lens} 0.88, v3blend8 0.85. **KNOWN** — Huang+2020: DESI-026.2679-04.9310, 0.13'' (lineage).

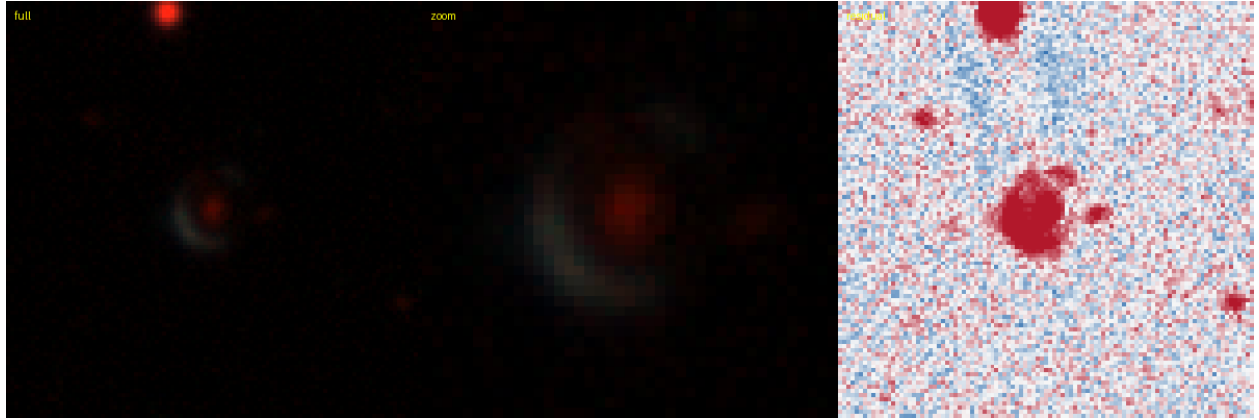


Figure 16: #5 s_181942.4122 — RA 61.6016, Dec -26.7733; grade **A**, tier-1 p_{lens} 0.87, v3blend8 0.86.
KNOWN — DES: DES J040624.49-264625.0, 1.74'' (SIMBAD).

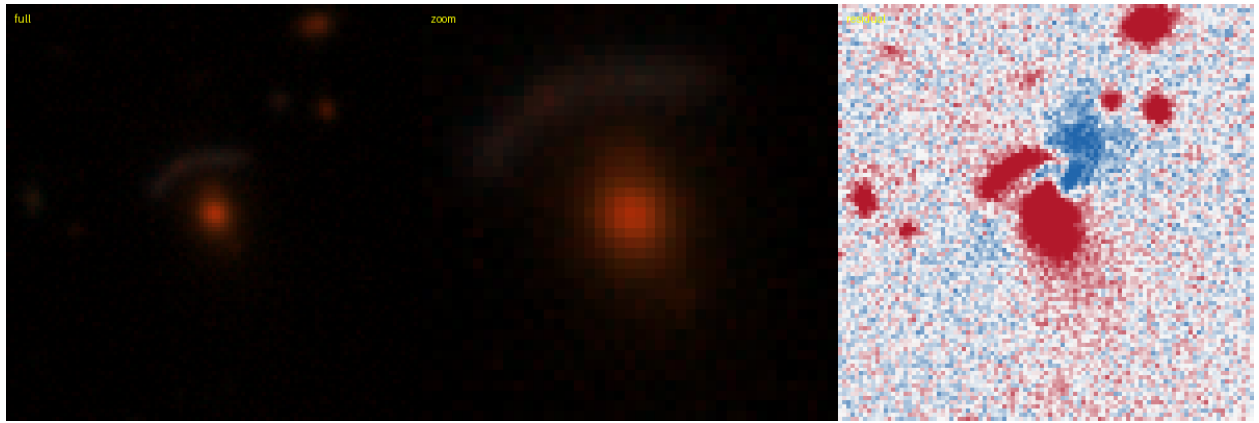


Figure 17: #6 s_137502.1047 — RA 26.4450, Dec -35.6910; grade **A**, tier-1 p_{lens} 0.87, v3blend8 0.80.
KNOWN — DES: DES J014546.78-354127.3, 0.20'' (SIMBAD).

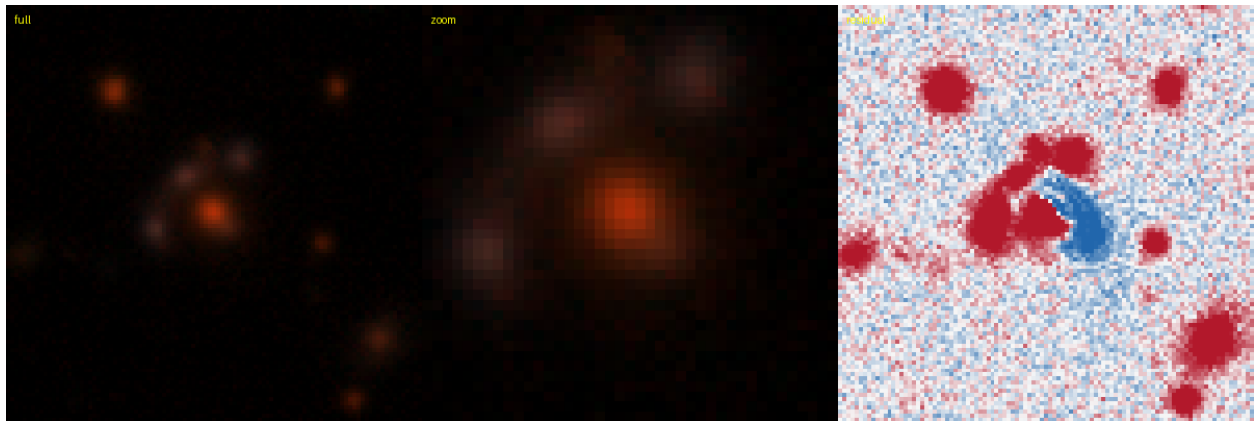


Figure 18: #8 s_71042.1653 — RA 20.1760, Dec -51.7314; grade **A**, tier-1 p_{lens} 0.85, v3blend8 0.82.
KNOWN — DES: [DBL2017] DES J0120-5143 (A), 0.19'' (SIMBAD).

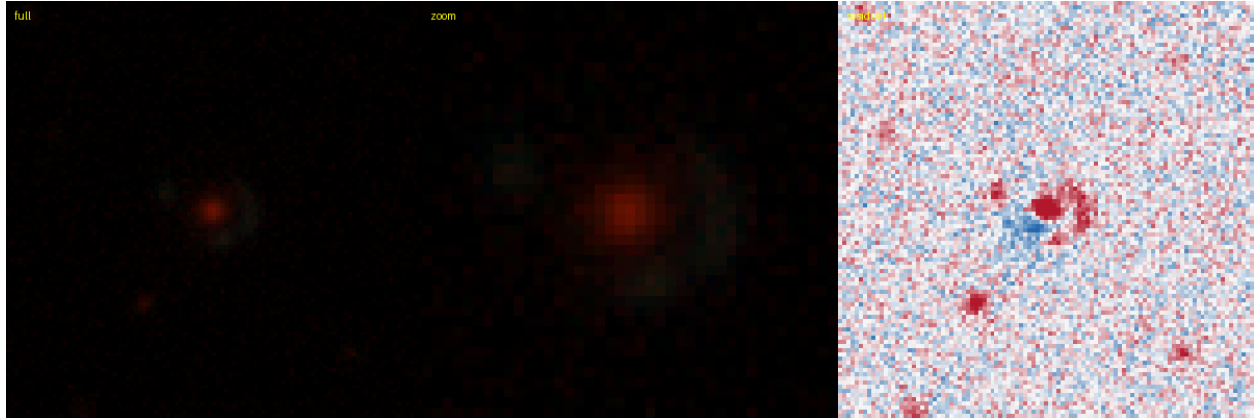


Figure 19: #9 s_180593.5604 — RA 44.0974, Dec -27.1218; grade **A**, tier-1 p_{lens} 0.85, v3blend8 0.88. **KNOWN** — DES: DES J025623.36-270718.5, 0.28'' (SIMBAD).



Figure 20: #11 s_304994.4662 — RA 133.5531, Dec -4.4026; grade **A**, tier-1 p_{lens} 0.82, v3blend8 0.84. **KNOWN** — Huang+2020: DESI-133.5531-04.4026, 0.14'' (lineage).



Figure 21: #12 s_61472.9399 — RA 359.5200, Dec -54.6765; grade **A**, tier-1 p_{lens} 0.80, v3blend8 0.88. **KNOWN** — DES: DES J235804.95-544034.4, 1.73'' (SIMBAD).

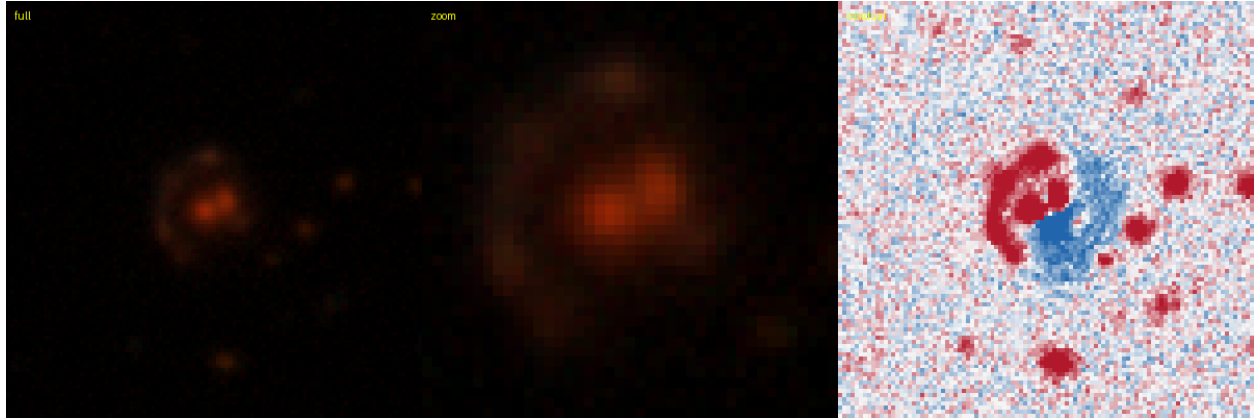


Figure 22: #13 s_91761_4679 — RA 350.3683, Dec -46.5138; grade **A**, tier-1 p_{lens} 0.80, v3blend8 0.86.
KNOWN — DES: DES J232128.37-463049.3, 0.25'' (SIMBAD).

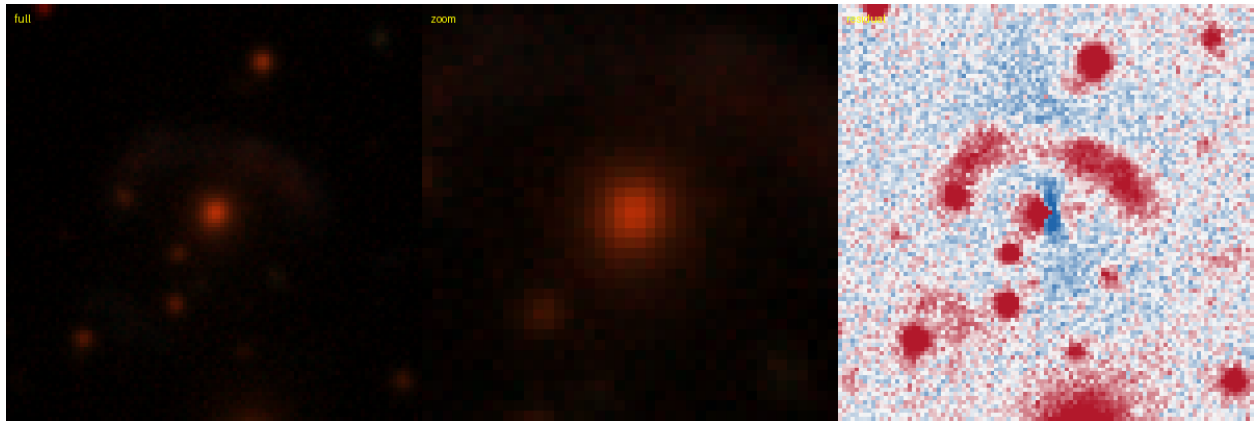


Figure 23: #14 s_91796_2958 — RA 2.9714, Dec -46.2395; grade **A**, tier-1 p_{lens} 0.78, v3blend8 0.79.
KNOWN — DES: [DBL2017] DES J0011-4614 (A), 0.15'' (SIMBAD).

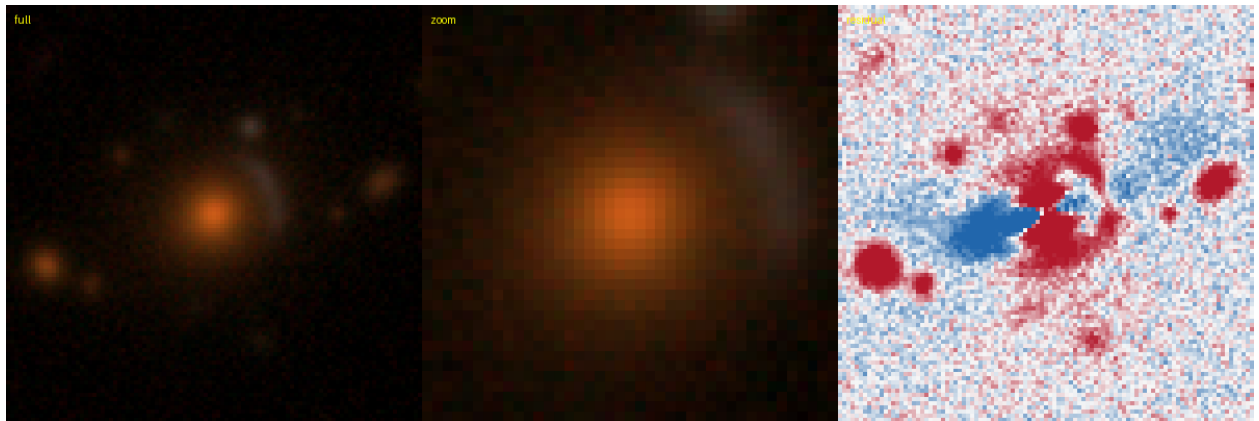


Figure 24: #16 s_292418_3907 — RA 216.8280, Dec -6.7541; grade **A**, tier-1 p_{lens} 0.78, v3blend8 0.89.
KNOWN — Huang+2020: DESI-216.8280-06.7541, 0.15'' (lineage).

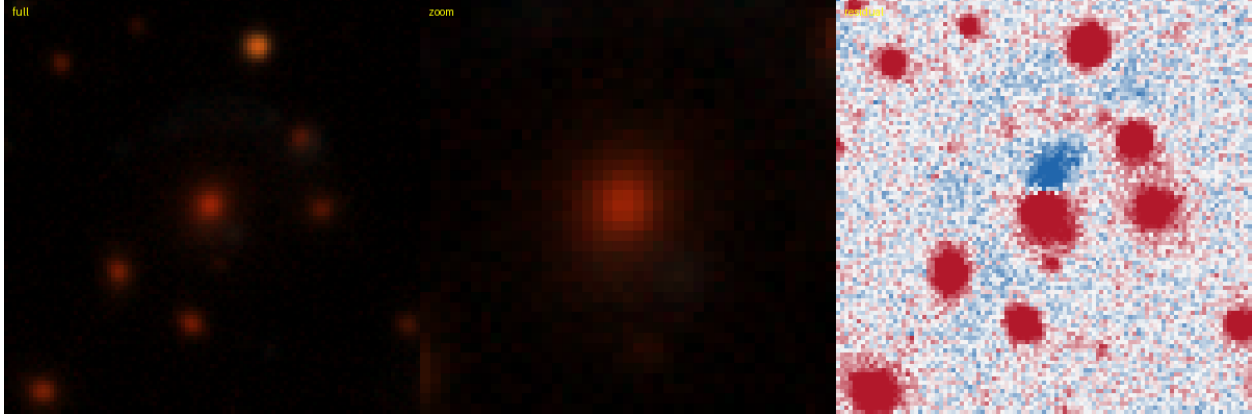


Figure 25: #17 s_81254.12458 — RA 57.3313, Dec -48.9592; grade **A**, tier-1 p_{lens} 0.76, v3blend8 0.84. **KNOWN** — DES: [DBL2017] DES J0349-4857 (A), 0.16'' (SIMBAD).

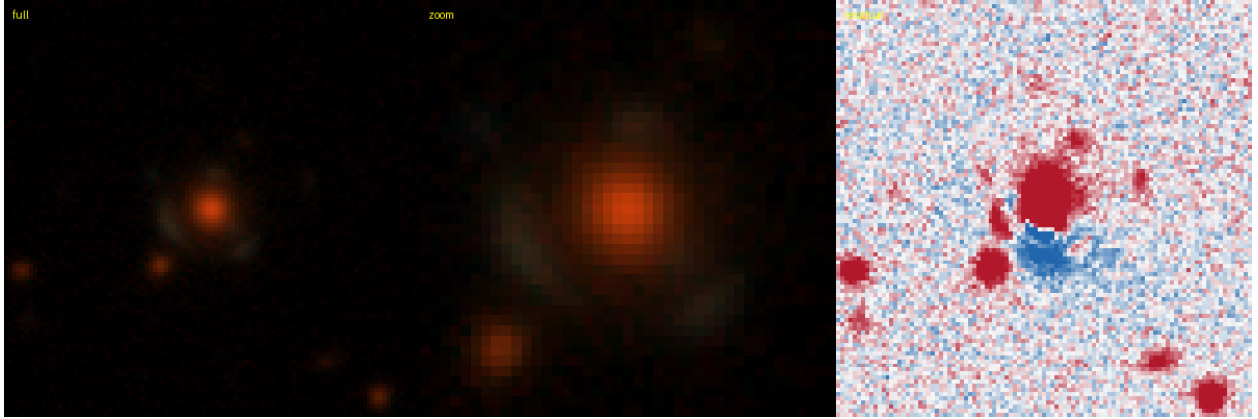


Figure 26: #18 s_124958.7349 — RA 58.1768, Dec -38.4292; grade **B**, tier-1 p_{lens} 0.68, v3blend8 0.89. **KNOWN** — DES: DES J035242.40-382544.9, 0.01'' (lineage).

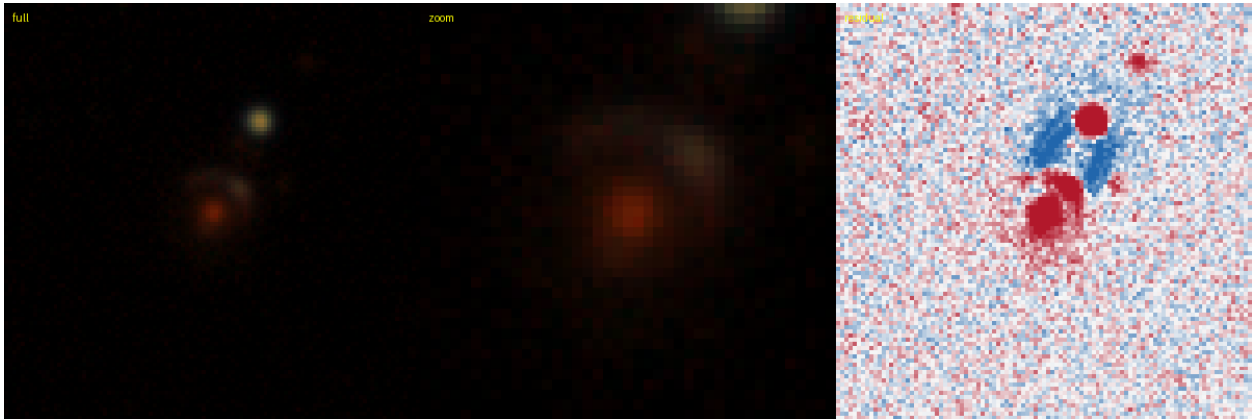


Figure 27: #19 s_59958.9730 — RA 64.5412, Dec -54.9597; grade **B**, tier-1 p_{lens} 0.65, v3blend8 0.83. **KNOWN** — DES: [DBL2017] DES J0418-5457 (A), 0.14'' (SIMBAD).

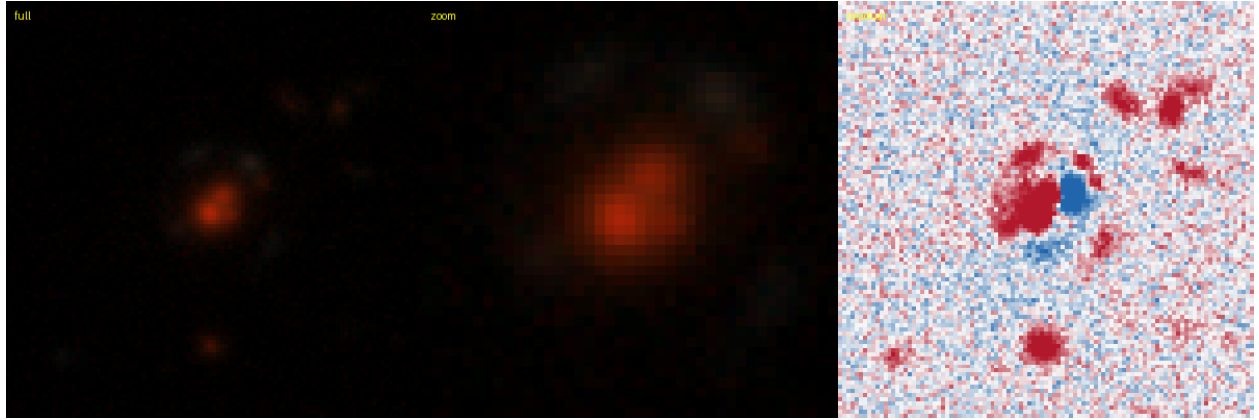


Figure 28: #20 s_197474_9756 — RA 30.7668, Dec -23.6341; grade **B**, tier-1 p_{lens} 0.65, v3blend8 0.81. **KNOWN** — DES: DES J020304.01-233802.5, 0.22'' (SIMBAD).

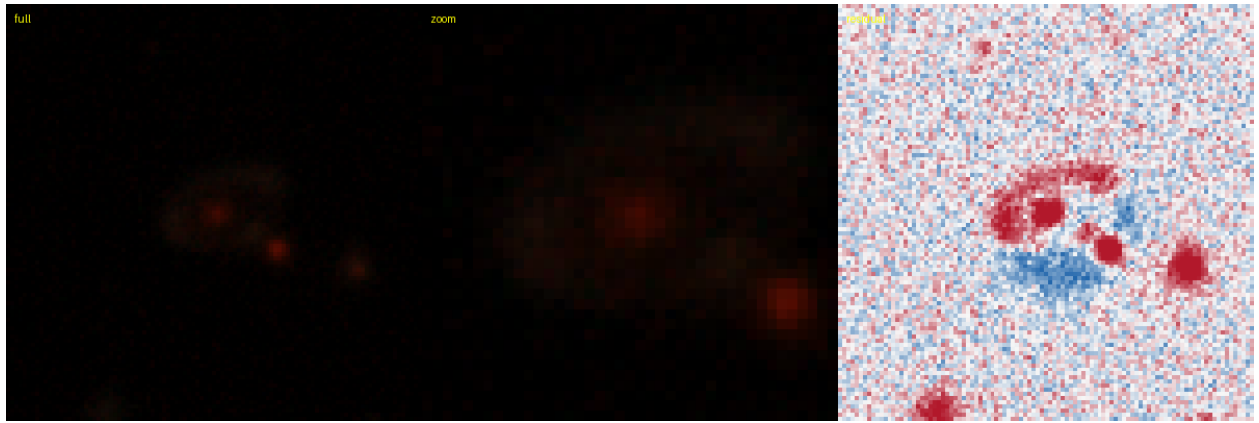


Figure 29: #21 s_313198_1471 — RA 27.5380, Dec -3.0773; grade **B**, tier-1 p_{lens} 0.65, v3blend8 0.81. **KNOWN** — AGEL: AGEL J015009-030438 system, 0.34'' (SIMBAD).



Figure 30: #22 s_232538_8716 — RA 25.2755, Dec -17.2232; grade **B**, tier-1 p_{lens} 0.65, v3blend8 0.85. **KNOWN** — Huang+2020: DESI-025.2755-17.2232, 0.19'' (lineage).

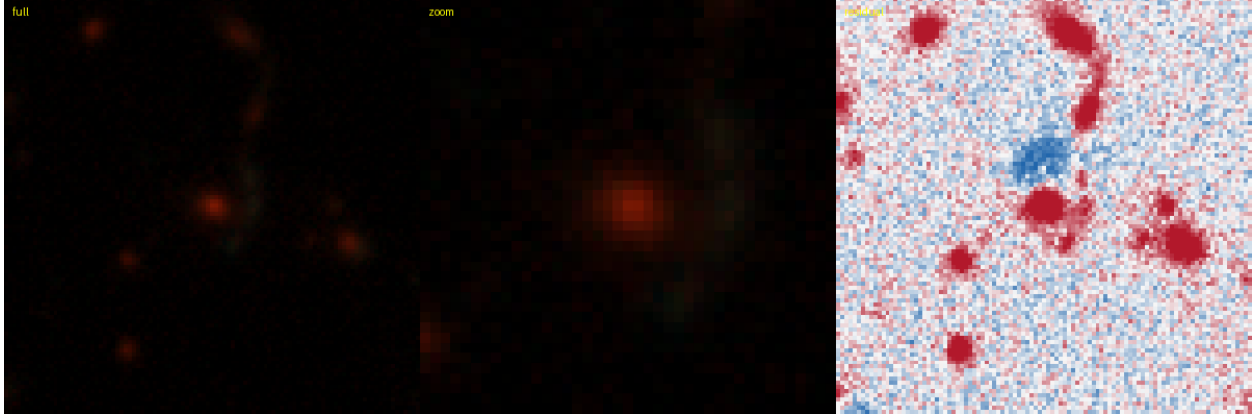


Figure 31: #23 s_280236.3353 — RA 25.8623, Dec -8.8393; grade **B**, tier-1 p_{lens} 0.65, v3blend8 0.84. **KNOWN** — AGEL: AGEL J014327-085021 system, 0.93'' (SIMBAD).

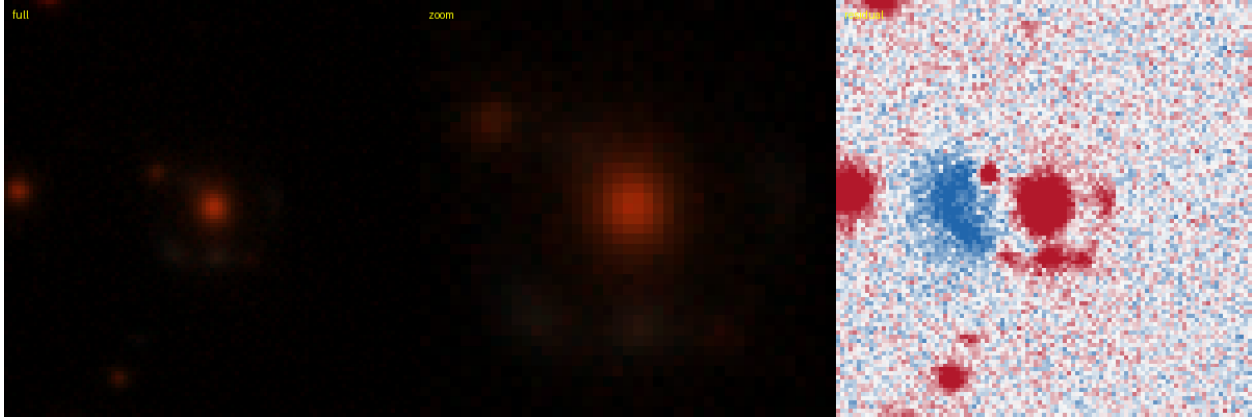


Figure 32: #26 s_94959.7228 — RA 56.8054, Dec -45.5850; grade **B**, tier-1 p_{lens} 0.62, v3blend8 0.84. **KNOWN** — DES: [DBL2017] DES J0347-4535 (A), 0.09'' (SIMBAD).



Figure 33: #27 s_119342.4136 — RA 50.7164, Dec -39.7694; grade **B**, tier-1 p_{lens} 0.62, v3blend8 0.90. **KNOWN** — DES: DES J032251.91-394609.6, 0.17'' (SIMBAD).

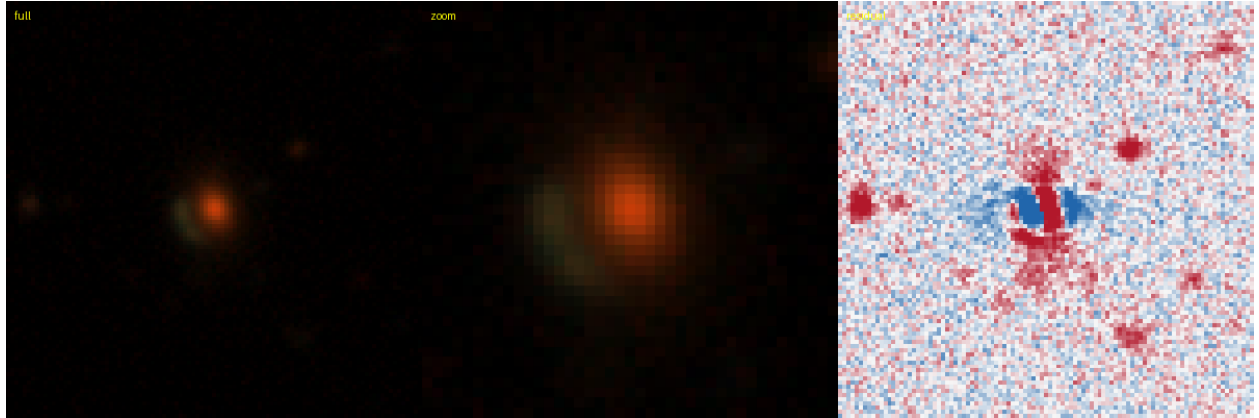


Figure 34: #29 s_240846.8850 — RA 30.2832, Dec -15.8547; grade **B**, tier-1 p_{lens} 0.62, v3blend8 0.80. **KNOWN** — Huang+2020: DESI-030.2832-15.8547, 0.15'' (lineage).

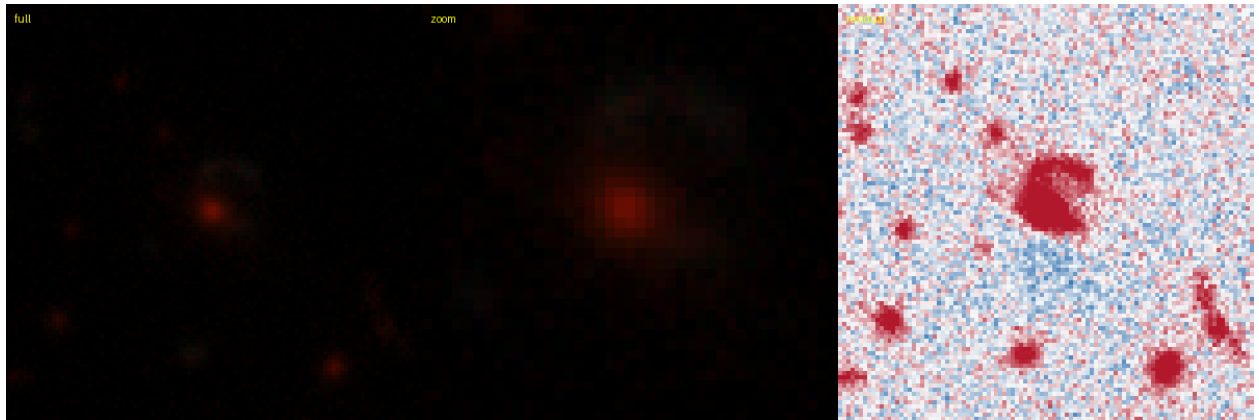


Figure 35: #30 s_188493.9685 — RA 83.4556, Dec -25.6151; grade **B**, tier-1 p_{lens} 0.62, v3blend8 0.88. **KNOWN** — DES: DES J053349.32-253654.3, 0.15'' (SIMBAD).



Figure 36: #31 s_94933.7964 — RA 47.5548, Dec -45.5743; grade **B**, tier-1 p_{lens} 0.60, v3blend8 0.78. **KNOWN** — DES: [DBL2017] DES J0310-4534 (A), 0.14'' (SIMBAD).

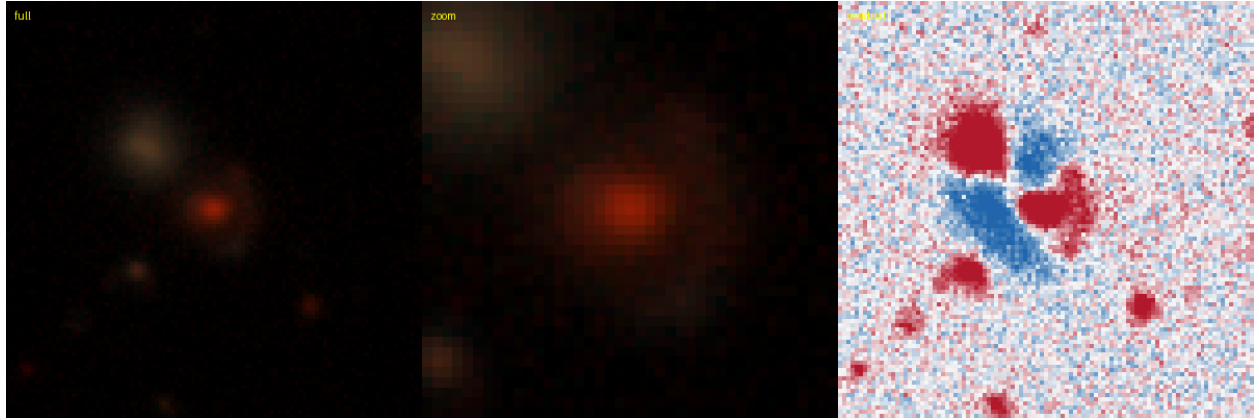


Figure 37: #32 s_95070.5466 — RA 96.2505, Dec -45.4358; grade **B**, tier-1 p_{lens} 0.60, v3blend8 0.78.
KNOWN — DES: [DBL2017] DES J0625-4526 (A), 0.08'' (SIMBAD).

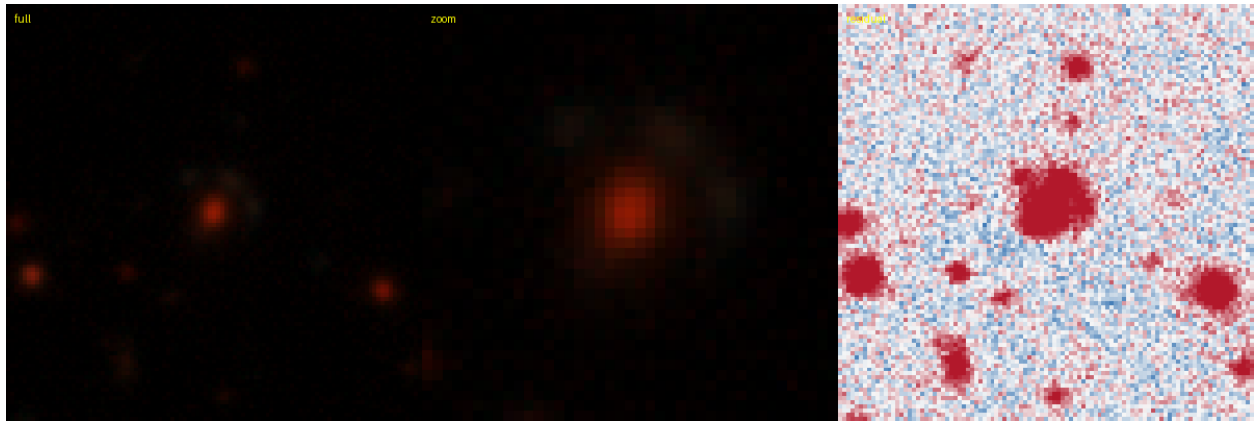


Figure 38: #34 s_34114.11363 — RA 353.7468, Dec -64.0687; grade **B**, tier-1 p_{lens} 0.58, v3blend8 0.90.
KNOWN — DES: DES J233459.19-640406.9, 0.32'' (SIMBAD).

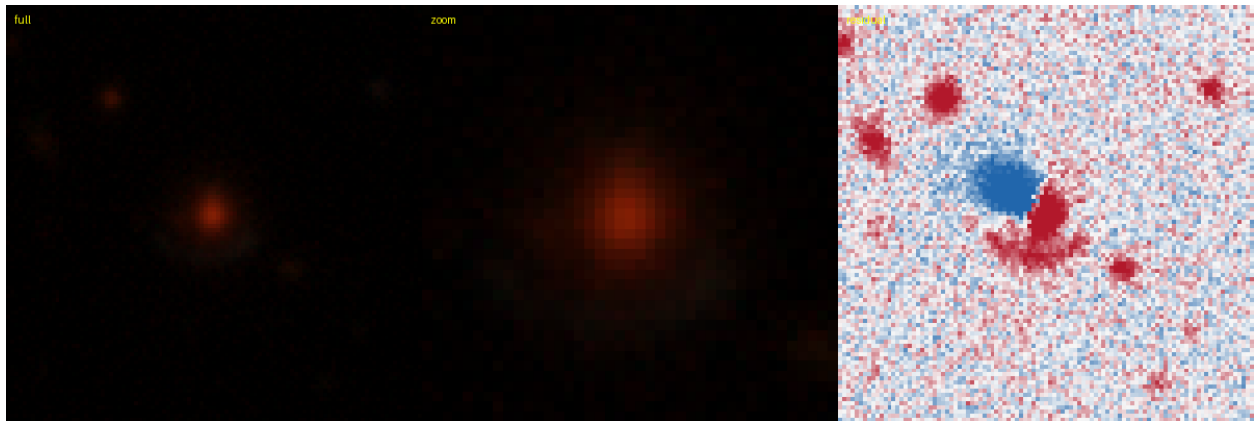


Figure 39: #35 s_102952.10164 — RA 334.8017, Dec -43.8098; grade **B**, tier-1 p_{lens} 0.58, v3blend8 0.85.
KNOWN — DES: DES J221912.39-434835.1, 0.01'' (lineage).

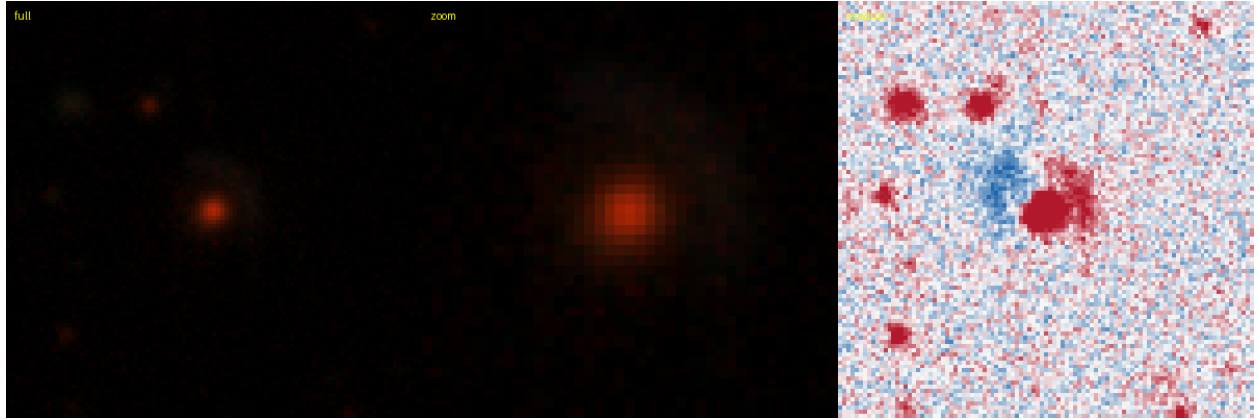


Figure 40: #36 s_191004.9637 — RA 56.9356, Dec -24.9088; grade **B**, tier-1 p_{lens} 0.58, v3blend8 0.88.
KNOWN — DES: DES J034744.53-245431.4, 0.20'' (SIMBAD).

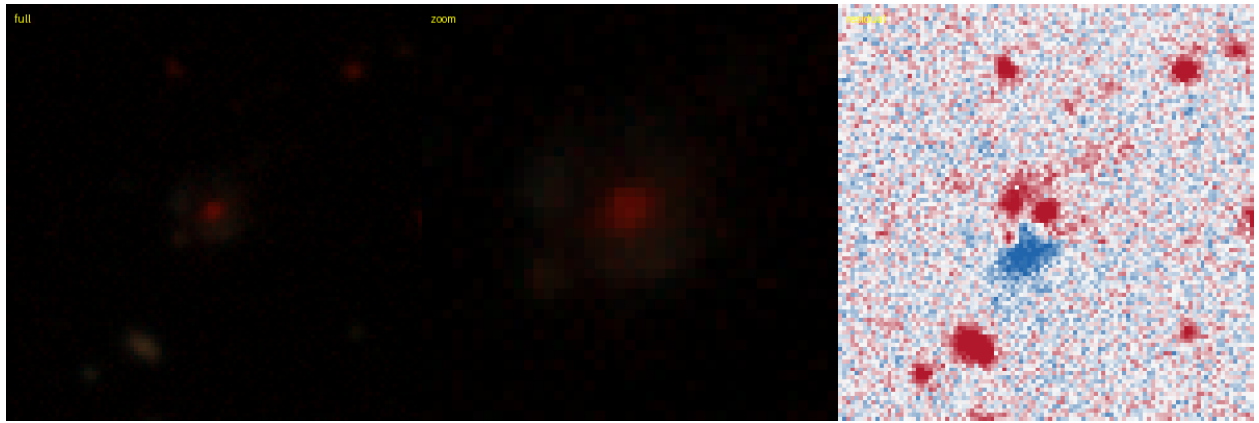


Figure 41: #37 s_43736.6056 — RA 67.3945, Dec -60.1905; grade **B**, tier-1 p_{lens} 0.58, v3blend8 0.83.
KNOWN — DES: DES J042934.66-601126.3, 0.35'' (SIMBAD).

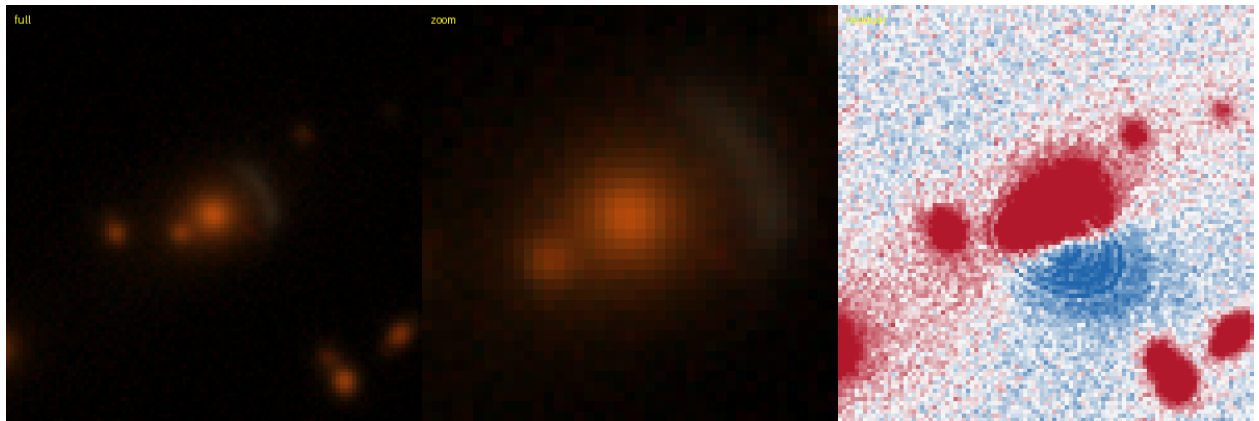


Figure 42: #38 s_137518.1133 — RA 31.3588, Dec -35.6632; grade **B**, tier-1 p_{lens} 0.58, v3blend8 0.84.
KNOWN — DES: DES J020526.09-353947.4, 0.14'' (SIMBAD).

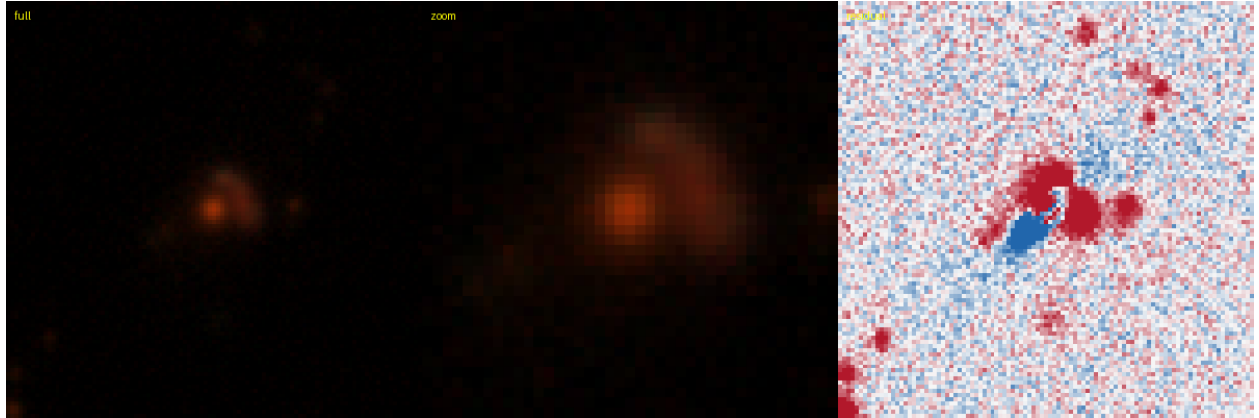


Figure 43: #39 s_129372_3768 — RA 16.4504, Dec -37.4286; grade **B**, tier-1 p_{lens} 0.58, v3blend8 0.82. **KNOWN** — DES: DES J010548.04-372542.4, 0.71'' (SIMBAD).

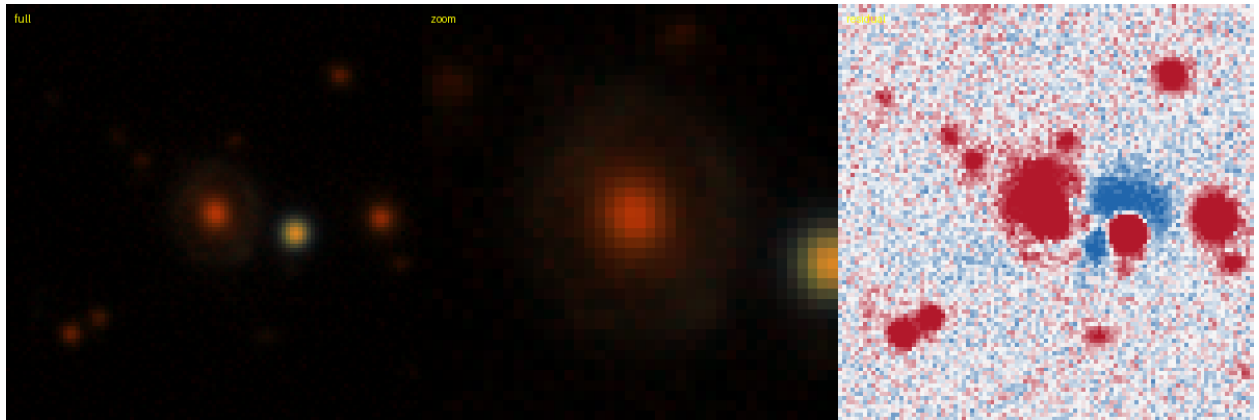


Figure 44: #40 s_98882_7820 — RA 1.6860, Dec -44.4974; grade **B**, tier-1 p_{lens} 0.58, v3blend8 0.80. **KNOWN** — DES: [DBL2017] DES J0006-4429 (A), 0.18'' (SIMBAD).



Figure 45: #43 s_381188_1199 — RA 144.1511, Dec +8.8633; grade **B**, tier-1 p_{lens} 0.55, v3blend8 0.90. **KNOWN** — Huang+2020: DESI-144.1511+08.8633, 0.20'' (lineage).

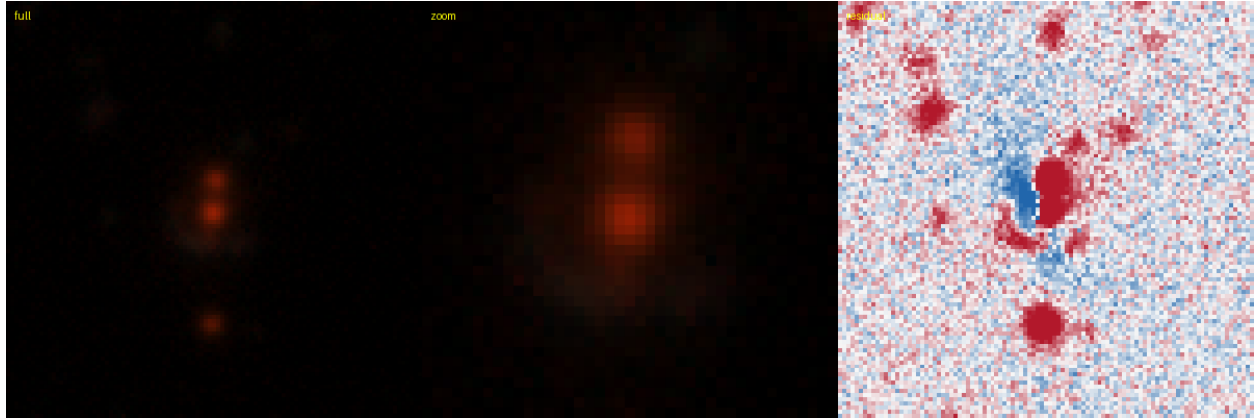


Figure 46: #44 s_114736_1046 — RA 343.5127, Dec -40.9303; grade **B**, tier-1 p_{lens} 0.55, v3blend8 0.86. **KNOWN** — DES: [DBL2017] DES J2254-4055 (A), $0.11''$ (SIMBAD).

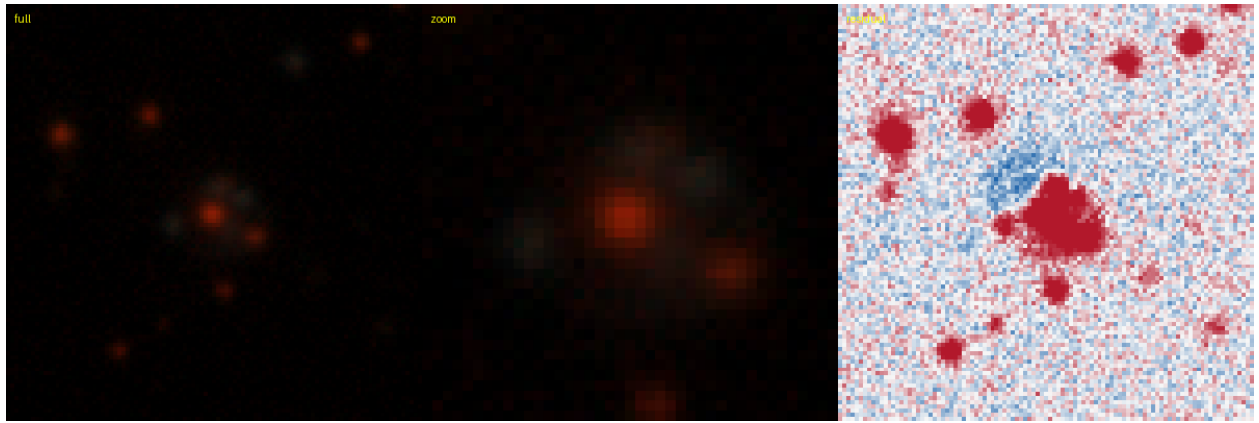


Figure 47: #45 s_87091_1800 — RA 84.5193, Dec -47.5872; grade **B**, tier-1 p_{lens} 0.55, v3blend8 0.89. **KNOWN** — DES: [DBL2017] DES J0538-4735 (A), $0.11''$ (SIMBAD).

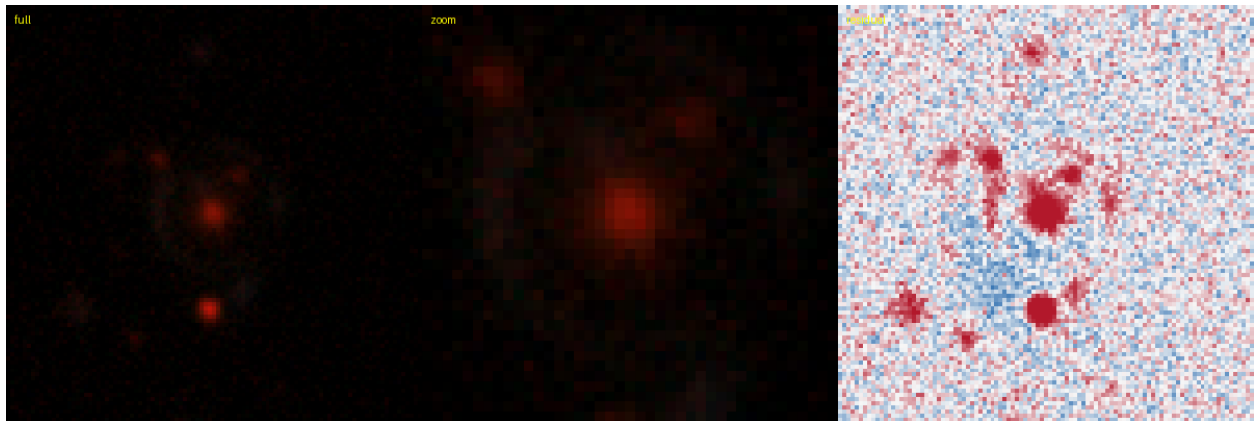


Figure 48: #48 s_384032_2232 — RA 143.3887, Dec +9.3219; grade **B**, tier-1 p_{lens} 0.55, v3blend8 0.84. **KNOWN** — Huang+2020: DESI-143.3887+09.3219, $0.17''$ (lineage).

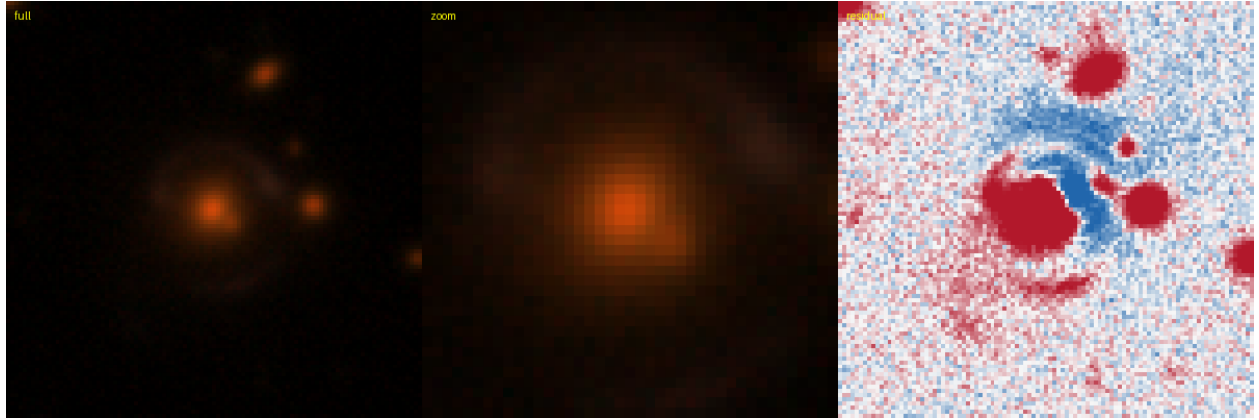


Figure 49: #49 s_73683.6069 — RA 357.3753, Dec -51.2276; grade **B**, tier-1 p_{lens} 0.55, v3blend8 0.83. **KNOWN** — DES: [DBL2017] DES J2349-5113 (A), 0.00'' (SIMBAD).

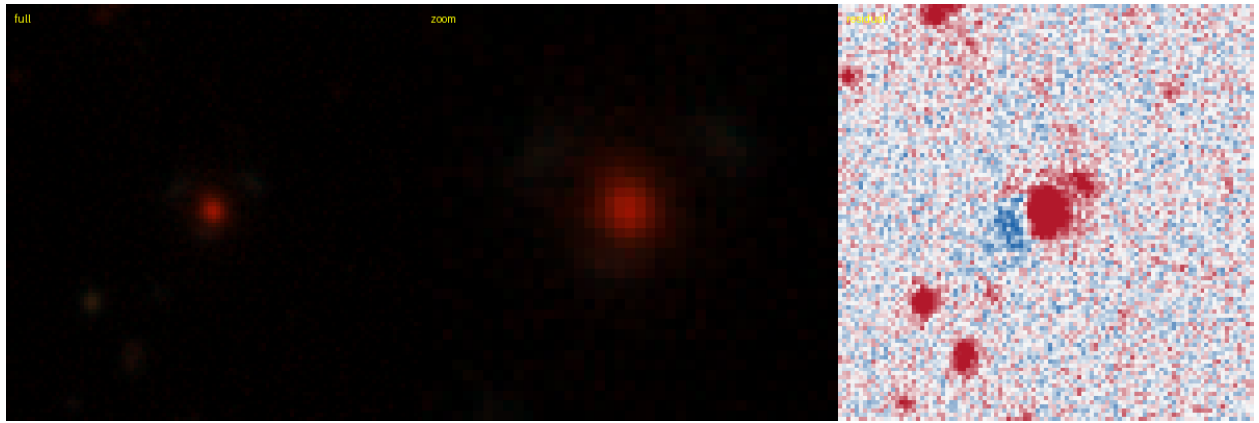


Figure 50: #50 s_80200.6872 — RA 15.4919, Dec -49.2940; grade **B**, tier-1 p_{lens} 0.55, v3blend8 0.88. **KNOWN** — DES: DES J010158.03-491738.1, 0.12'' (SIMBAD).

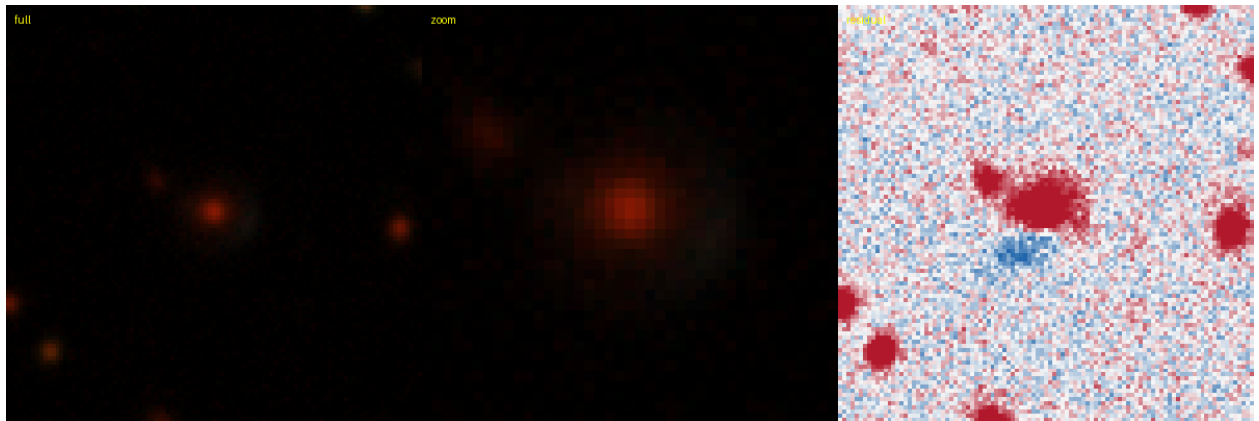


Figure 51: #51 s_58263.3751 — RA 42.7179, Dec -55.4032; grade **B**, tier-1 p_{lens} 0.55, v3blend8 0.88. **KNOWN** — DES: DES J025052.27-552411.7, 0.22'' (SIMBAD).

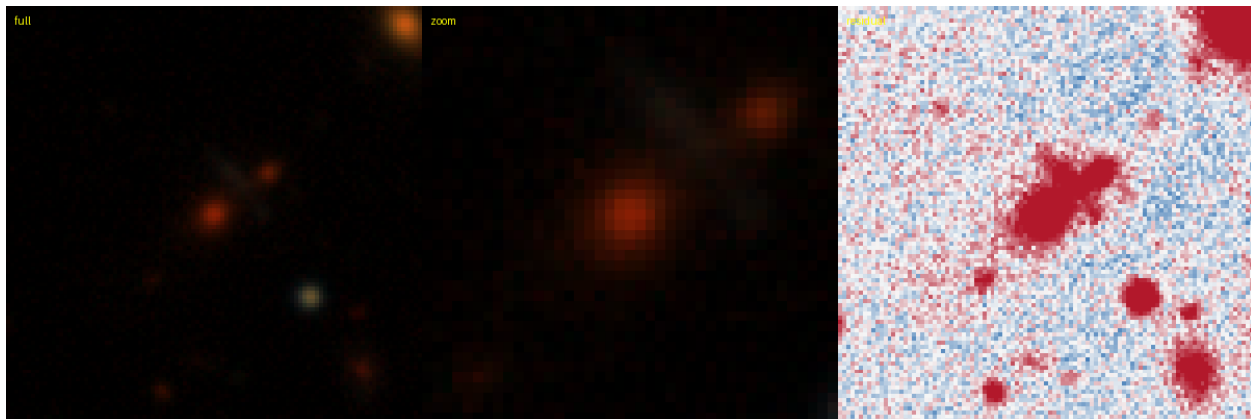


Figure 52: #53 s_164140_7181 — RA 41.2060, Dec -30.3214; grade **B**, tier-1 p_{lens} 0.50, v3blend8 0.79.
KNOWN — DES: DES J024449.42-301916.7, 0.24'' (SIMBAD).

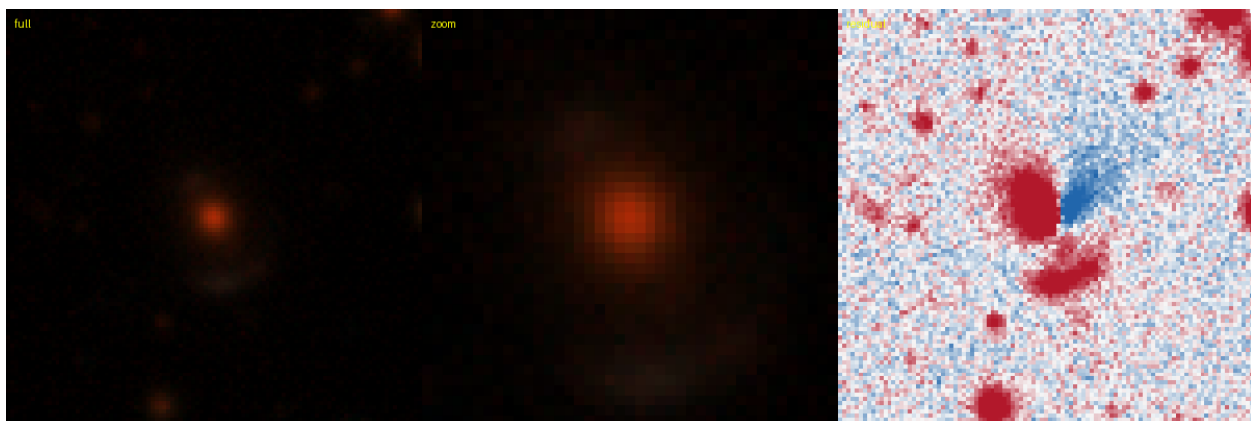


Figure 53: #54 s_71871_246 — RA 353.9664, Dec -51.8716; grade **B**, tier-1 p_{lens} 0.50, v3blend8 0.76.
KNOWN — AGEL: AGEL J233552-515218 system, 0.95'' (SIMBAD).



Figure 54: #55 s_61626_1035 — RA 65.1782, Dec -54.3770; grade **B**, tier-1 p_{lens} 0.50, v3blend8 0.87.
KNOWN — DES: DES J042042.76-542237.4, 0.18'' (SIMBAD).

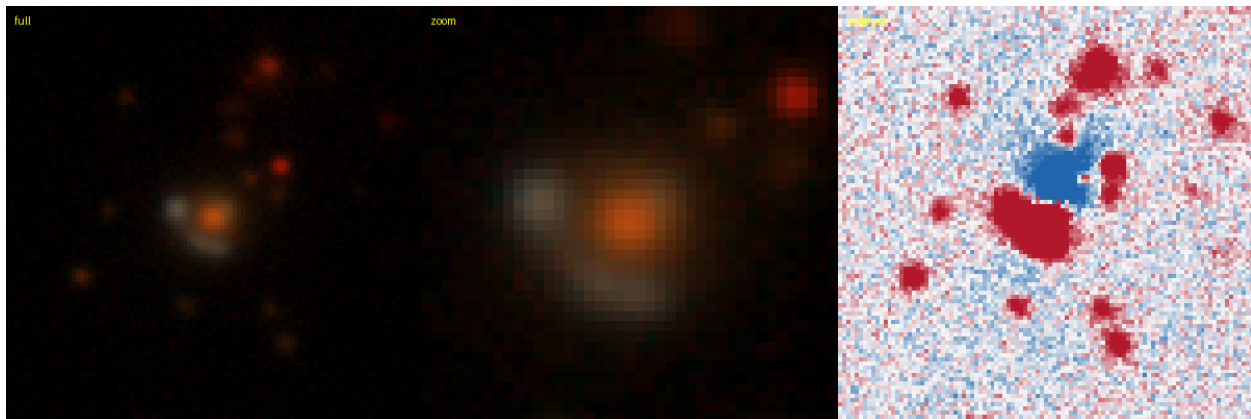


Figure 55: #57 s_107312_5223 — RA 23.8451, Dec -42.5398; grade **B**, tier-1 p_{lens} 0.50, v3blend8 0.85. **KNOWN** — DES: [DBL2017] DES J0135-4232 (A), 0.18'' (SIMBAD).

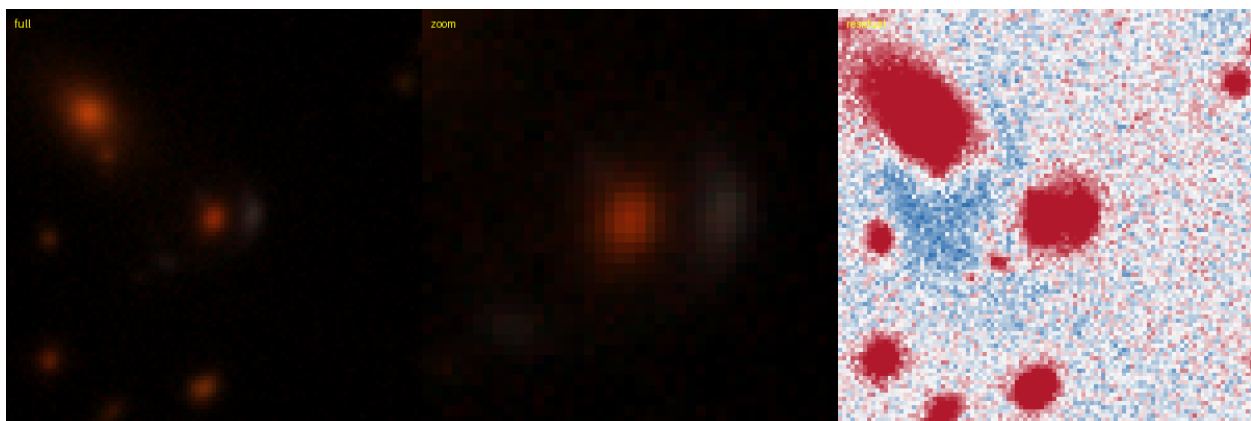


Figure 56: #58 s_90920_9419 — RA 45.9517, Dec -46.4407; grade **B**, tier-1 p_{lens} 0.50, v3blend8 0.85. **KNOWN** — DES: [DBL2017] DES J0303-4626 (A), 0.15'' (SIMBAD).

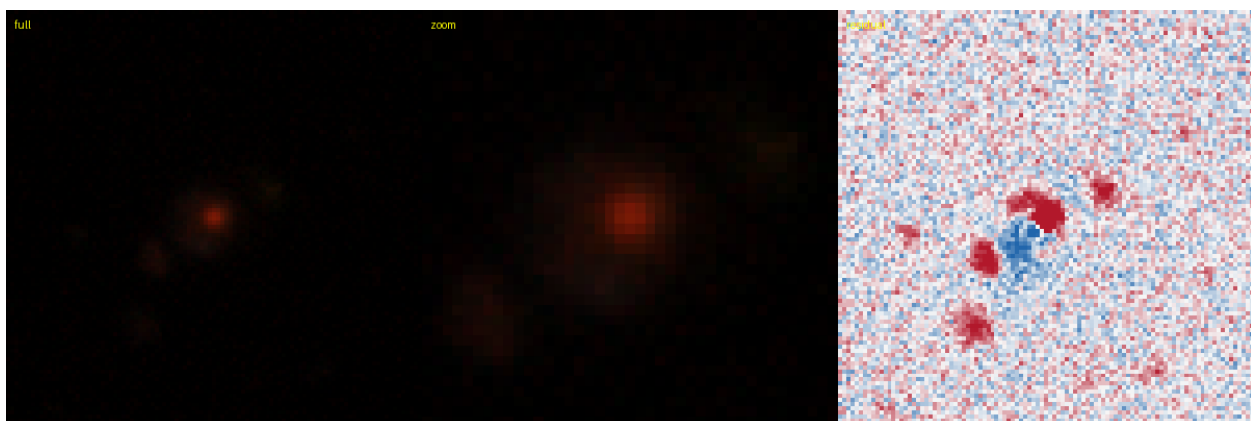


Figure 57: #62 s_162970_1777 — RA 62.5554, Dec -30.6238; grade **B**, tier-1 p_{lens} 0.48, v3blend8 0.82. **KNOWN** — DES: DES J041013.30-303725.8, 0.09'' (SIMBAD).

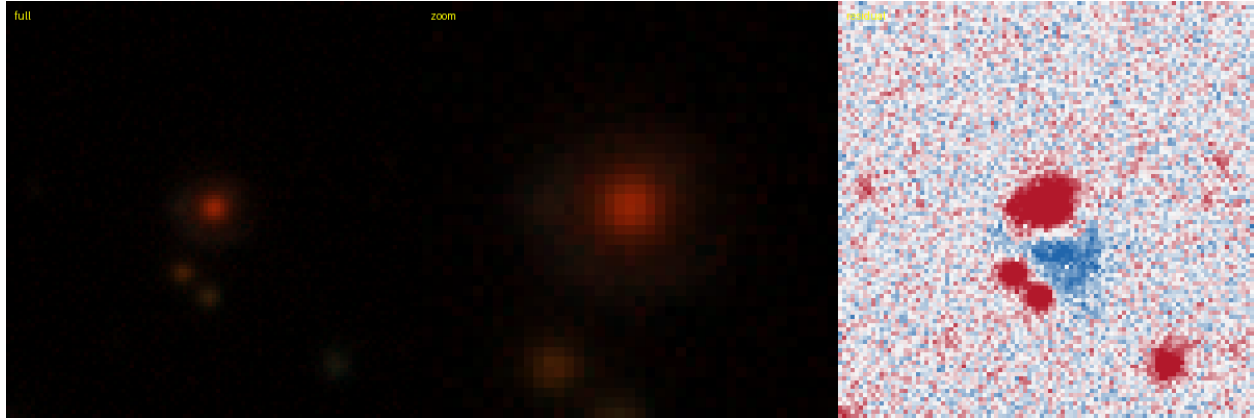


Figure 58: #63 s_234043_6467 — RA 58.4427, Dec -17.1109; grade **B**, tier-1 p_{lens} 0.47, v3blend8 0.88.
KNOWN — AGEL: AGEL J035346-170639 l_{lens} , 0.04'' (SIMBAD).

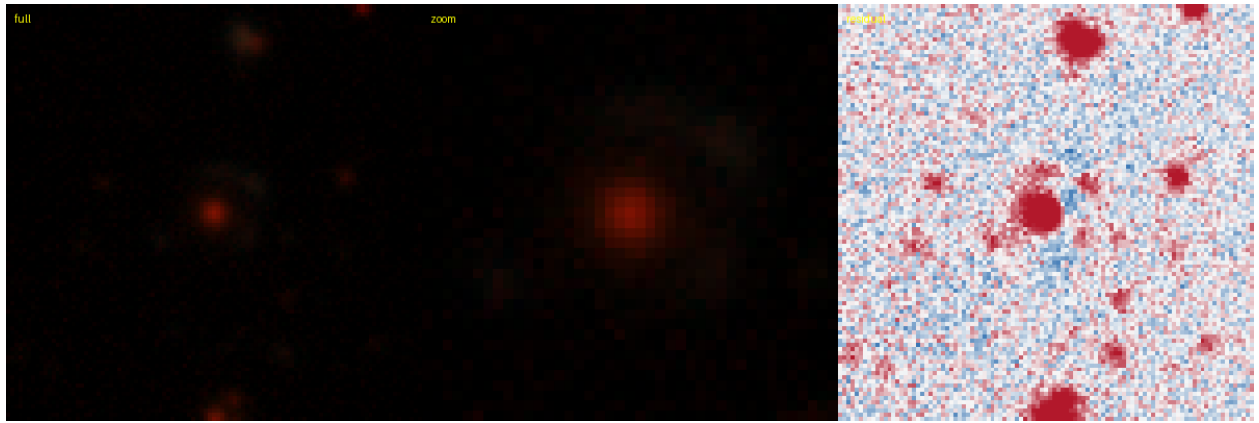


Figure 59: #65 s_50907_11900 — RA 303.5808, Dec -57.9504; grade **B**, tier-1 p_{lens} 0.45, v3blend8 0.74.
KNOWN — DES: DES J201419.38-575701.4, 0.15'' (SIMBAD).

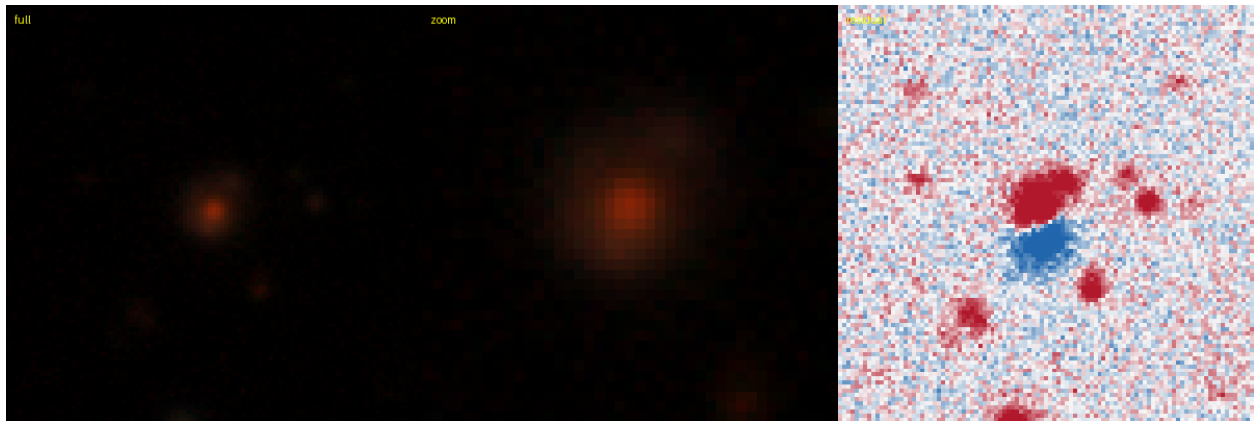


Figure 60: #66 s_61411_4481 — RA 332.9866, Dec -54.6445; grade **B**, tier-1 p_{lens} 0.45, v3blend8 0.76.
KNOWN — DES: DES J221156.77-543839.9, 0.22'' (SIMBAD).



Figure 61: #67 s_165468.9648 — RA 64.7771, Dec -29.9937; grade **B**, tier-1 p_{lens} 0.45, v3blend8 0.78. **KNOWN** — DES: DES J041906.50-295937.7, 0.49'' (SIMBAD).

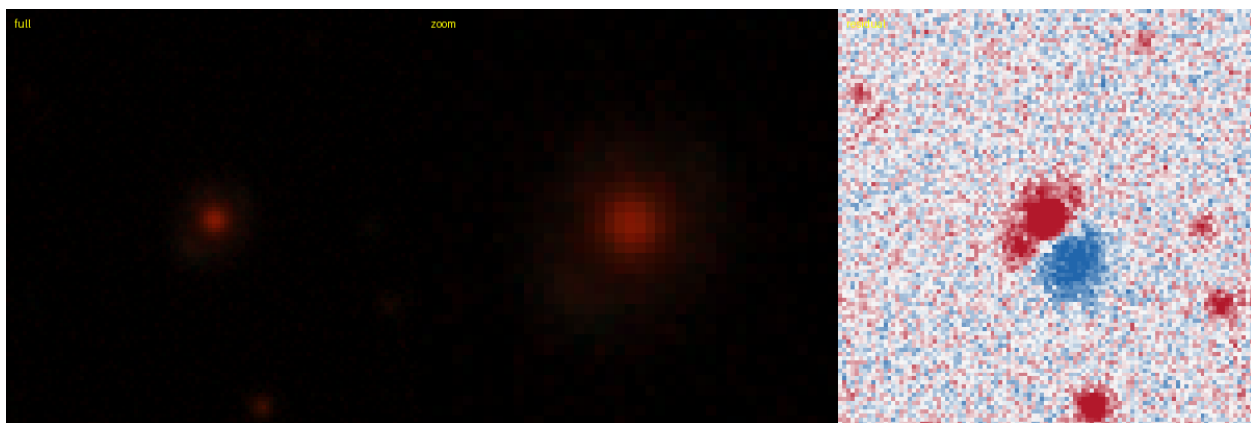


Figure 62: #69 s_280264.9147 — RA 33.1051, Dec -8.8697; grade **B**, tier-1 p_{lens} 0.45, v3blend8 0.84. **KNOWN** — literature (SIMBAD): [MVM2016] SW45, 0.12'' (SIMBAD).

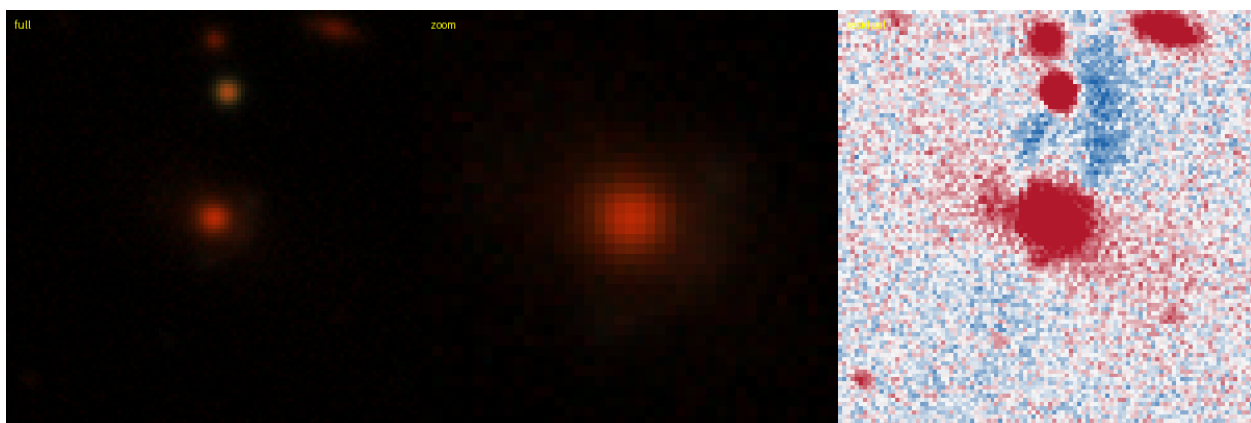


Figure 63: #72 s_52008.8091 — RA 96.5963, Dec -57.5083; grade **B**, tier-1 p_{lens} 0.45, v3blend8 0.89. **KNOWN** — DES: DES J062623.11-573030.1, 0.09'' (SIMBAD).

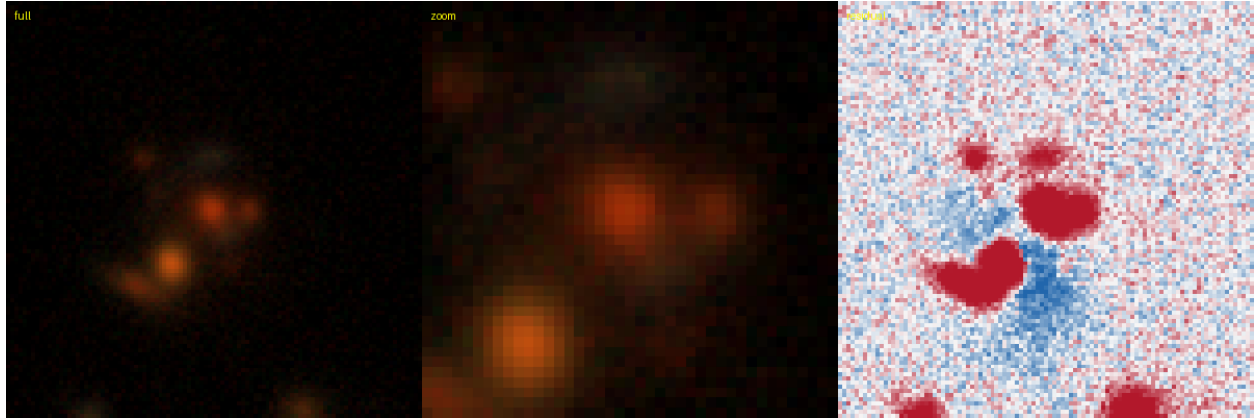


Figure 64: #74 s_504880_4027 — RA 152.0763, Dec +31.7005; grade **B**, tier-1 p_{lens} 0.45, v3blend8 0.84. **KNOWN** — Huang+2020: DESI-152.0763+31.7005, 0.12'' (lineage).

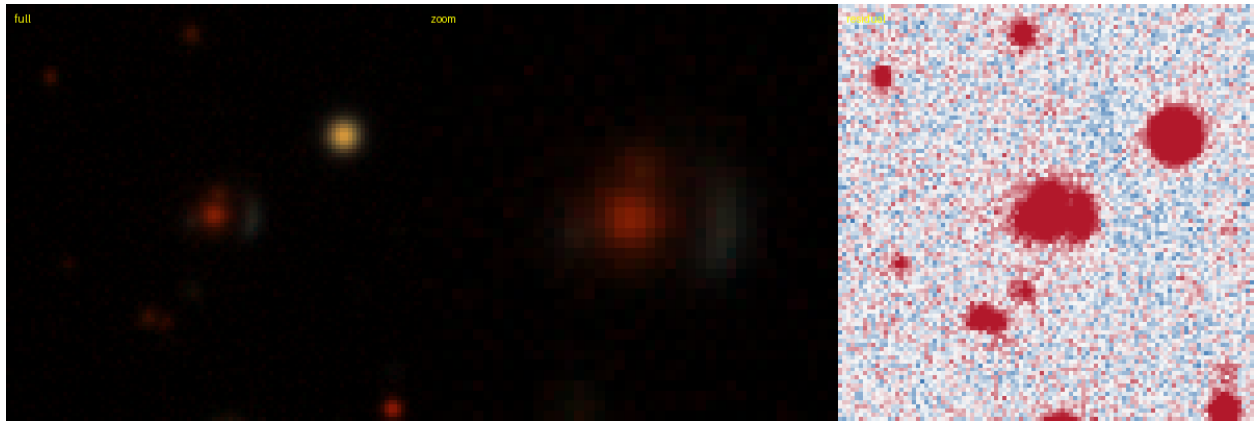


Figure 65: #76 s_69087_268 — RA 307.2325, Dec -52.5218; grade **B**, tier-1 p_{lens} 0.43, v3blend8 0.81. **KNOWN** — DES: DES J202855.79-523118.3, 0.12'' (SIMBAD).

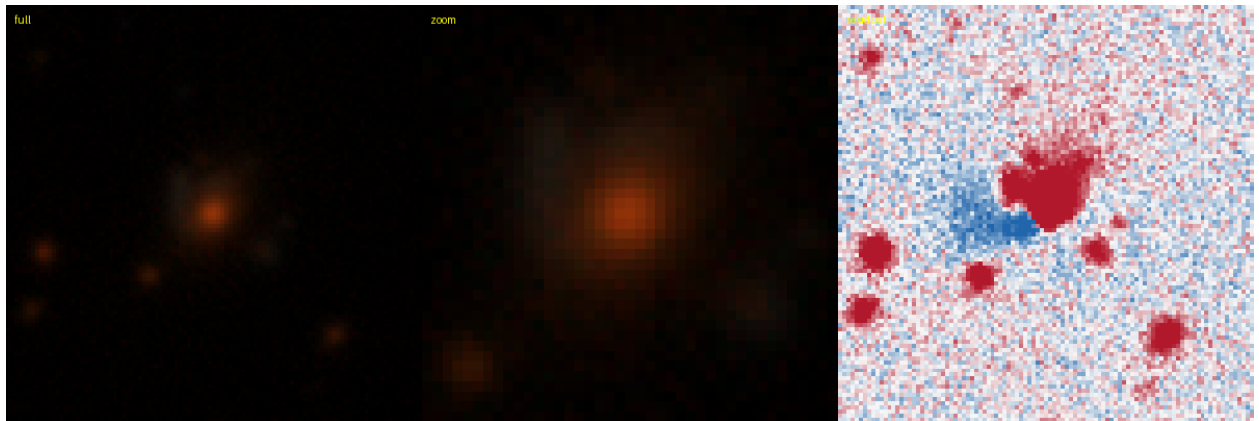


Figure 66: #77 s_256199_9097 — RA 29.6819, Dec -13.0258; grade **B**, tier-1 p_{lens} 0.42, v3blend8 0.78. **KNOWN** — DES: DES J015843.66-130132.8, 0.18'' (SIMBAD).

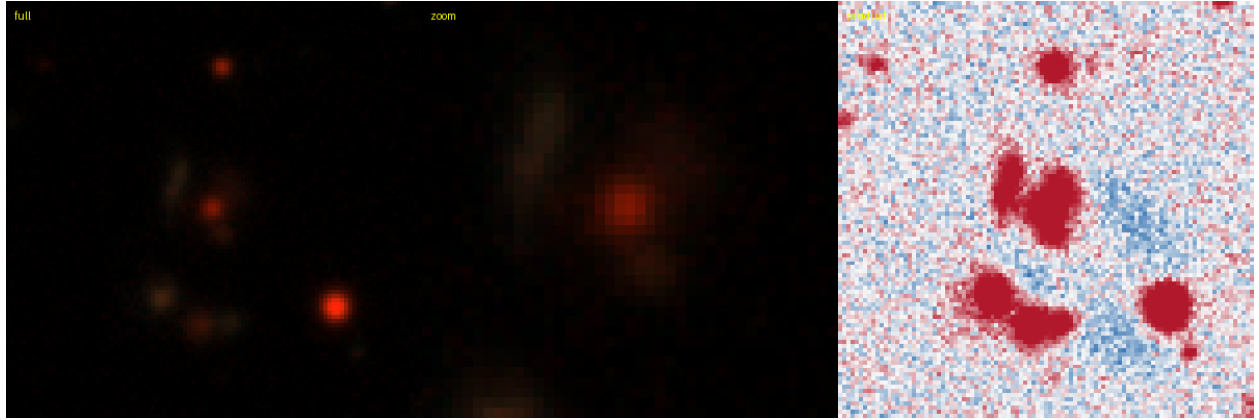


Figure 67: #78 s_36700.530 — RA 347.9904, Dec -63.1176; grade **B**, tier-1 p_{lens} 0.42, v3blend8 0.78.
KNOWN — literature (SIMBAD): DES J2311-6307: [OWD2022] 1, 2.80'' (NED).

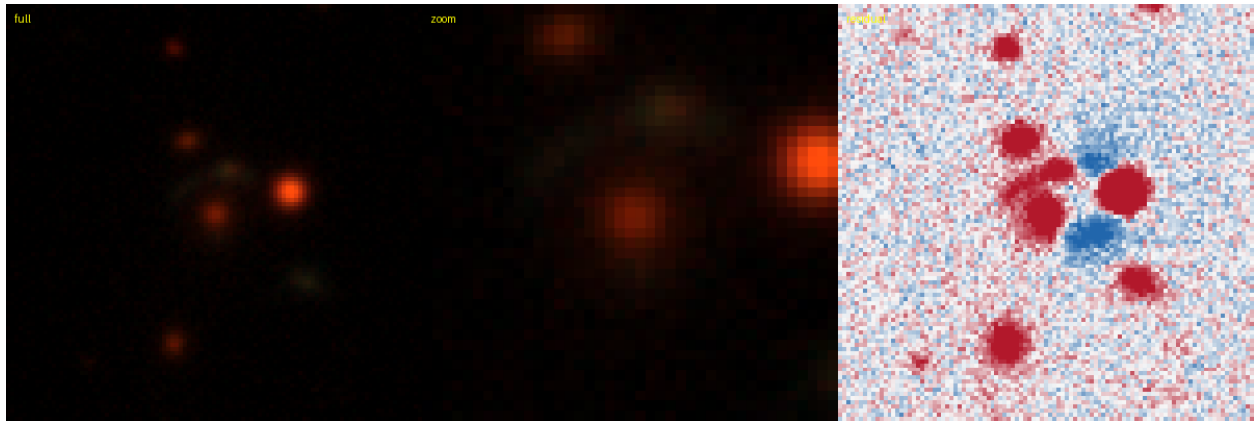


Figure 68: #81 s_204169.8722 — RA 49.1618, Dec -22.6093; grade **B**, tier-1 p_{lens} 0.42, v3blend8 0.77.
KNOWN — DES: DES J031638.82-223633.3, 0.15'' (SIMBAD).



Figure 69: #83 s_139825.11098 — RA 18.9226, Dec -35.3386; grade **B**, tier-1 p_{lens} 0.42, v3blend8 0.76.
KNOWN — literature (SIMBAD): DES J0115-3520: [OWD2022] 1, 4.88'' (NED).

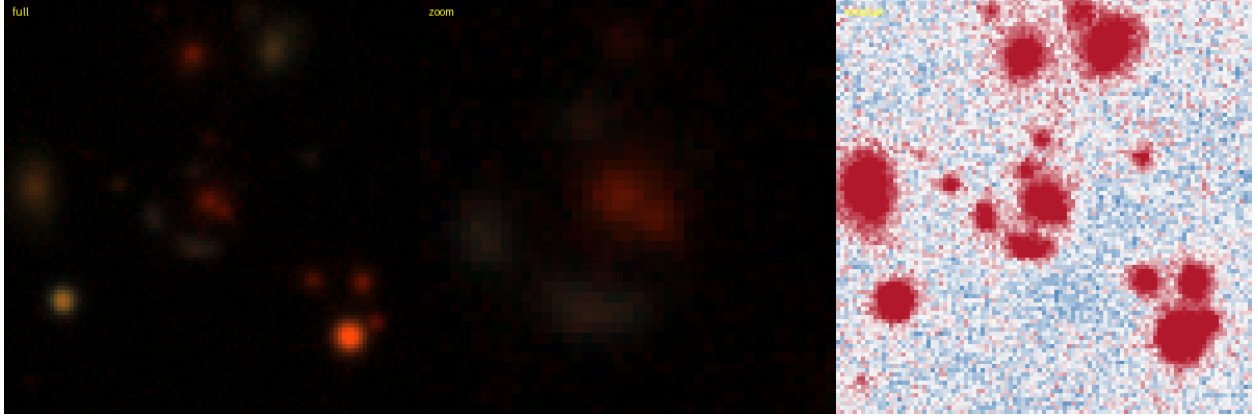


Figure 70: #84 s_60524_3113 — RA 309.7973, Dec -54.9952; grade **B**, tier-1 p_{lens} 0.42, v3blend8 0.81. **KNOWN** — DES: [DBL2017] DES J2039-5459 (A), 0.16'' (SIMBAD).

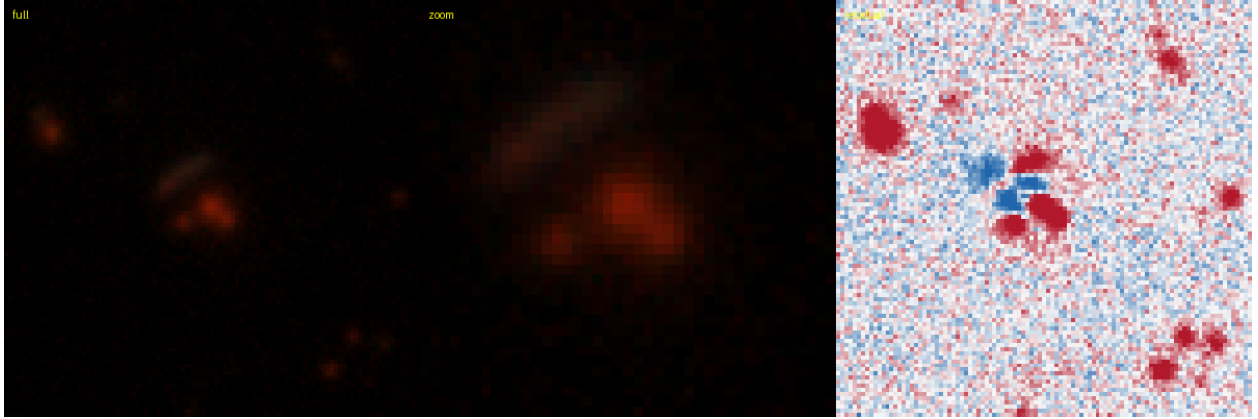


Figure 71: #85 s_38835_314 — RA 56.6543, Dec -61.9792; grade **B**, tier-1 p_{lens} 0.42, v3blend8 0.84. **KNOWN** — DES: [DBL2017] DES J0346-6158 (A), 0.23'' (SIMBAD).

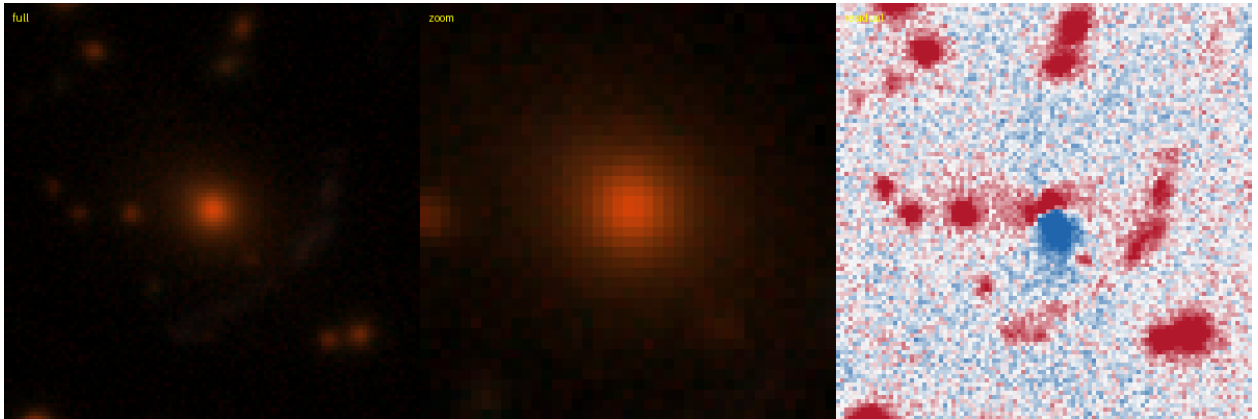


Figure 72: #87 s_68464_11443 — RA 52.7373, Dec -52.4703; grade **B**, tier-1 p_{lens} 0.42, v3blend8 0.88. **KNOWN** — DES: [DBL2017] DES J0330-5228, 0.11'' (SIMBAD).

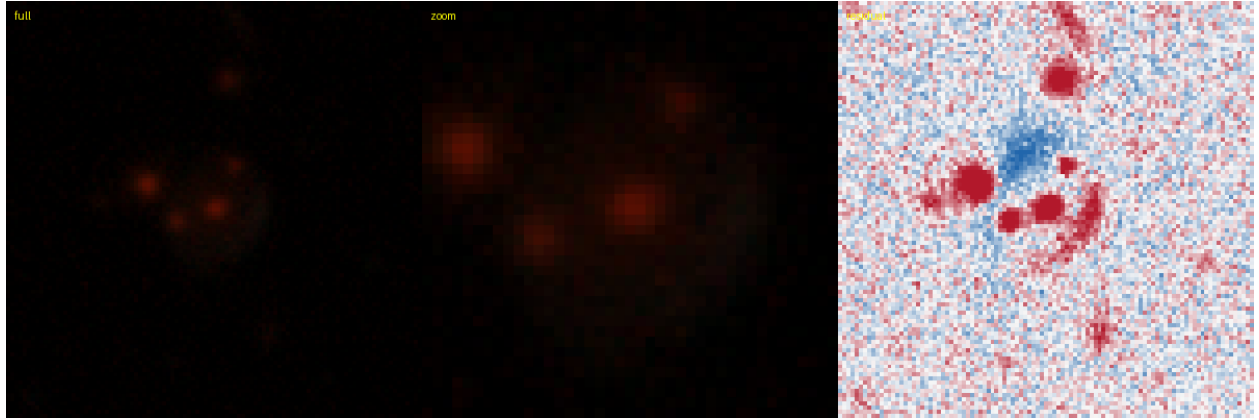


Figure 73: #88 s_206743_10995 — RA 24.2120, Dec -22.0076; grade **B**, tier-1 p_{lens} 0.42, v3blend8 0.85.
KNOWN — DES: DES J013650.85-220027.2, 0.23'' (SIMBAD).

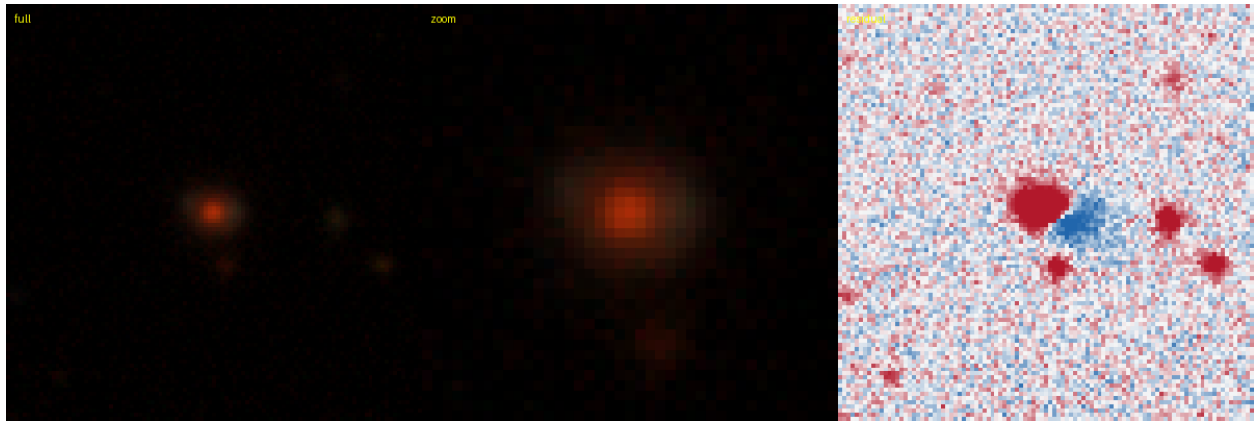


Figure 74: #89 s_41562.4180 — RA 42.2402, Dec -60.9010; grade **B**, tier-1 p_{lens} 0.40, v3blend8 0.87.
KNOWN — DES: DES J024857.61-605403.4, 0.19'' (SIMBAD).

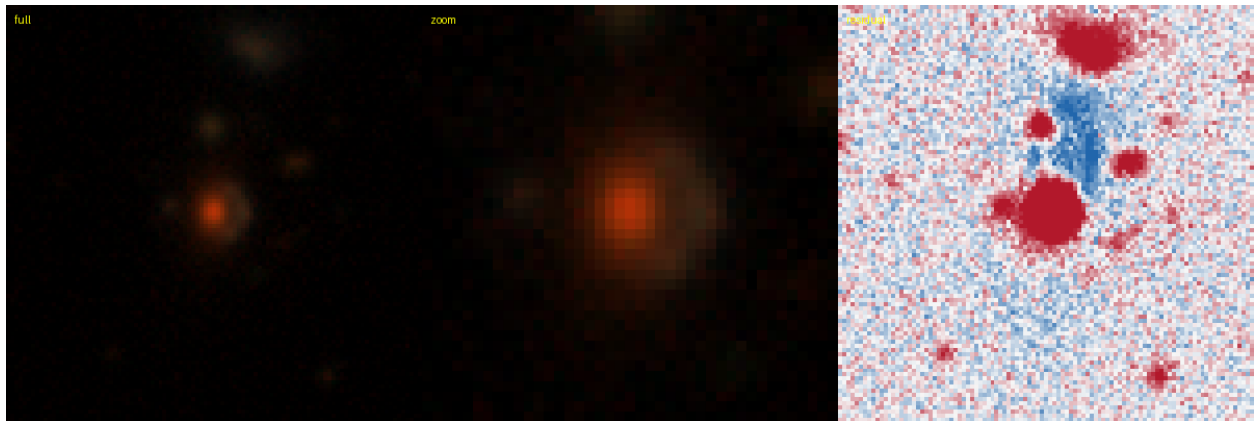


Figure 75: #97 s_156771_3446 — RA 54.3219, Dec -31.8704; grade **B**, tier-1 p_{lens} 0.38, v3blend8 0.89.
KNOWN — AGEL: AGEL J033717-315214 system, 0.99'' (SIMBAD).

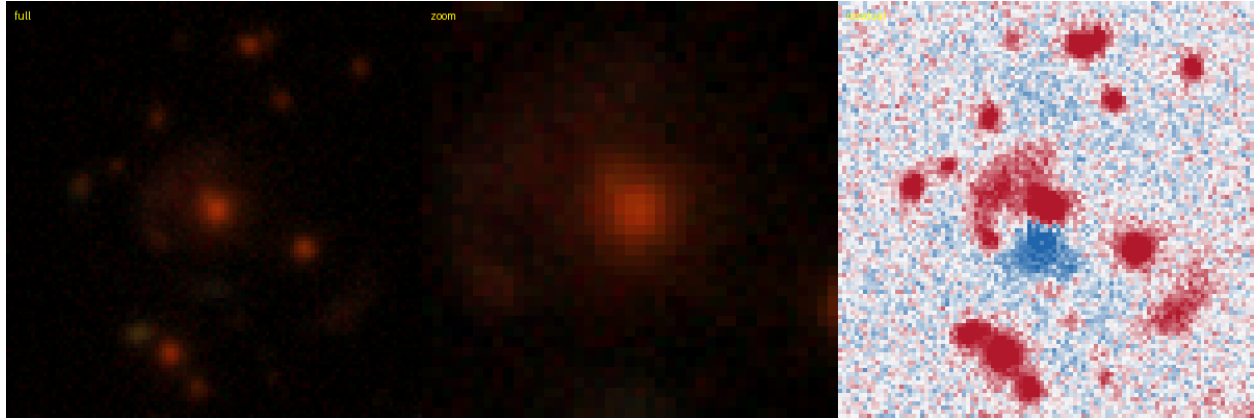


Figure 76: #99 s_416616_6090 — RA 189.5370, Dec +15.0309; grade **B**, tier-1 p_{lens} 0.38, v3blend8 0.93.
KNOWN — Huang+2020: DESI-189.5370+15.0309, 0.06'' (lineage).

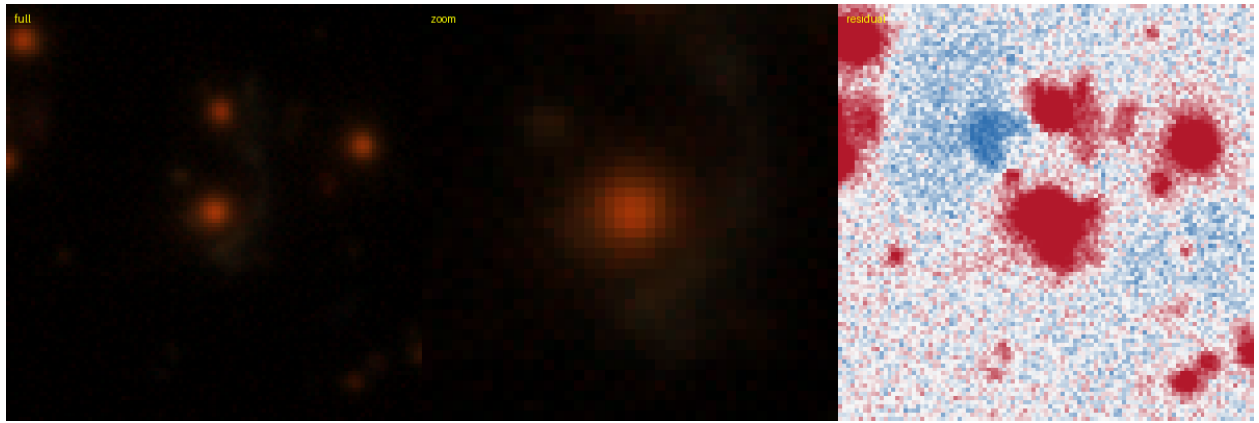


Figure 77: #104 s_194944_4494 — RA 59.2045, Dec -24.1447; grade **B**, tier-1 p_{lens} 0.35, v3blend8 0.85.
KNOWN — DES: DES J035649.05-240841.1, 0.28'' (SIMBAD).

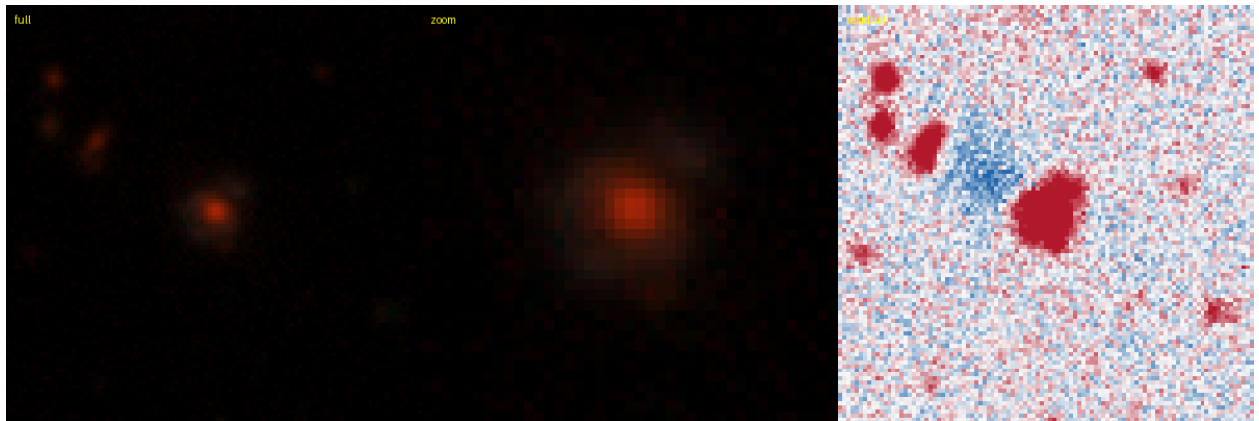


Figure 78: #105 s_94861_9433 — RA 21.9716, Dec -45.5428; grade **B**, tier-1 p_{lens} 0.32, v3blend8 0.87.
KNOWN — DES: DES J012753.17-453233.8, 0.17'' (SIMBAD).

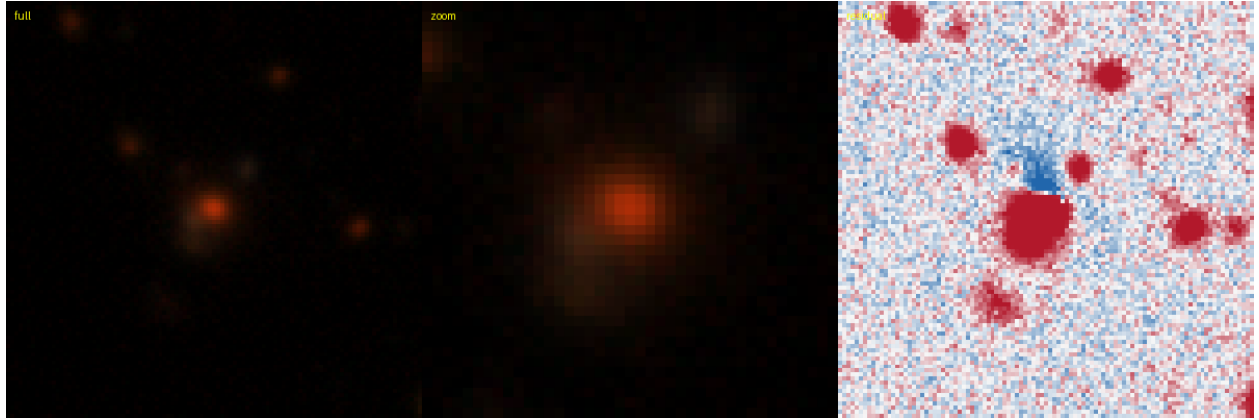


Figure 79: #106 s_235253_4230 — RA 14.4452, Dec -16.7457; grade **B**, tier-1 p_{lens} 0.32, v3blend8 0.76.
KNOWN — Huang+2020: DESI-014.4452-16.7457, 0.17'' (lineage).

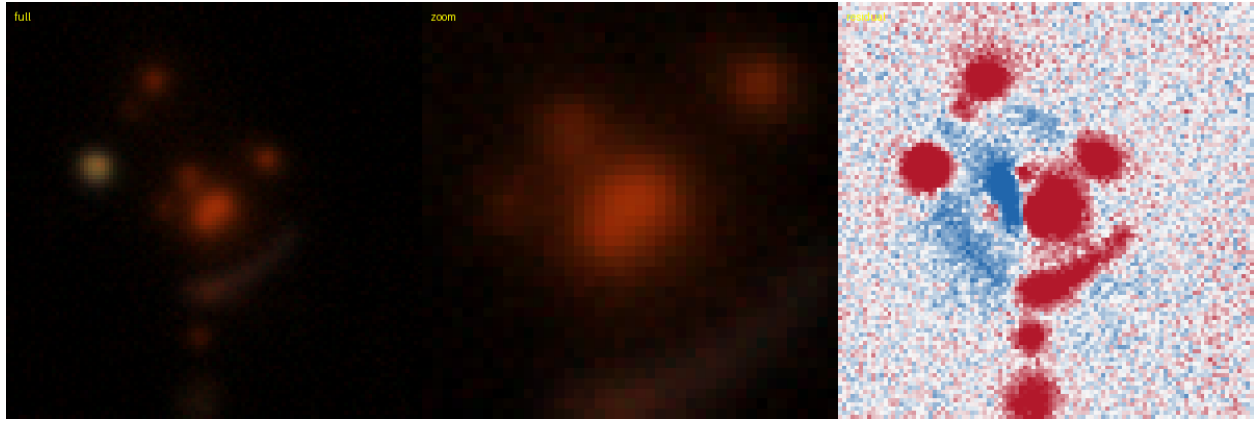


Figure 80: #107 s_394431_6727 — RA 261.7637, Dec +11.0021; grade **B**, tier-1 p_{lens} 0.30, v3blend8 0.77.
KNOWN — AGEL: AGEL J172703+110008 system, 0.61'' (SIMBAD).

New — candidates new to the literature (39 of 108)

These are absent from the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature (3 grade A and 36 grade B); at DECaLS 1'' an unknown (likely large) fraction are lrg+companion mimics. Ranked by tier-1 p_{lens} .

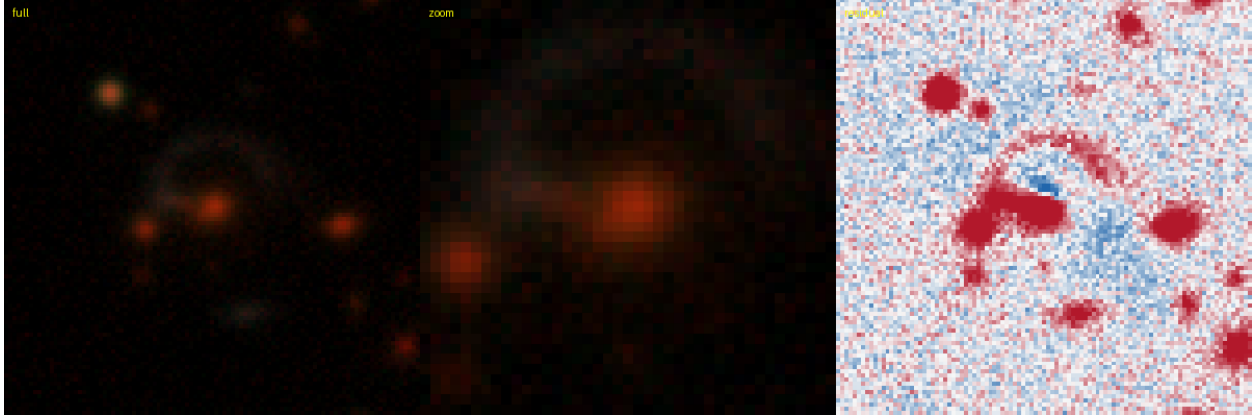


Figure 81: #7 s_230539_4159 — RA 222.0715, Dec -17.8489; grade **A**, tier-1 p_{lens} 0.85, v3blend8 0.84. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

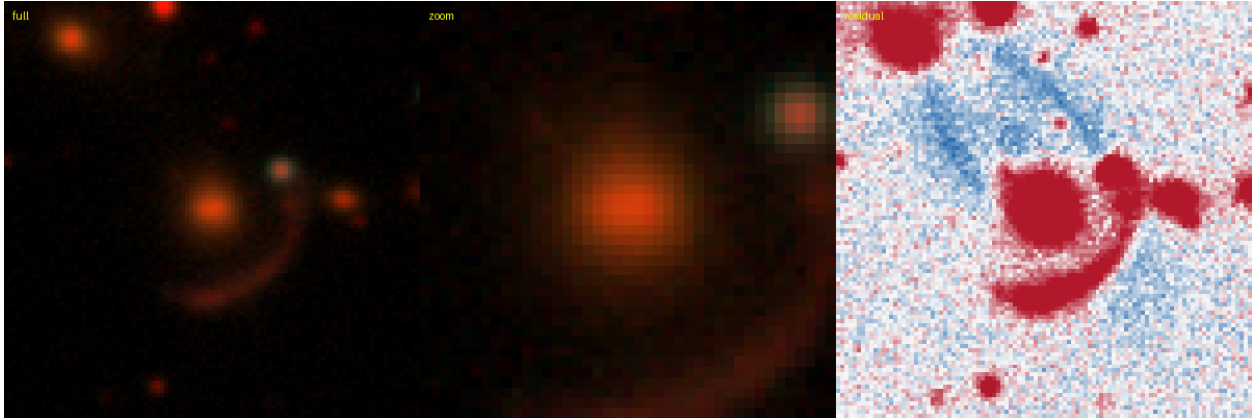


Figure 82: #10 s_194300_2796 — RA 242.6052, Dec -24.4155; grade **A**, tier-1 p_{lens} 0.83, v3blend8 0.74. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

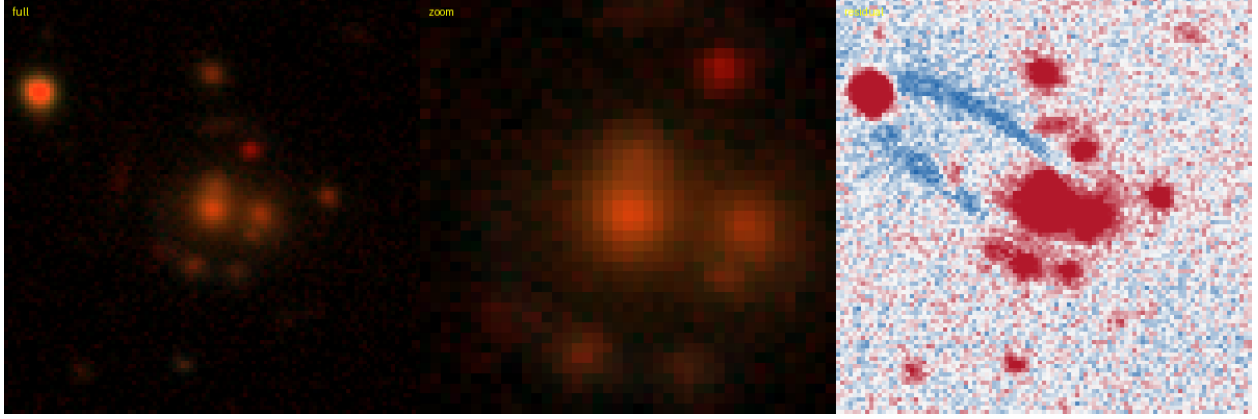


Figure 83: #15 s_320069_835 — RA 305.2638, Dec -1.9376; grade **A**, tier-1 p_{lens} 0.78, v3blend8 0.89. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

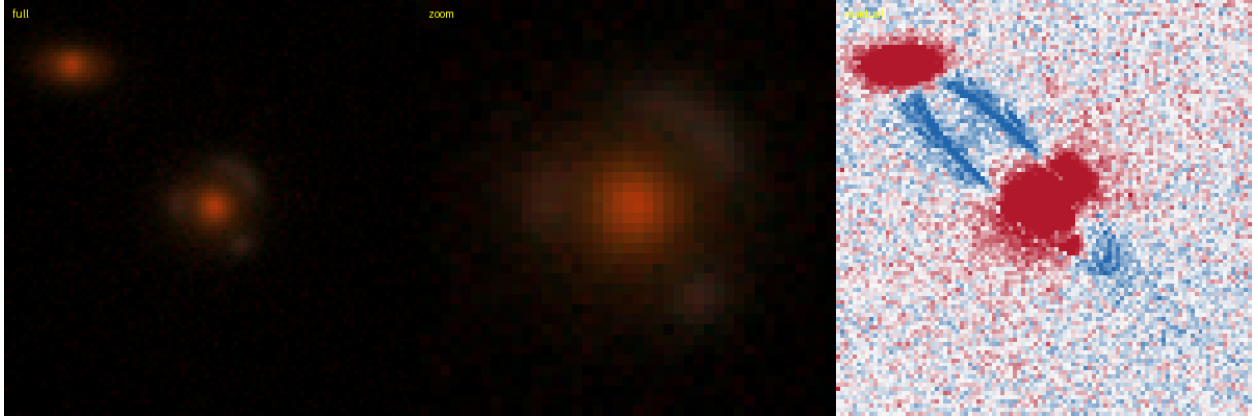


Figure 84: #24 s_234689_542 — RA 227.0427, Dec -16.8771; grade **B**, tier-1 p_{lens} 0.65, v3blend8 0.80. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

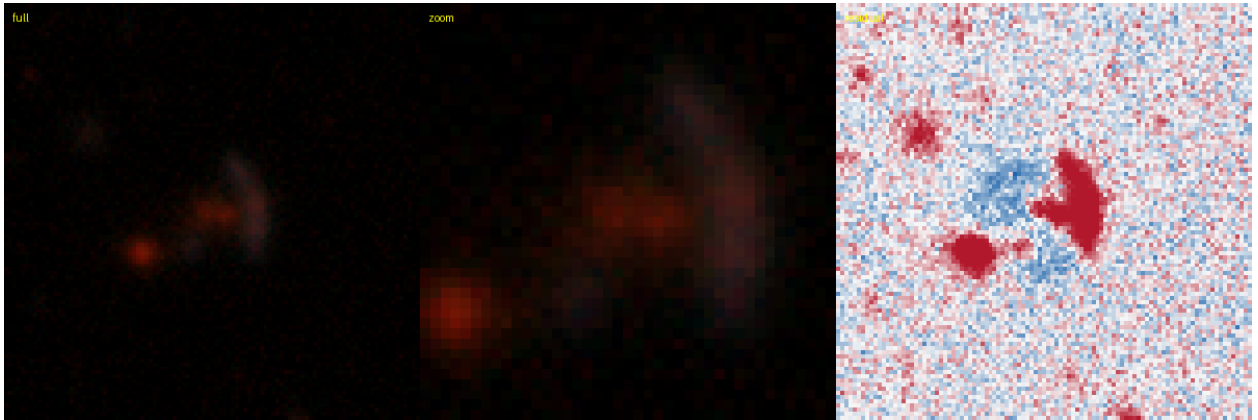


Figure 85: #25 s_502450_1680 — RA 161.1145, Dec +31.2344; grade **B**, tier-1 p_{lens} 0.63, v3blend8 0.79. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

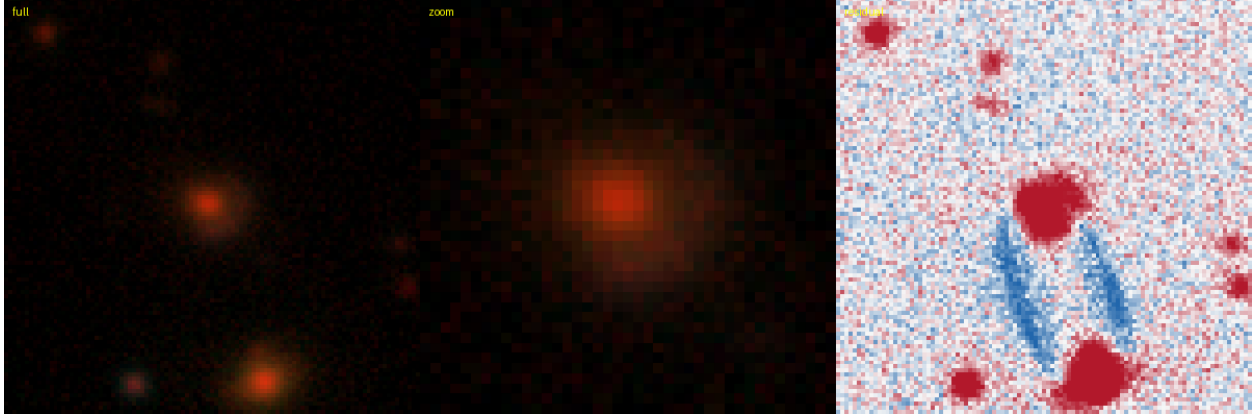


Figure 86: #28 s_243378_3831 — RA 326.5170, Dec -15.4964; grade **B**, tier-1 p_{lens} 0.62, v3blend8 0.84. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

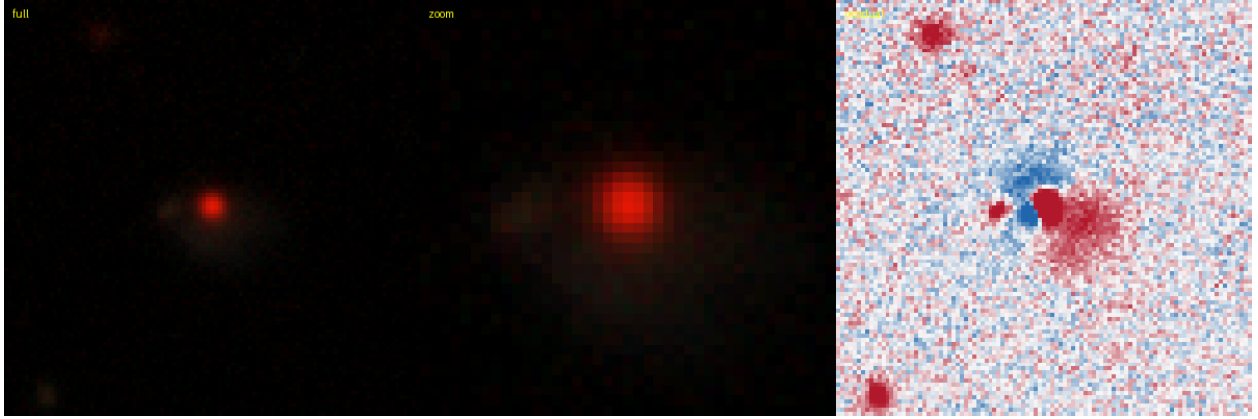


Figure 87: #33 s_369309_653 — RA 30.4465, Dec +6.6551; grade **B**, tier-1 p_{lens} 0.60, v3blend8 0.74. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.



Figure 88: #41 s_218735_721 — RA 350.2028, Dec -20.0497; grade **B**, tier-1 p_{lens} 0.58, v3blend8 0.79. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.



Figure 89: #42 s_360639_3636 — RA 13.4868, Dec +5.3590; grade **B**, tier-1 p_{lens} 0.55, v3blend8 0.85. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

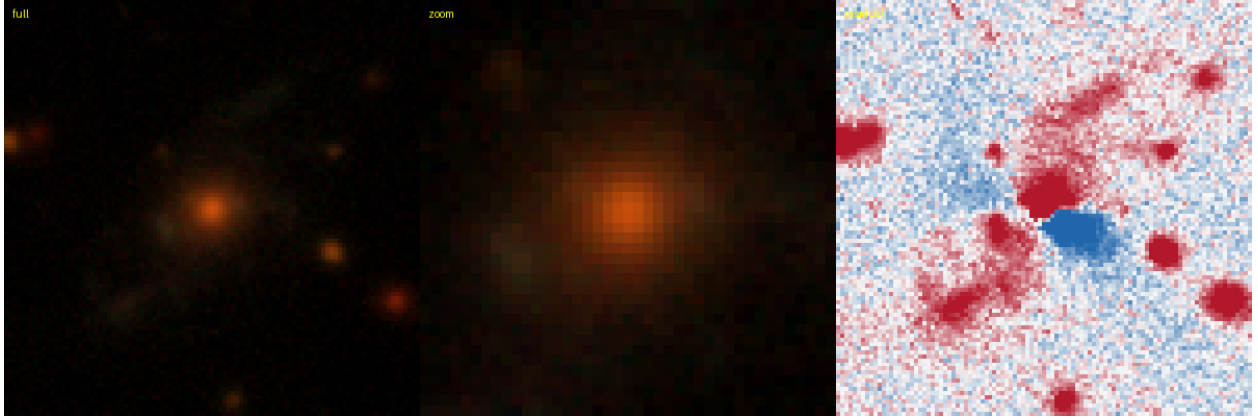


Figure 90: #46 s_327710_4950 — RA 55.6649, Dec -0.3820; grade **B**, tier-1 p_{lens} 0.55, v3blend8 0.74. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

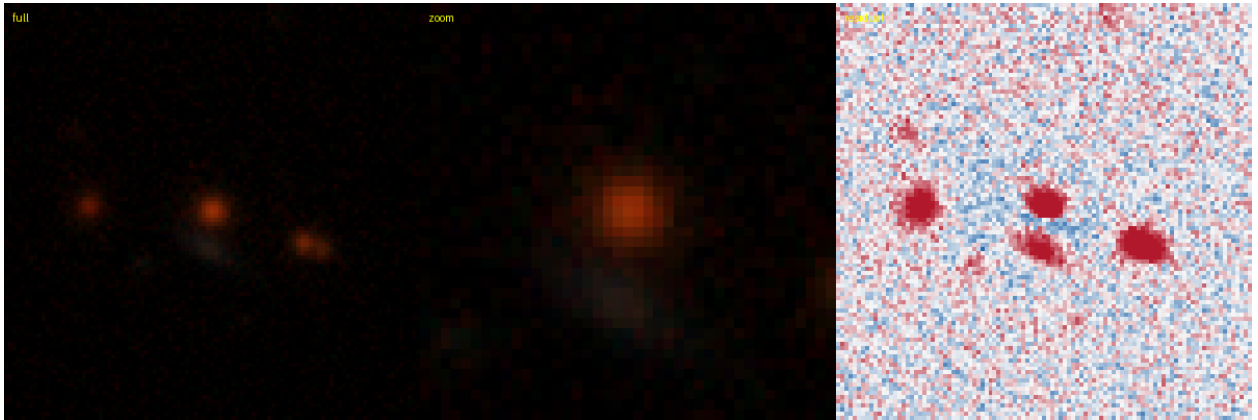


Figure 91: #47 s_289350_3716 — RA 164.8051, Dec -7.1449; grade **B**, tier-1 p_{lens} 0.55, v3blend8 0.81. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

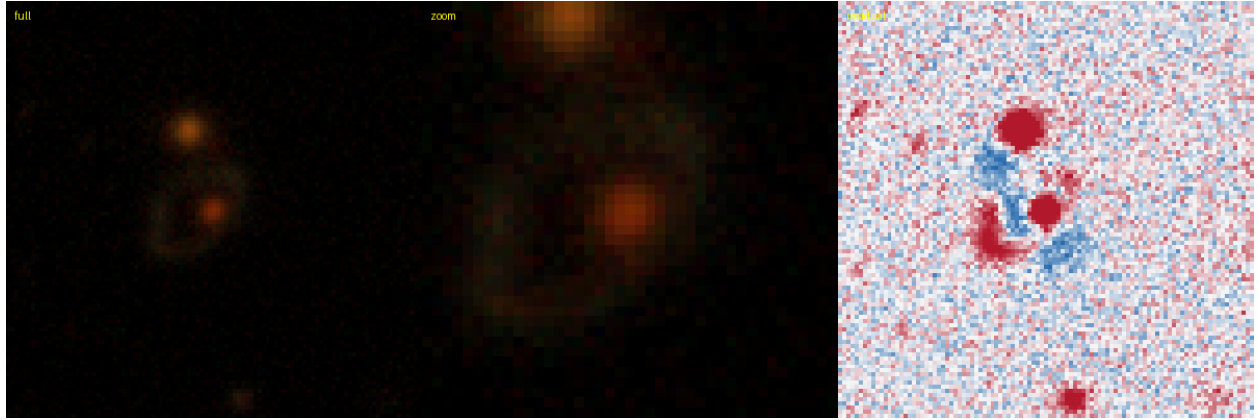


Figure 92: #52 s_491286_1438 — RA 180.3616, Dec +29.0116; grade **B**, tier-1 p_{lens} 0.52, v3blend8 0.74. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

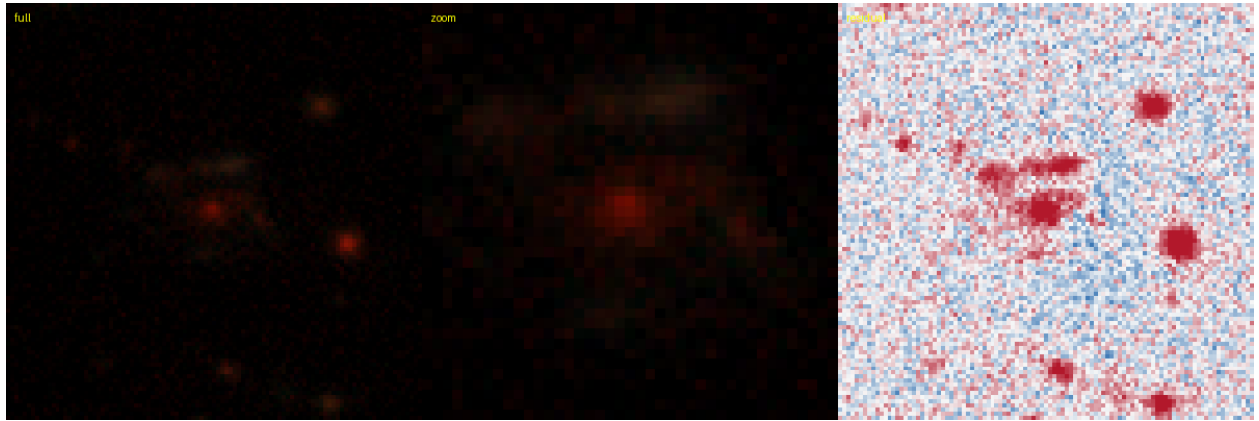


Figure 93: #56 s_262449_2840 — RA 188.7861, Dec -11.9798; grade **B**, tier-1 p_{lens} 0.50, v3blend8 0.87. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

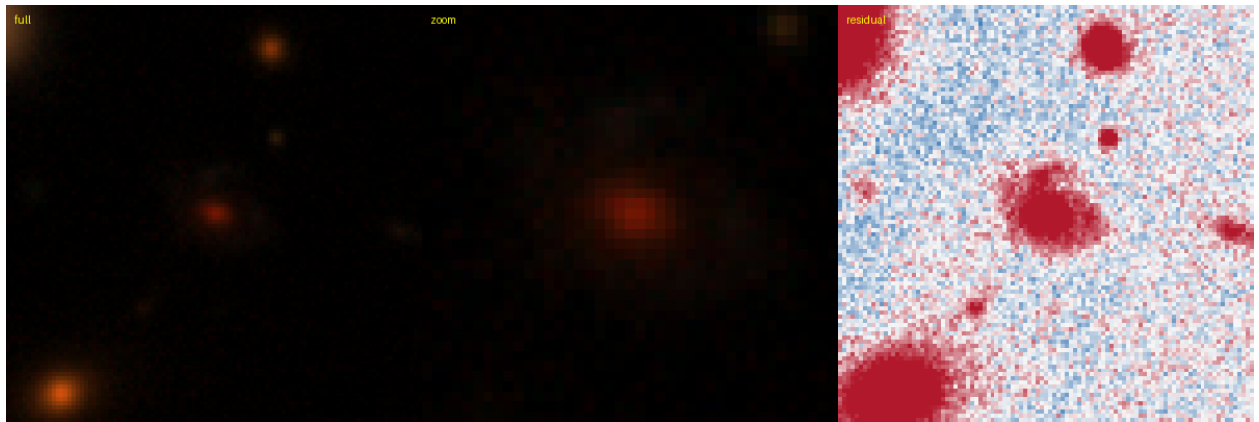


Figure 94: #59 s_217461_3974 — RA 12.0451, Dec -19.9206; grade **B**, tier-1 p_{lens} 0.50, v3blend8 0.78. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.



Figure 95: #60 s_260341_325 — RA 9.9798, Dec -12.2097; grade **B**, tier-1 p_{lens} 0.50, v3blend8 0.84. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

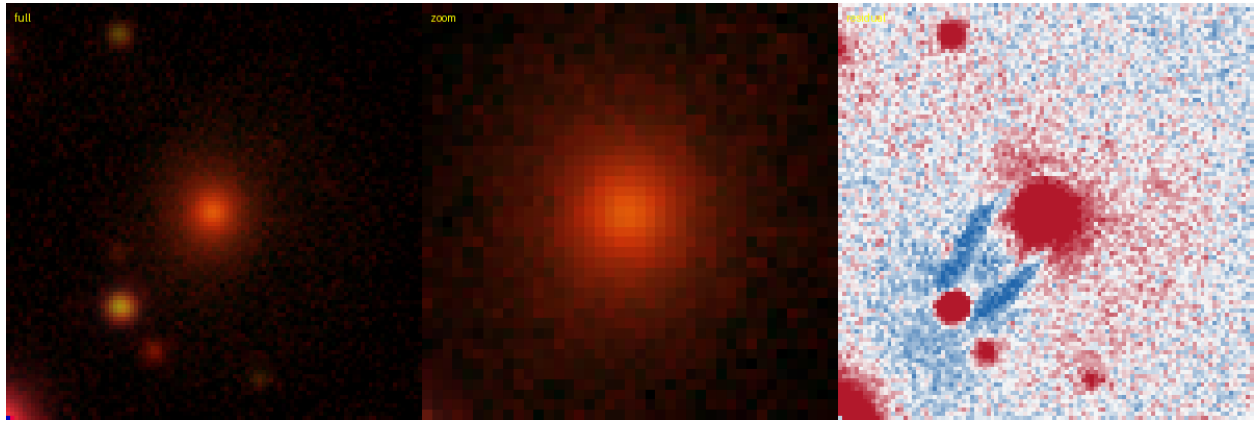


Figure 96: #61 s_207576_2272 — RA 248.1195, Dec -22.0355; grade **B**, tier-1 p_{lens} 0.48, v3blend8 0.76. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.



Figure 97: #64 s_283917_2408 — RA 235.6200, Dec -8.3490; grade **B**, tier-1 p_{lens} 0.45, v3blend8 0.89. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

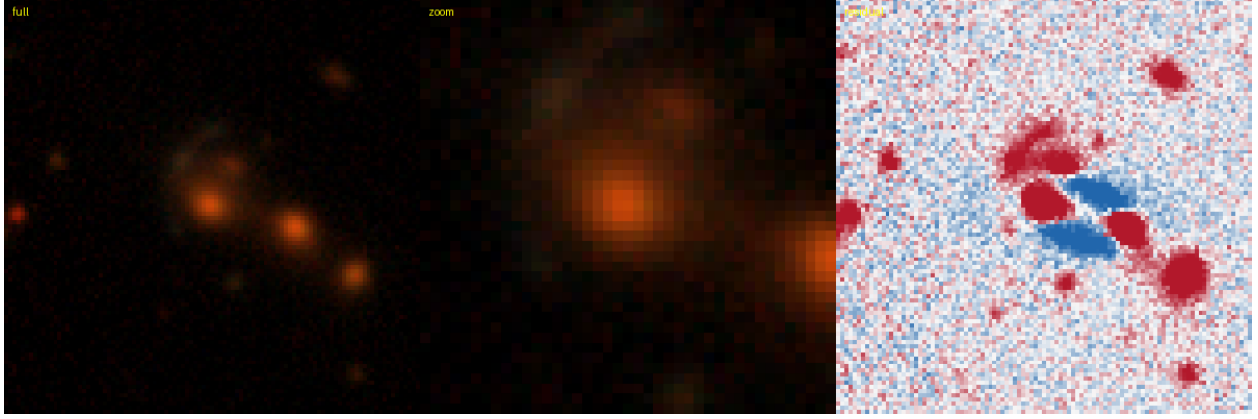


Figure 98: #68 s_148704_6732 — RA 183.8176, Dec -33.6006; grade **B**, tier-1 p_{lens} 0.45, v3blend8 0.80. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

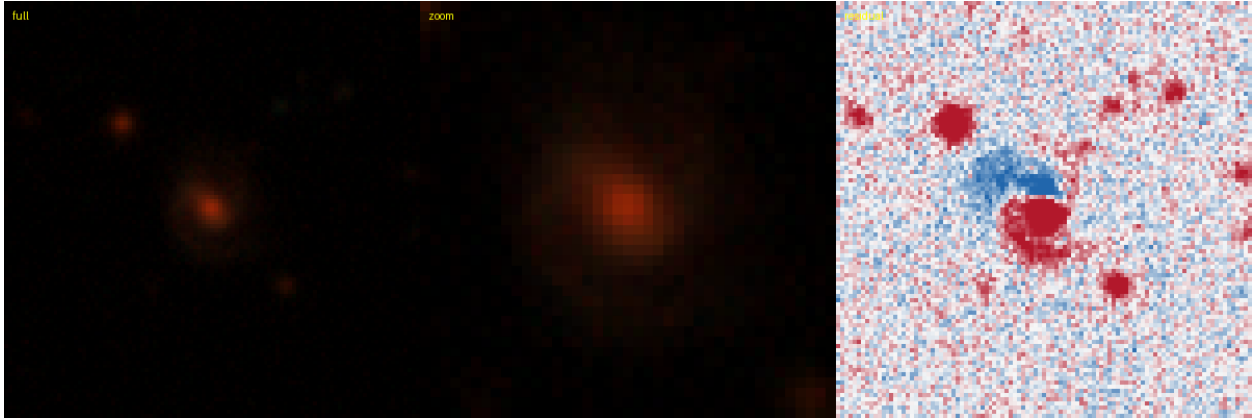


Figure 99: #70 s_266046_1185 — RA 25.9989, Dec -11.2770; grade **B**, tier-1 p_{lens} 0.45, v3blend8 0.76. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

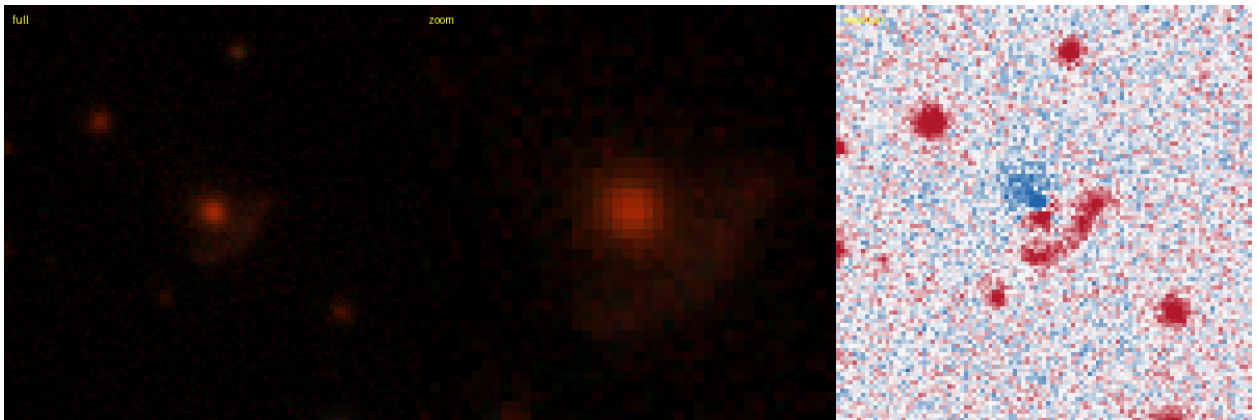


Figure 100: #71 s_251013_4202 — RA 137.4230, Dec -13.9601; grade **B**, tier-1 p_{lens} 0.45, v3blend8 0.79. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

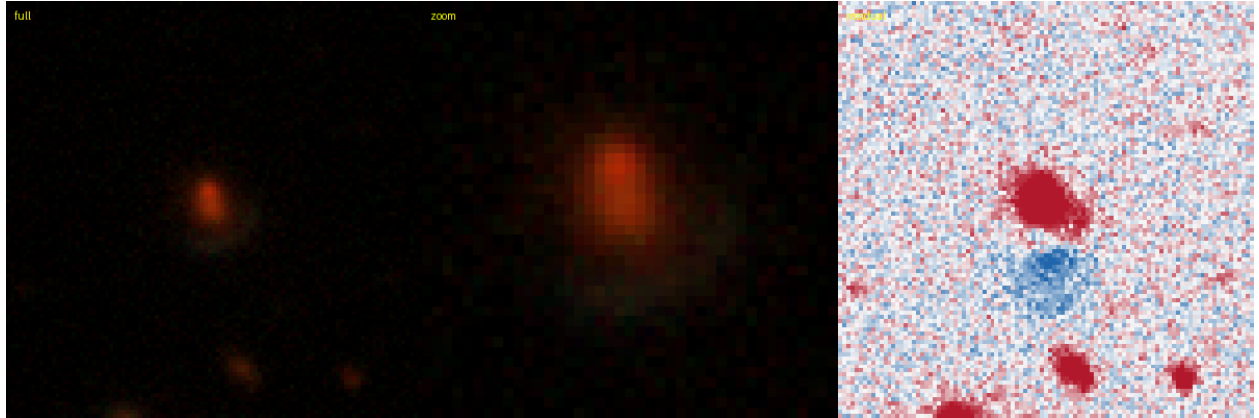


Figure 101: #73 s_209853_1103 — RA 139.8053, Dec -21.4604; grade **B**, tier-1 p_{lens} 0.45, v3blend8 0.85. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

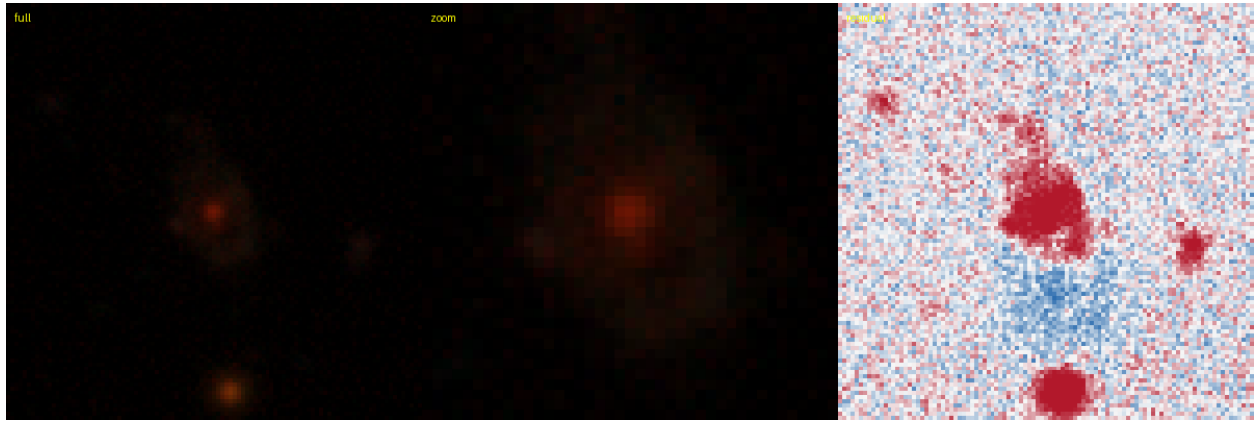


Figure 102: #75 s_181762_9592 — RA 11.4347, Dec -26.6294; grade **B**, tier-1 p_{lens} 0.45, v3blend8 0.86. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

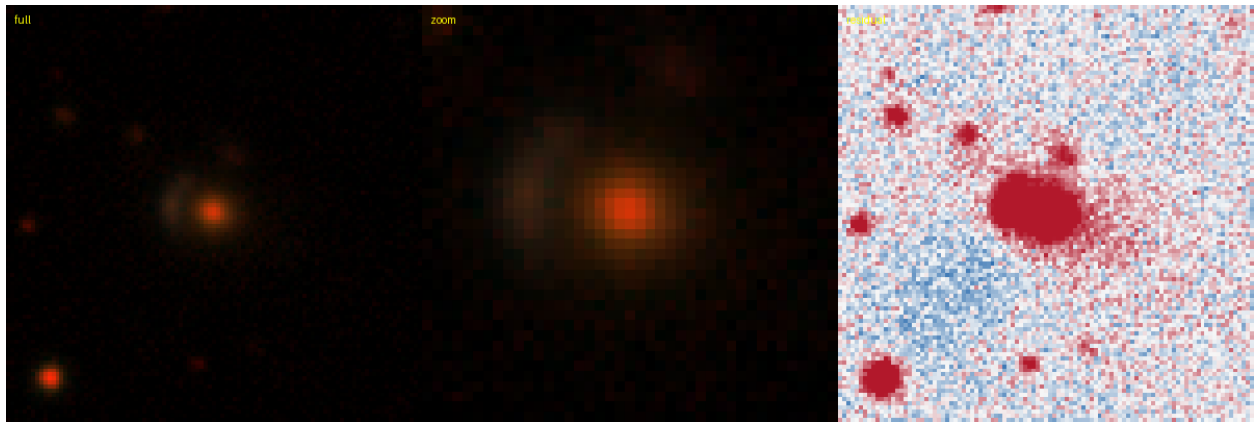


Figure 103: #79 s_23764_4161 — RA 132.6603, Dec -68.1876; grade **B**, tier-1 p_{lens} 0.42, v3blend8 0.79. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

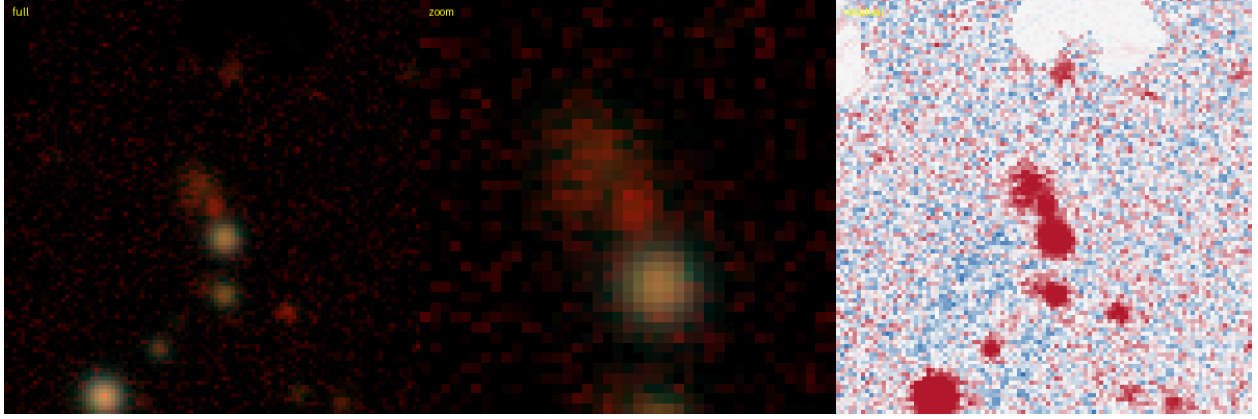


Figure 104: #80 s_29019_24056 — RA 248.9430, Dec -66.0747; grade **B**, tier-1 p_{lens} 0.42, v3blend8 0.74. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

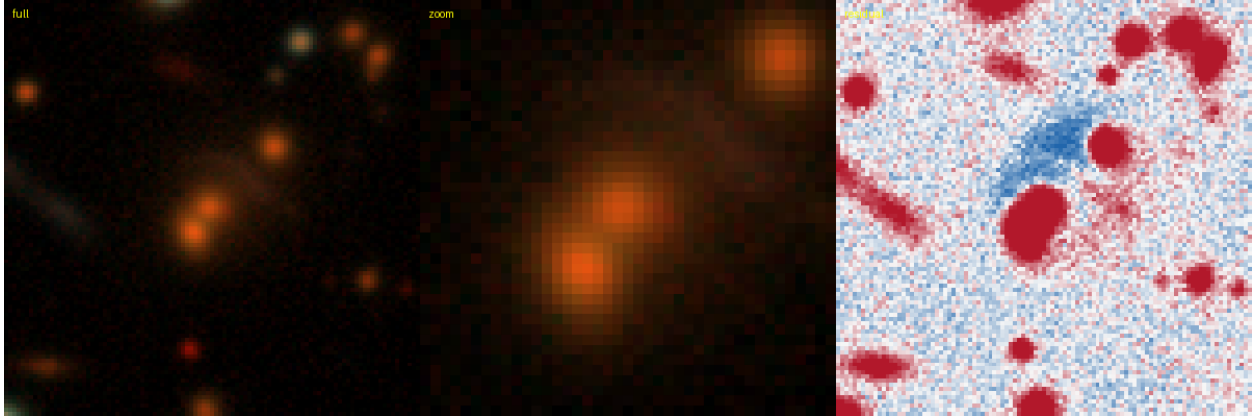


Figure 105: #82 s_112158_2867 — RA 209.6741, Dec -41.4932; grade **B**, tier-1 p_{lens} 0.42, v3blend8 0.83. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

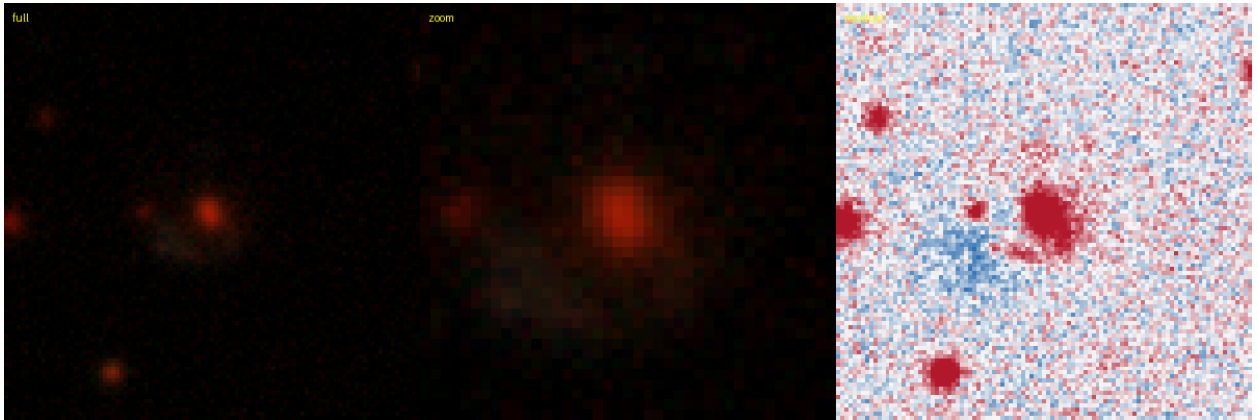


Figure 106: #86 s_405220_1292 — RA 136.4655, Dec +13.0307; grade **B**, tier-1 p_{lens} 0.42, v3blend8 0.89. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

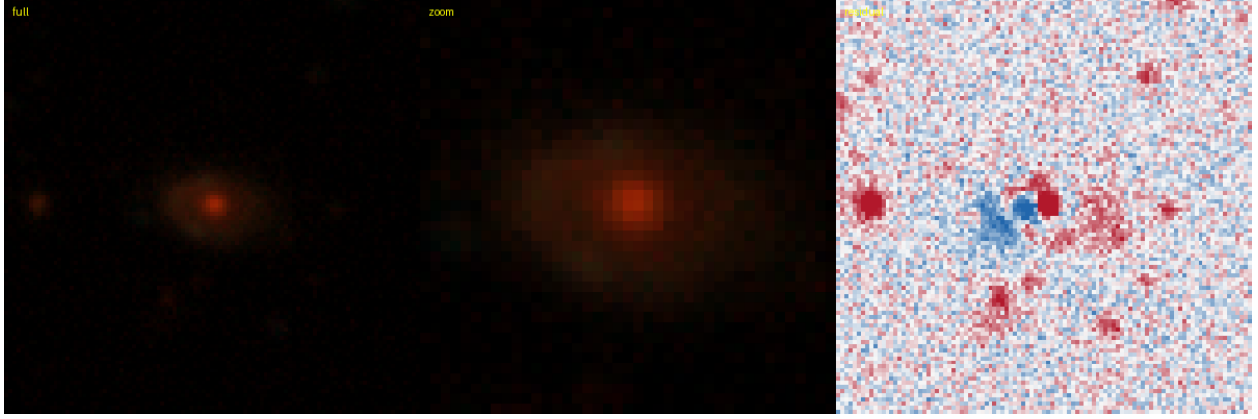


Figure 107: #90 s_188312_3322 — RA 33.2537, Dec -25.3945; grade **B**, tier-1 p_{lens} 0.40, v3blend8 0.76. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

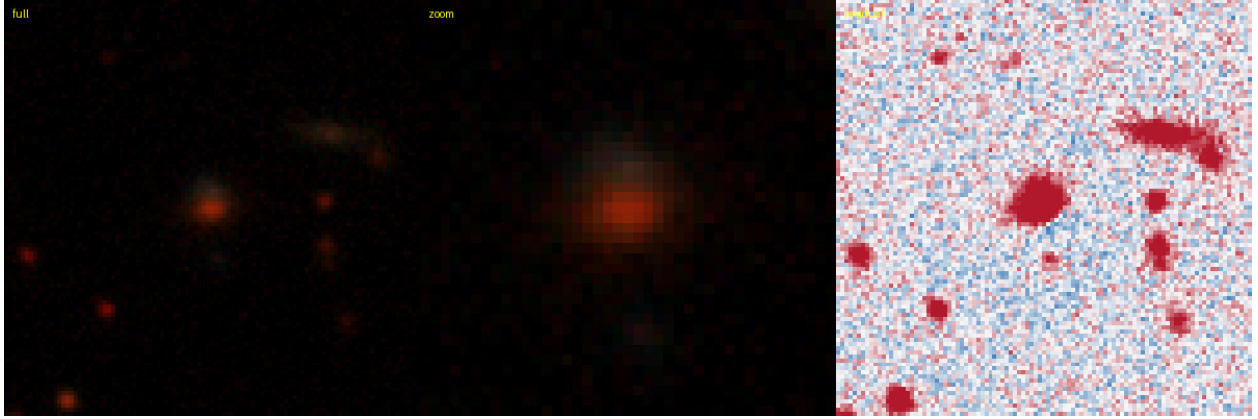


Figure 108: #91 s_120781_3317 — RA 156.8525, Dec -39.6093; grade **B**, tier-1 p_{lens} 0.40, v3blend8 0.83. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

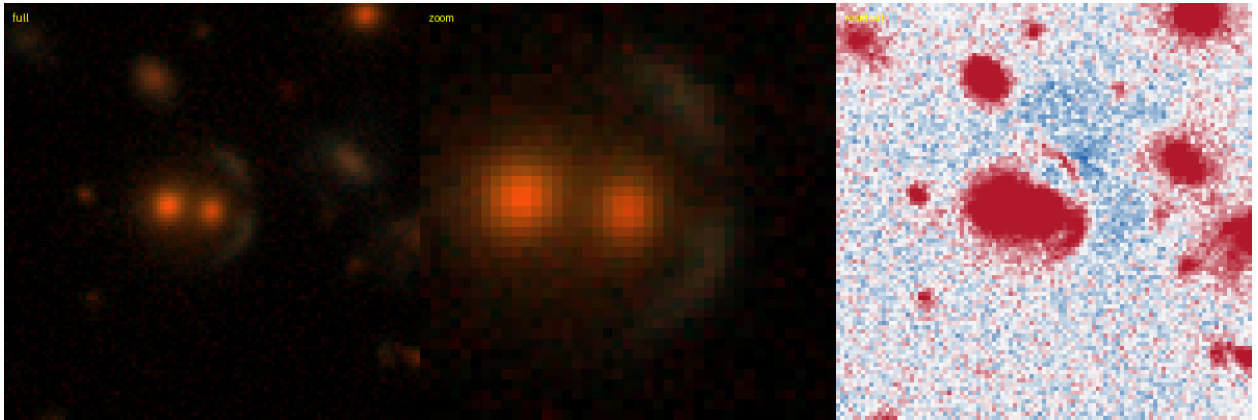


Figure 109: #92 s_266620_918 — RA 172.1320, Dec -11.3108; grade **B**, tier-1 p_{lens} 0.40, v3blend8 0.80. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.



Figure 110: #93 s_410935_2097 — RA 164.1204, Dec +14.0606; grade **B**, tier-1 p_{lens} 0.38, v3blend8 0.83. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

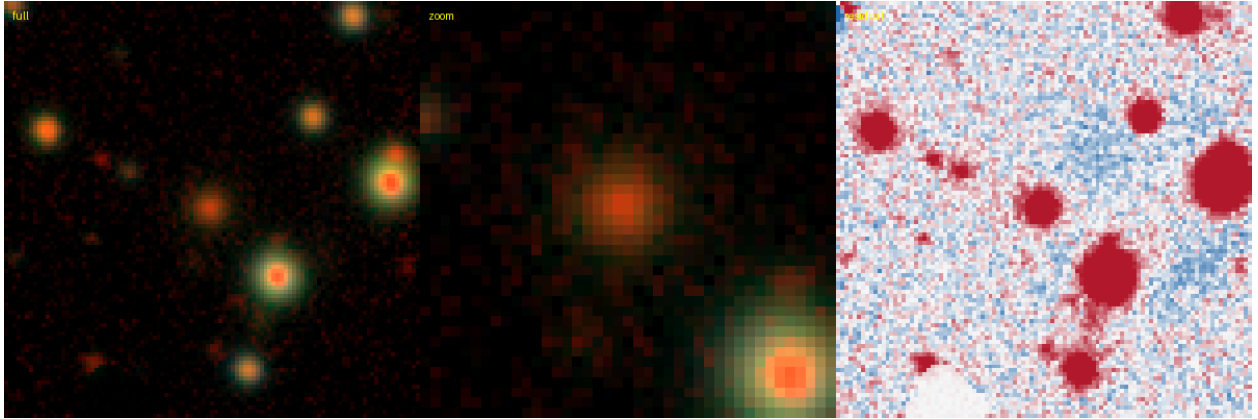


Figure 111: #94 s_32666_26056 — RA 248.6090, Dec -64.5569; grade **B**, tier-1 p_{lens} 0.38, v3blend8 0.73. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

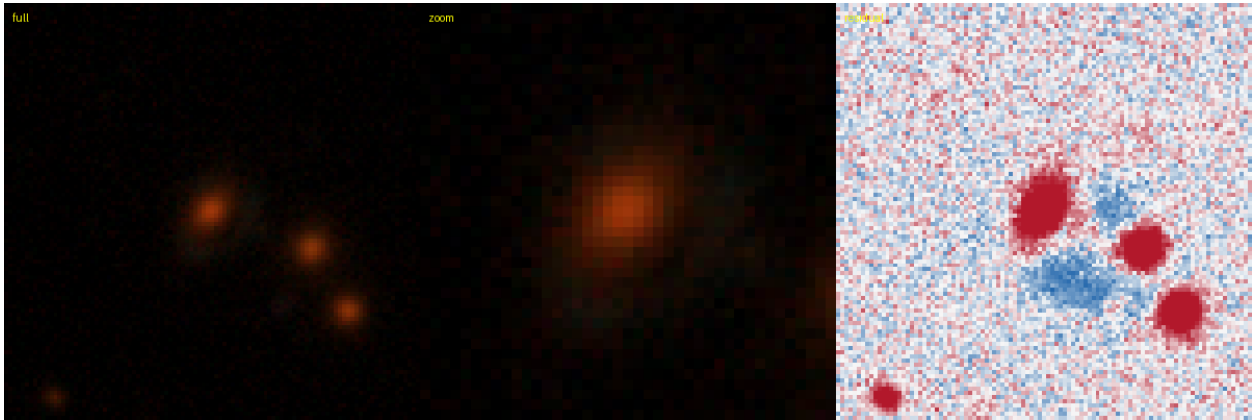


Figure 112: #95 s_244220_4253 — RA 184.3042, Dec -15.2610; grade **B**, tier-1 p_{lens} 0.38, v3blend8 0.80. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

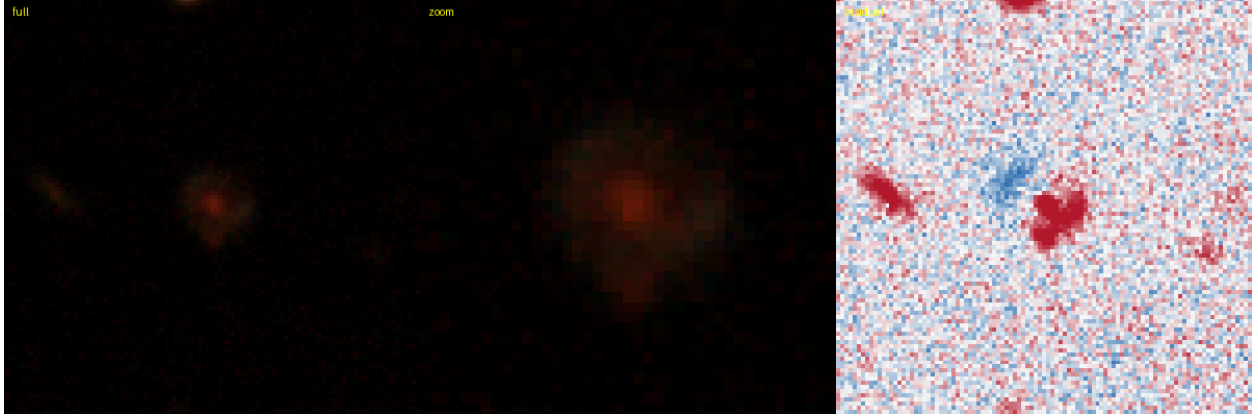


Figure 113: #96 s_119499_8020 — RA 101.8093, Dec -39.8359; grade **B**, tier-1 p_{lens} 0.38, v3blend8 0.75. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

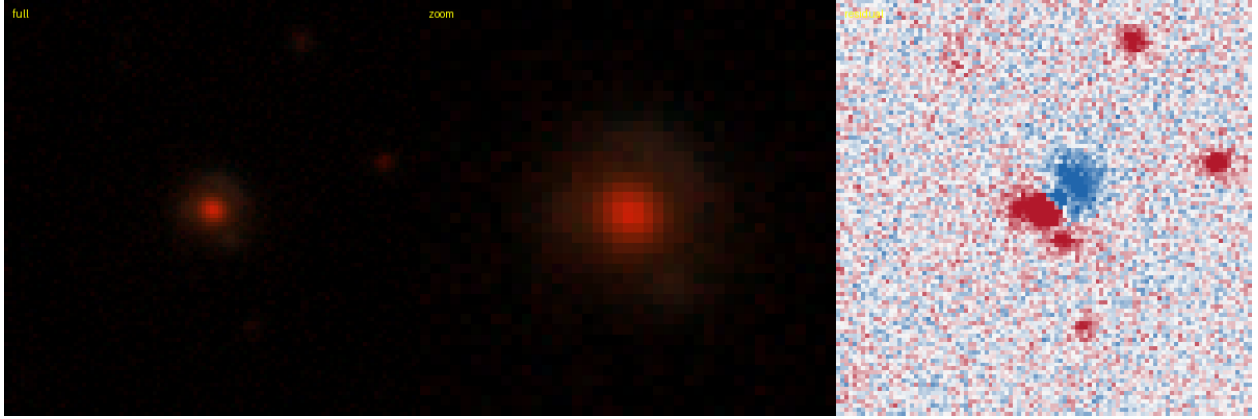


Figure 114: #98 s_205319_6061 — RA 359.9772, Dec -22.4596; grade **B**, tier-1 p_{lens} 0.38, v3blend8 0.88. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

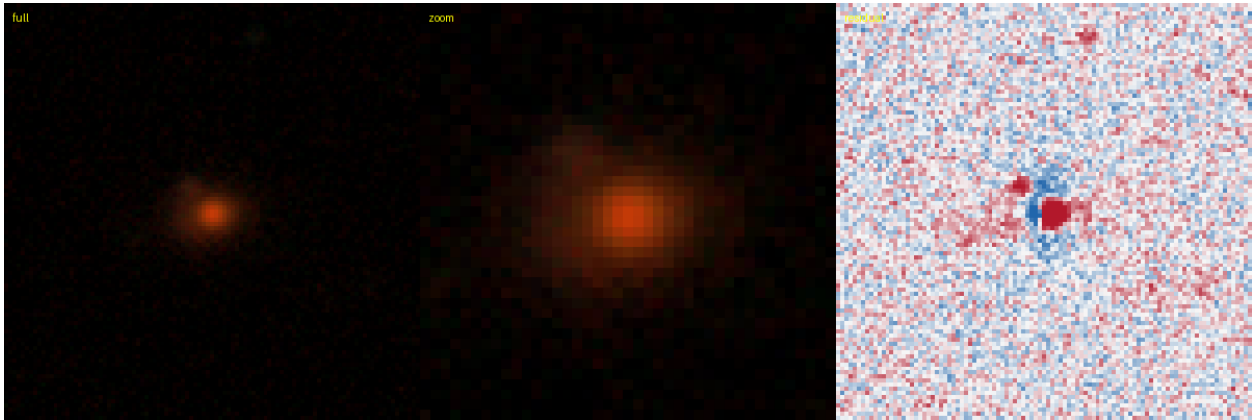


Figure 115: #100 s_364281_4525 — RA 207.3133, Dec +5.6308; grade **B**, tier-1 p_{lens} 0.35, v3blend8 0.90. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

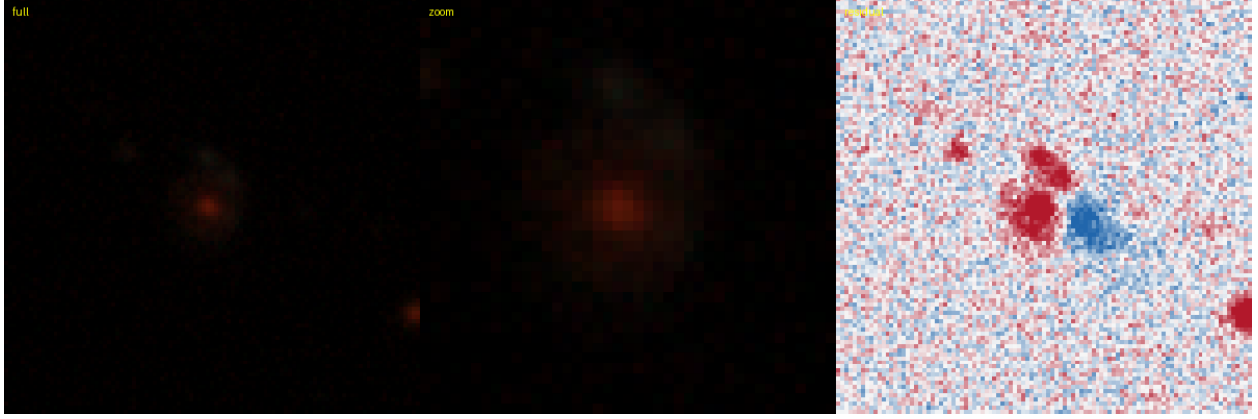


Figure 116: #101 s_250611_2161 — RA 33.7898, Dec -14.1185; grade **B**, tier-1 p_{lens} 0.35, v3blend8 0.77. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

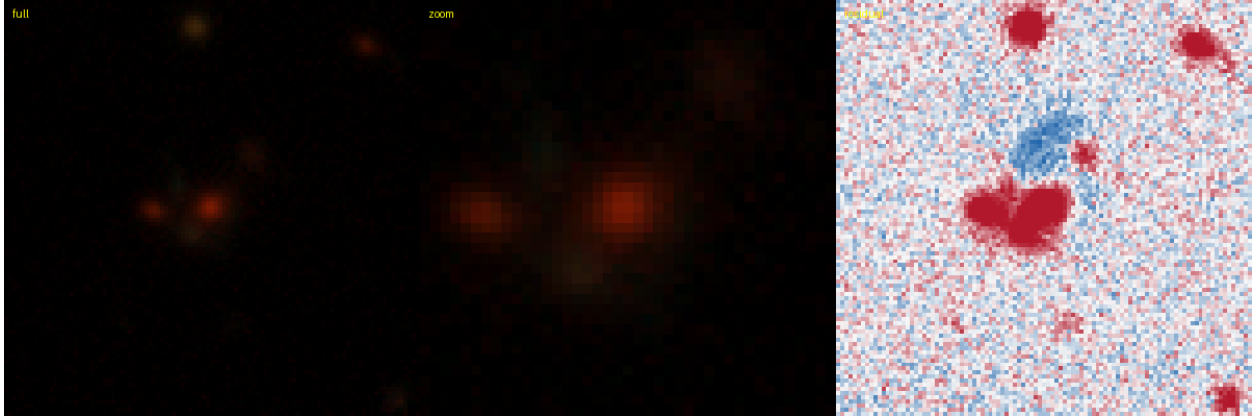


Figure 117: #102 s_212207_5081 — RA 50.6597, Dec -20.9567; grade **B**, tier-1 p_{lens} 0.35, v3blend8 0.84. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

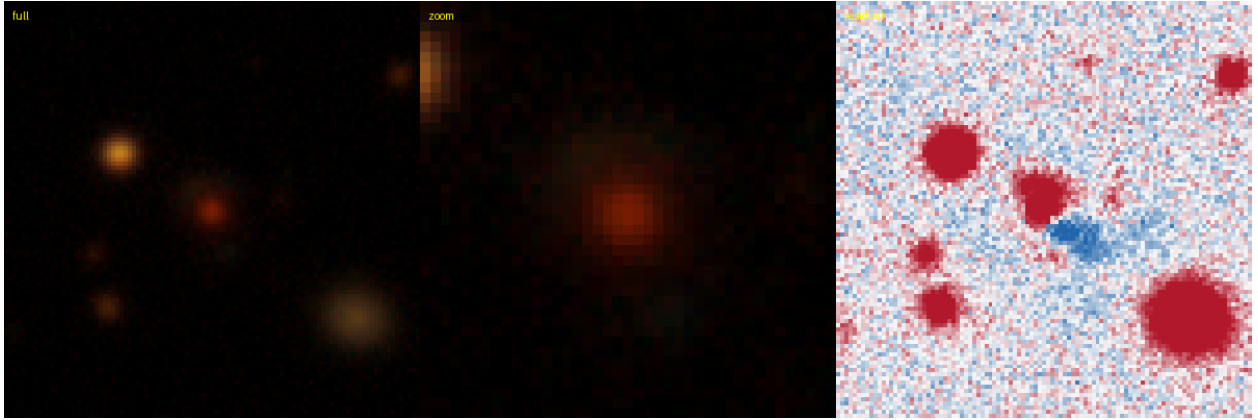


Figure 118: #103 s_128091_3374 — RA 332.6355, Dec -37.9263; grade **B**, tier-1 p_{lens} 0.35, v3blend8 0.85. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

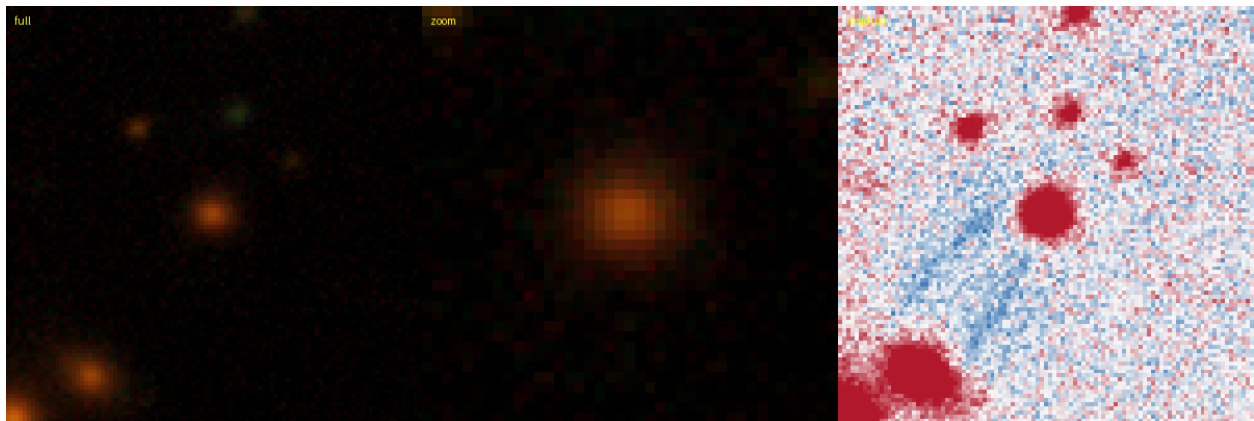


Figure 119: #108 s_369977_4434 — RA 198.5174, Dec +6.6887; grade **B**, tier-1 p_{lens} 0.28, v3blend8 0.75. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

Table 5: The 24 confirmed tier-2 grade-A/B systems tracked across the three residual arms (grade / p_{lens}). **All 9/9 new-class campaign discoveries survive as A/B under the honest residual;** 13/15 SuGOHI cross-matches do likewise. $\downarrow$ marks the two SuGOHI lenses the honest grader scores below A/B (one already lost to nondeterminism in the legacy-now control); both are independently-confirmed lenses, i.e. conservative grading, not campaign candidates.

name	class	prior	legacy-now	new (χ)
s_341950_6798	SuGOHI	A/0.97	A/0.95	A/0.96
s_340513_3688	NEW	A/0.97	A/0.95	A/0.97
s_328891_7514	SuGOHI	A/0.97	A/0.93	A/0.93
s_331753_3591	SuGOHI	A/0.95	A/0.97	A/0.97
s_336034_1750	SuGOHI	A/0.95	A/0.97	A/0.97
s_336034_1788	SuGOHI	A/0.95	A/0.95	A/0.95
s_318982_1451	SuGOHI	A/0.93	A/0.93	A/0.93
s_327526_9059	NEW	A/0.92	A/0.88	A/0.80
s_351303_461	NEW	A/0.92	A/0.85	A/0.88
s_345385_1515	NEW	A/0.92	A/0.90	A/0.92
s_323786_3896	SuGOHI	A/0.91	D/0.04	A/0.88
s_327403_2882	SuGOHI	A/0.91	A/0.90	A/0.92
s_340971_2860	NEW	A/0.91	A/0.88	A/0.85
s_345385_1514	NEW	A/0.87	A/0.88	A/0.87
s_351084_1943 $\downarrow$	SuGOHI	A/0.87	A/0.83	C/0.22
s_332668_5796	SuGOHI	A/0.87	B/0.72	A/0.83
s_326089_1785	NEW	A/0.82	D/0.04	B/0.65
s_325163_2981	NEW	A/0.82	A/0.82	A/0.82
s_300291_2747	SuGOHI	A/0.82	A/0.75	A/0.87
s_325489_4248	SuGOHI	B/0.68	A/0.80	B/0.58
s_329032_842	SuGOHI	B/0.65	B/0.58	A/0.78
s_336801_4965	SuGOHI	B/0.65	D/0.05	B/0.65
s_318043_1037	NEW	B/0.45	B/0.40	B/0.48
s_346996_4322 $\downarrow$	SuGOHI	B/0.40	D/0.03	C/0.08

C Residual-robustness A/B: the confirmed catalogue across residual arms

Table 5 tracks all 24 confirmed tier-2 grade-A/B systems (§3: the 9 new-class detections of Table 2 plus the 15 SuGOHI cross-matches) across the three arms of the paired residual re-run (§3, “Residual robustness”): the stored legacy vetting run, a legacy-residual rerun (the LLM-nondeterminism control), and the new honest- χ run. All 9 new-class campaign discoveries survive as A/B under the honest residual. Reproduced by `compare_residual_ab.py` from the three cascade parquets.